

**DYNAMICS OF MITOCHONDRIAL GENOME EVOLUTION IN
THE LOWER METAZOAN PHyla PORIFERA, CNIDARIA
AND CTENOPHORA**

by

Russell Floyd Watkins

B.Sc., University of Victoria, 1988

**THESIS SUBMITTED IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
DOCTOR OF PHILOSOPHY**

**in the Department
of
Molecular Biology and Biochemistry**

**© Russell F. Watkins 2001
SIMON FRASER UNIVERSITY**

June 2001

**All rights reserved. This work may not be
reproduced in whole or in part, by photocopy
or other means, without permission of the author.**



**National Library
of Canada**

**Acquisitions and
Bibliographic Services**

**385 Wellington Street
Ottawa ON K1A 0N4
Canada**

**Bibliothèque nationale
du Canada**

**Acquisitions et
services bibliographiques**

**385, rue Wellington
Ottawa ON K1A 0N4
Canada**

Your file Votre référence

Our file Notre référence

The author has granted a non-exclusive licence allowing the National Library of Canada to reproduce, loan, distribute or sell copies of this thesis in microform, paper or electronic formats.

L'auteur a accordé une licence non exclusive permettant à la Bibliothèque nationale du Canada de reproduire, prêter, distribuer ou vendre des copies de cette thèse sous la forme de microfiche/film, de reproduction sur papier ou sur format électronique.

The author retains ownership of the copyright in this thesis. Neither the thesis nor substantial extracts from it may be printed or otherwise reproduced without the author's permission.

L'auteur conserve la propriété du droit d'auteur qui protège cette thèse. Ni la thèse ni des extraits substantiels de celle-ci ne doivent être imprimés ou autrement reproduits sans son autorisation.

0-612-61692-4

Canada

Abstract

The dynamics of mitochondrial genome evolution in the basal metazoan taxa and its relationship to that of the bilateral animals has not been fully characterized. This information is critical to placing the characteristics of the higher metazoan mitochondrial genomes into context with the history of the metazoan radiation. The rate of sequence evolution, the frequency of genome rearrangements, and the genome content of the basal metazoan phyla must be characterized in order to understand the constraints which have been functional in shaping the nature of the mitochondrial genomes of the higher metazoan taxa.

In this thesis I isolate and sequence the *rrnL*, *cox2*, *atp8*, *atp6*, and *cox3* genes from several demosponges and calcareous sponges, representing two of the three poriferan classes and *cox2* and *rrnL* from two ctenophoran species. When combined with published data available for all cnidarian classes, this represents the first full mitochondrial data sets available for comparison of the evolution of mitochondrial genomes in the basal metazoan taxa. As well, a 2.5 kb to 3.5 kb orthologous section of the mitochondrial genomes of several demosponges and one calcareous sponge are isolated and fully sequenced, allowing for preliminary characterization of the frequency of gene rearrangements in sponge mitochondrial genomes relative to other published metazoan mitochondrial genomes in an ancient and highly conserved section of the genome.

I show that in terms of nucleotide bias, codon usage, and gene content and structure the mitochondrial genomes of the sponges fall within the typical spectrum of variation found in the metazoan radiation. The sponges nonetheless cluster with the Cnidaria much more strongly and on shorter internal and terminal branches than might be expected a

priori. This clustering is supported at high bootstrap levels for both nucleotide and amino acid data, and is also supported by surprisingly low levels of transitional saturation in all diploblast-to-diploblast pairwise comparisons.

Careful analysis of rates of sequence evolution in the Porifera, Cnidaria, and Ctenophora reveals that the Demospongiae, Calcarea, and the cnidarian class Anthozoa together share very low rate of sequence evolution in their mitochondrial genes relative to the triploblast animals. The *rrnL* gene is very widely sampled in both the Cnidaria and the triploblasts, and my addition of sequences from several sponges and one ctenophoran thus allows analysis of a full metazoan data set. Despite being by itself uninformative for phylogeny inference, the data set reveals that the sponges and the Anthozoa together cluster on a set of short internal and terminal branches, and that the triploblast metazoa cluster at the end of an extremely long branch, implying both a period of accelerated sequence evolution and a rate which likely remains currently high. A more detailed analysis of sequence evolution for the Poriferan and the Anthozoa can be made using protein-coding gene sequences, and these reveal an identical and strongly supported set of trends for both nucleotide and amino acid sequence evolution. Support is found for a low mitochondrial rate in the sponges and Anthozoa, for a long internal branch leading to the triploblasts, and for a currently high rate in the triploblasts relative to the sponges and Anthozoa. The causative factors for this low diploblast rate are not apparent, but preliminary analysis weakly supports no strong difference in purifying selection between the groups.

Acknowledgments

I want to thank my wife Heather for her patience and support, and Stephanie just for being there.

My parents, Marion Watkins and Floyd Watkins for their help and support.

Margot Tubman, Thomas Broeren, Deborah Tubman, Shaun and Christine Tubman, Bob and Dorothy Tubman, Evelyn Galvin.

Andrew Nimmo and Tom Torrie for increasingly hot curry.

My supervisor, Andy Beckenbach, and my supervisory committee, Michael J. Smith and Felix Breden.

For help and assistance in my laboratory work I want to thank Karen Beckenbach, Cam Muir, Allan Arndt, Denise Reichow, Jim Stewart, Andrea Scouras, Bryan Crawford, Alison Cloutier, Michael Vankoevinge, Jeff Bryer, Graham Corely-Smith, Steve Halford, and Doug Swanston.

For discussion and assistance I'd also like to acknowledge Jeff Boore, Timothy Beagley, Jean Vacelet, Bill Austin, Gert Woerheide and Dave Baillie

I want to gratefully acknowledge Bill Johnson, Glenna Kari, and McDonald's Restaurants of Canada Limited., for their support and help when it was needed.

Table of Contents

Approval	ii
Abstract	iii
Acknowledgments	v
Table of Contents	vi
List of Tables	ix
List of Figures	x
Chapter 1 Introduction.....	1
The Question of Metazoan Origins.....	1
Evolution of the Mitochondrial Genome.....	12
Goals of this Study	21
Chapter 2 Methods and Materials.....	26
Sources of Materials	26
DNA Extraction.....	26
Primer Design	30
PCR	31
DNA Cloning.....	35
DNA Sequencing.....	36
Southern Blotting	36
Sequence Analysis	37
Chapter 3 Isolation and Characterization of a 3.5 kb Demosponge mitochondrial DNA fragment	39
Gross Characterization of the Sequence	39
Verification of Amplified DNA Source.....	40
Protein-Coding Genes	53

Codon Usage, Nucleotide Bias, Genetic Code.....	58
The tRNA-K and tRNA-R Genes	66
The lrRNA Gene	70
Discussion.....	71
Chapter 4 Variability in Mitochondrial Genome Organization Within the Diploblast Radiation.....	73
Introduction	73
Methods.....	75
Genome Size	76
Restriction Mapping.....	79
Genome Organization	80
The tRNA Gene Sequences	86
Noncoding Sequences	86
Discussion.....	96
Chapter 5 Mitochondrial Sequence Evolution Rate Variation Across the Diploblast Radiation I: 16S rRNA.....	99
Introduction	99
Amplification and Sequencing.....	103
Sequence Alignment	106
Sequence Evolution Across the Metazoan Radiation	114
Construction of Phylogenetic Trees	114
Transitional Saturation	123
Pairwise Genetic Distances	128
Relative Rate Tests	136
Relative Rates.....	145
Sequence Evolution Within the Diploblast Radiation	146
Discussion.....	154

Chapter 6 Mitochondrial Sequence Evolution Rate Variation Across the	
Diploblast Radiation II: Protein-Coding Genes	161
Introduction	161
Amplification, Alignment, Nucleotide Composition.....	164
Nucleotide Bias.....	170
Construction of Phylogenetic Trees.....	178
Relative Rates of Sequence Evolution	189
Synonymous and Nonsynonymous Rates	204
Discussion.....	210
Bibliography.....	218

List of Tables

1	Sample Organisms	27
2	Taxonomic classification of species used in this study	29
3	PCR primers used in this study	33
4	Comparison of mitochondrial protein-coding genes at the amino acid level	60
5	Nucleotide composition of <i>Tetilla</i> sp protein-coding genes	62
6	Codon composition and relative synonymous codon usage values	63
7	Aligned poriferan tRNA genes from all sequence mitochondrial genome segments	88
8	Compilation of intergenic noncoding sequences for all sponge mitochondrial genome sequences	91
9	Number of parsimoniously informative sites in mitochondrial 16S rDNA alignments	113
10	Average pairwise gamma-corrected Kimura 2-parameter transitional + transversional distances between sets of taxa	130
11	Average pairwise gamma-corrected Kimura 2-parameter transversional distances between sets of taxa	132
12	Results of the 2-cluster relative rate test as calculated with the PHYLTEST package comparing diploblast to triploblast lineages	141
13	Relative rates of mitochondrial <i>mtL</i> sequence evolution between diploblast and triploblast taxa	144
14	Pairwise gamma-corrected Kimura 2-parameter transitional + transversional distances between diploblast taxa	147
15	Results of the 2-cluster relative rate test as calculated with the PHYLTEST package comparing diploblast to triploblast lineages with the reduced data set	153
16	Relative rates of mitochondrial <i>mtL</i> sequence evolution between diploblast taxa	155
17	Species, genes, and accession numbers utilized in this study	166
18	Nucleotide composition of mitochondrial protein-coding genes	169
19	Relative synonymous codon usage table for mitochondrial protein-coding genes	175
20	Incomplete codon adaption indices for mitochondrial protein-coding genes	177
21	Results of 2-cluster relative rate tests for mitochondrial protein-coding genes	199
22	Results of constrained 2-cluster relative rate tests for nucleotide and amino acid sequences for mitochondrial protein-coding genes	201
23	Relative rates of mitochondrial protein-coding gene evolution in the metazoa	203

List of Figures

1	Issues in timing metazoan origins	6
2	Results of numerous studies on the basal metazoan radiation	10
3	Nucleotide sequence of the 3550-bp <i>Tetilla</i> sp. band	44
4	Phylogenetic trees constructed from <i>cox2</i> data	47
5	Phylogenetic trees constructed from <i>atp6</i> data	48
6	Phylogenetic trees constructed from <i>cox3</i> data	49
7	Neighbour-joining tree derived from cnidarian, echinoderm, and poriferan 16S rRNA sequences	50
8	Bootstrap 50% majority-rule consensus tree derived by maximum parsimony from mitochondrial 16S rRNA sequences	51
9	Aligned full <i>cox2</i> amino acid sequences	54
10	Aligned full <i>atp6</i> amino acid sequences	55
11	Aligned full <i>cox3</i> amino acid sequences	56
12	Kyte-Doolittle hydropathy plot of <i>atp8</i> and <i>atp6</i> sequences	57
13	Secondary structure of tRNA-Lys	68
14	Secondary structure of tRNA-Arg	69
15	Southern blot of restriction-digested whole CsCl-purified <i>Tetilla</i> sp. genomic DNA extraction probed with a mitochondrial band	78
16	Restriction map of whole <i>Tetilla</i> sp. mitochondrial genome	82
17	Relative organizations of orthogous sections of the mitochondrial genomes of four demosponges and one calcisponge	85
18	RNA foldings of noncoding intergenic mitochondrial sequences	93
19	RNA foldings of noncoding intergenic mitochondrial sequences	95
20	Mitochondrial 16S rDNA gene alignments across diploblast and triploblast radiations	110
21	Neighbour-joining tree derived from the aligned 16S rDNA sequences	116
22	Metazoan constrained backbone tree used for maximum likelihood reconstructions	120
23	Constrained maximum likelihood tree derived from 16S rDNA gene data	122
24	Transitional saturation graph for aligned triploblast and diploblast mt 16S rDNA genes	125
25	Histogram of pairwise Kimura 2-parameter gamma distances between sets of taxa	135
26	Constrained maximum likelihood tree generated from a reduced diploblast and triploblast data set	139

27	Neighbour-joining tree derived from aligned diploblast 16S rDNA gene data	148
28	Constrained maximum likelihood tree derived from aligned diploblast 16S rDNA gene data	150
29	Correlations between purines and pyrimidines in mitochondrial protein-coding genes	172
30	Phylogenetic trees derived from mitochondrial protein-coding gene data	184
31	Neighbour-joining tree derived from mitochondrial atp6 dna data	186
32	Constrained maximum likelihood trees derived from translated amino acid sequence data	188
33	Intertaxon distances between early-diverging primary metazoan lineages	197
34	Concordance between nonsynonymous changes per nonsynonymous site for mitochondrial protein-coding genes as estimated by a number of methods	209
35	Pairwise nonsynonymous changes per nonsynonymous site plotted against pairwise Kimura 2-parameter gamma distance	213

Chapter 1

Introduction

The question of metazoan origins

If we wish to understand the broad-scale pattern of evolution of the metazoan mitochondrial genome, and particularly its earliest evolution and the significance of any differences between it and the structure and organization of protist mitochondrial genomes, then the precise relationships within and between the primary metazoan lineages is a question of some significance. A phylogenetic tree represents a hypothesis onto which we map character states in an attempt to understand their origin and the constraints under which they are maintained. The structure of the tree itself must be meaningful for those inferences to be meaningful.

Despite over a century of inquiry into the earliest origins of the metazoa, some doubts remain as to the relationship of the lower phyla, the Porifera, Cnidaria, Ctenophora, and Placozoa, to one another or to the higher metazoan phyla. Morphological and embryological evidence have brought some light to bear upon the rough outlines of the earliest divergences within the metazoan kingdom, but there is no single outline of early metazoan phylogeny which has gained universal acceptance (Wilmer 1990). Recent investigations of relationships within the metazoa at the level of macromolecular sequence data have begun to shed new light upon the early evolution of the animals and upon the Precambrian evolution of the first multicellular animals as revealed in the fossil

record¹. Major reorganizations of the relationships between phyla within the protostome lineages have been made as a consequence of studies into ribosomal RNA sequence data (Halanych *et al.* 1995, Aguinaldo *et al.* 1997), shedding new light upon the evolution of higher-level morphological features. Some relationships, however, are if anything more obscure now than they were understood to be prior to the advent of molecular phylogenetics. The earliest evolution of the animal kingdom is such a case (Conway-Morris 2000, Adoutte *et al.* 2000).

The most general consensus tree for primary phylogeny of the metazoan kingdom, and one which has not changed substantially for over sixty years despite being based upon the smallest amount of supporting data (Nielsen 1996) can be summarized in the following Newick-format tree:

(Protists, (Porifera, (Placozoa, (Cnidaria, Ctenophora), Triploblasts))));

This is an intuitively obvious tree, and represents a parsimonious interpretation of a gradual increase in complexity of organization of organisms leading up to the triploblast radiation. Recent analyses of molecular sequence data obtained from these phyla, to be discussed below, have not strictly disproved this tree but have provided no clear support for any particular phylogeny. Indeed, the dearth of phylogenetically informative morphological characters in these taxa

¹ I will continue to refer to the Porifera, Cnidaria, Ctenophora, and Placozoa as “diploblast” phyla and the higher metazoans as “triploblasts” for convenience, although the somewhat artificial nature of this distinction has been recognized for over fifty years (Hyman 1940, Willmer 1990), and in any case the Porifera do not fit neatly into this system.

appears to be matched by a lack of informative nucleotide variation as well.

The results of analysis of molecular data must clearly be integrated with our understanding of the fossil record, but the fossil record itself merely provides an upper limit to the time of origin of the metazoa. While there is undoubtedly a major transition in the history of the metazoa marked by the Cambrian Explosion some 543 million years ago (Grotzinger *et al.* 1995) and characterized by a massive increase in the apparent diversity of animal life, recent discoveries have begun to shed light on an extended period of metazoan evolution prior to this point. The widely discussed Ediacaran fauna, which are found in strata ranging from 570 to almost 700 million years in age, have long provided material for an active debate about an extended period of Precambrian metazoan evolution (i.e., Gehling 1991, Seilacher 1992, Retallack 1994, Waggoner 1996). Recent work (Conway-Morris 2000) seems to leave little doubt that at least some of these problematic taxa do indeed have cnidarian affinities. Much more significant is the discovery of Precambrian sponges complete with resolvable cellular ultrastructure (Li *et al.* 1998) along with what appear to be the fossilized embryos of bilateral animals (Xiao *et al.* 1998, Chen *et al.* 2000) and cnidarians (Chen *et al.* 2000). The age of the deposits in which these phosphatized remains are found is about 570 +/- 20 million years, and thus there is strong evidence for an extended Precambrian period of animal evolution. That this period must exist in some form has been long recognized, but the discovery of the apparent embryos of bilateral animals demonstrates that the primary metazoan lineages themselves were well established prior to the Cambrian Explosion itself.

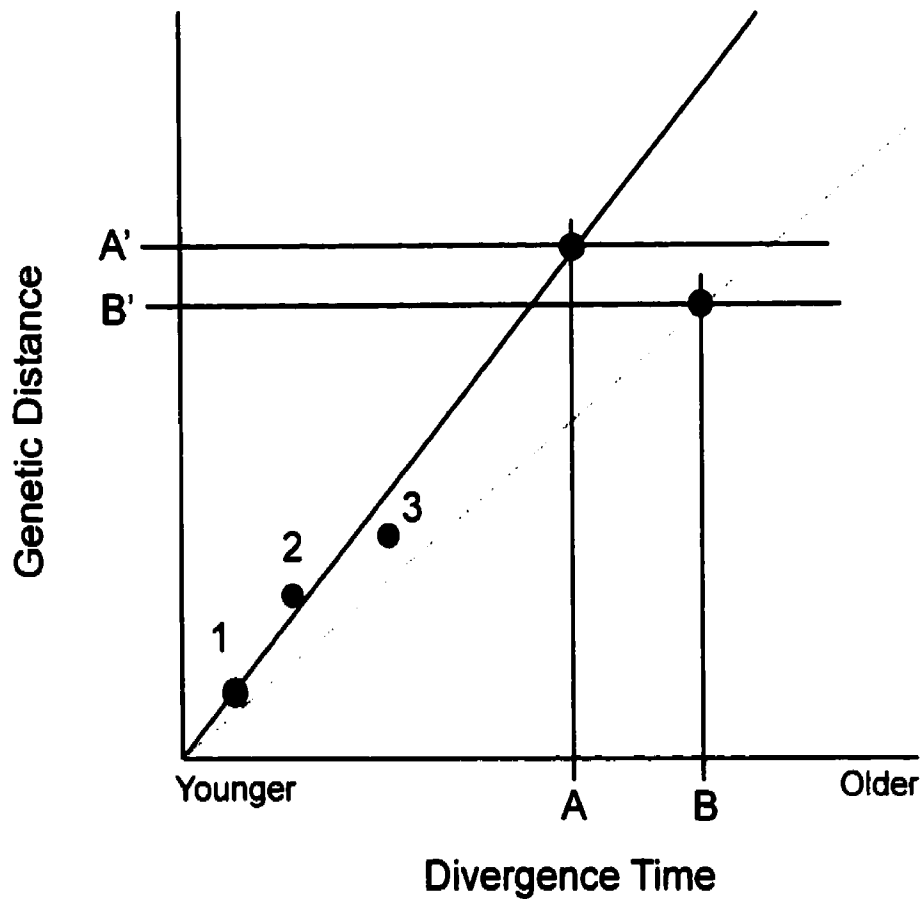
Just how far this period extends into Precambrian history is an open question. Much work has been performed recently in an attempt to infer the timing of the origin of the metazoa as a whole. Wray *et al.* (1996) estimated the age of the protostome/deuterostome split with several mitochondrial protein sequences and found a divergence time in excess of 1200 million years, a strikingly high number. However, this result was criticized (Ayala *et al.* 1998) as being systematically biased and a re-analysis including more data found a lower figure of some 700 million years. In any event the age of the metazoa itself must be greater than this number. Numerous additional studies have been performed using various combinations of sequence data and analysis methods.

Doolittle *et al.* (1996) addressed the question of the origin time for the metazoa itself using an extensive analysis of 57 different protein sequences correlated with known divergences in the fossil record and found a value for the protostome/deuterostome divergence of about 670 million years, compatible with the results of Ayala *et al.* (1998), and a specific age for the metazoan radiation itself of approximately 1000 million years. A further analysis of this data (Feng *et al.* 1997) moved the age of the animals back to some 1200 million years. Additional analysis by Nikoh *et al.* (1997) of just two clocklike genes dated the age of the metazoa to over 900 million years. More recently Wang *et al.* (1999) engaged in an extensive analysis of 50 genes selected for their clocklike behavior and found a metazoan age of over 1200 million years, a much higher number. Nonetheless their data set, as with all of the other studies cited here, did not include sequence data from any diploblast taxa. In light of the long branch on which the triploblast taxa are typically found in studies of the metazoan radiation (see below), the issue

Figure 1

A. Inference of the metazoan origin time using fossil-record/genetic divergence relationships from a set of taxa with a consistently high rate biases the inferred time of divergence towards a lower value. Points 1, 2, and 3 are divergences correlated with the fossil record, whereas point A' is the calculated divergence time between the metazoa and the fungi using the biased, higher rate taxon. Point B' is the genetic divergence between the metazoa and the fungi using the lower-rate basally diverged metazoan taxon. It can be seen that the bias towards a higher rate causes the age of the metazoan root to be underestimated from time B to the lower figure of time A.

B. The results of several studies on the timing of the divergences between metazoan phyla. Not all of these studies examined the age of the metazoan root, but it is obvious that the root age must be greater than that between any higher-level group (typically protostome/deuterostome). All studies examined amino acid divergences, and the table indicates whether or not mitochondrial gene data was included. The study of Wray *et al.* (1996) has been widely criticized for systematic statistical biases.

A**B**

Study	Metazoan Age	Mitochondrial Genes?
Wray <i>et al.</i> 1996	>1200 Myr	Yes
Doolittle <i>et al.</i> 1996	1000 Myr	No
Feng <i>et al.</i> 1997	>1100 Myr	No
Nikoh <i>et al.</i> 1997	>900 Myr	No
Ayala <i>et al.</i> 1998	>700 Myr	Yes
Lynch 1999	>900 Myr	Yes
Wang <i>et al.</i> 1999	>1200 Myr	No
Cutler 2000	>680 Myr	Yes

Note: Cambrian Explosion = 547 +/- 3 Myr (Grotzinger et al. 1995)

of a consistent bias in these estimates has not been addressed. In particular, as is shown in Table 1a, a consistent rate bias can significantly lower the inferred estimate for the age of a taxon.

Mitochondrial sequence data has also been utilized in the studies of Lynch (1999) and Cutler (2000) to estimate the age of the metazoan radiation. These studies found values of over 900 and at least 680 million years, respectively. As will be discussed below, there has been significant indication of rate biases in the triploblast relative to the diploblast radiation, and this in particular could complicate interpretation of metazoan age estimates based on mitochondrial data (see Figure 1a). The methods of Cutler (2000) were developed specifically to address non-clocklike behavior in sequence data, but the effect of consistent, systematic bias upon these methods is not clear. In any even there is still a gap of at least 100 million years between the estimates of metazoan age and the fossil record itself, regardless of the data set used.

Attempts to work out the true relationships in the basal branches of the metazoa have been no more definitive. A range of possible basal metazoan phylogenies has been proposed, often with widely different phylogenies derived from data from the same gene, depending upon the taxonomic sampling and analysis methods employed.

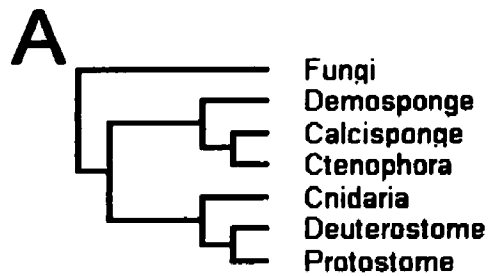
Christen *et al.* (1991) and Lafay *et al.* (1992) examined the metazoan radiation using 28S rRNA sequences and found the first instances of a “diploblast monophyly” tree as shown in Figure 2e and 2f. In each case the split between the diploblast and triploblast radiations is very deep, but is simply attributable to the specific rooting of nonmetazoan outgroups. Lafay *et al.* (1992) also found evidence for strong polyphyly of the sponges themselves, although at low bootstrap

values. A much more robust tree was provided by Wainwright *et al.* (1993) who examined 18S rDNA sequences from a very wide range of metazoan and nonmetazoan taxa. Although the phylogeny they produced (Figure 1b) was compatible with a fairly classical view of metazoan origins, and specifically did not imply the “diploblast monophyly” tree, it was nonetheless somewhat unusual in implying a more basal placement for the Ctenophora than was expected. More importantly, the precise portion of the tree which was most uncertain in the studies of Christen *et al.* (1991) and Lafay *et al.* (1992) was equally uncertain using 18S rDNA sequence – bootstrap values were extremely low within the basal metazoan radiation, and no one precise branching order could be confidently inferred.

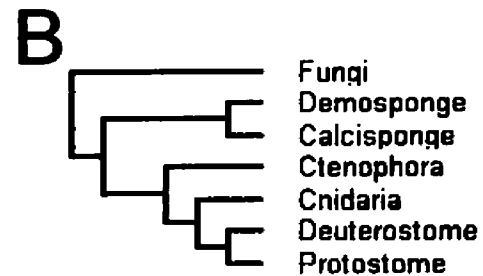
This result was further supported by the work of Kobayashi *et al.* (1993) on a similar data set. Using 18S rDNA data from a smaller subset of taxa, the tree shown in Figure 2A was inferred, once again implying diploblast monophyly, and in this case a closer link between the Calcarea and the Ctenophora, a result which has since been replicated in a number of studies. Nonetheless, in spite of their publication of a diploblast monophyly tree, the bootstrap values were once again extremely low for the basal metazoan divergences, which is in concert with the results of Wainwright *et al.* (1993). Kobayashi *et al.* (1996) went on to examine a highly conserved nuclear protein-coding gene, elongation factor 1-alpha in hopes of providing a more robust early metazoan phylogeny, but once again demonstrated a lack of informative variation between the diploblast lines. The tree shown in Figure 2E was published, but bootstrap values within and between the diploblast phyla.

Figure 2

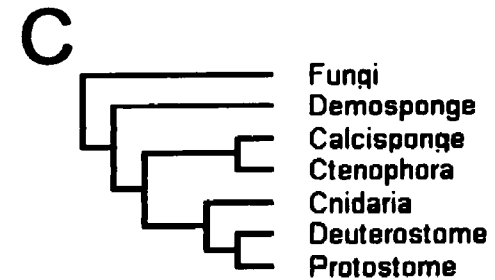
Phylogenetic trees representing the results of numerous studies of the basal phylogeny of the metazoa. The studies which supported each tree are listed below that tree, as is the particular nuclear gene which was examined. Tree B is likely the best supported in terms of maximum likelihood bootstrap support and methodology used for phylogeny inference. Tree E is an “inverted” tree in the sense that the root has been placed between the diploblast and triploblast phyla, and the diploblast phyla appear to diverge in an opposite orientation from that of tree B. The study of Kobayashi *et al.* (1996) did not include the calcisponges and ctenophores but is compatible with tree E. As well, the study of Lafay *et al.* (1992) found paraphyly of the sponges in that some particular lineages rooted inside the Cnidarian/Ctenophoran radiation, but the vast majority of sequences clustered in the manner of tree E.



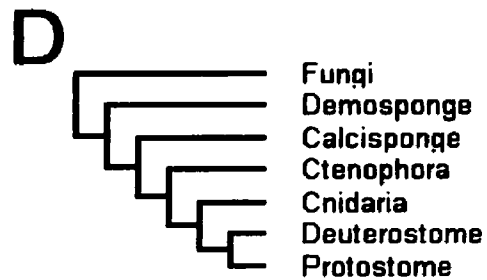
Kobayashi *et al* 1993
18S



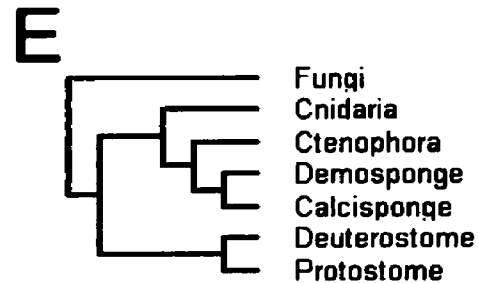
Wainwright *et al* 1993
Kim *et al* 1993
18S



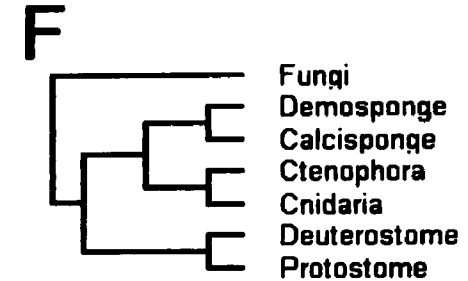
Collins 1998
18S



Cavalier-Smith *et al* 1996
18S



Lafay *et al* 1992
28S
Kobayashi *et al* 1996
EF1-alpha
Borchiellini *et al* 1998
HSP70



Christen *et al* 1991
28S

and the link to the triploblast phyla rarely rose above 50%, once again withdrawing confidence in any particular branching order

Cavalier-Smith *et al.* (1996) sensibly attributed the diploblast monophyly results published to that time to rate variation between the diploblast and triploblast phyla, and based upon a reanalysis of much previously published data produced a tree in strict concordance with a classical view of metazoan evolution as shown in Figure 2D. Bootstrap values were low for many primary metazoan divergences in this study as well. Borchiellini *et al.* (1998) analyzed another conserved nuclear protein-coding gene, HSP70, and found yet another diploblast monophyly tree with low bootstrap support for the internal nodes, as shown in Figure 2E. Collins (1998) also re-analyzed the previously published 18S rDNA data with an eye to establishing a realistic framework for the origin of the bilateral animal phyla and found strong support for a poriferan/ctenophoran clade to the exclusion of a cnidarian/bilateral animal one. This result does not imply an origin of the Ctenophora strictly closer to the Calcarea however, as the results are not strong and rather suggests a somewhat closer deep branching of the Calcarea with the Ctenophora. Nonetheless the results of Collins are broadly consistent with the trees shown in Figures 2a, 2b, 2c, and 2d in implying that the earliest metazoan divergences include a closer relationship of the ctenophores to the sponges than was previously expected, and a closer branching of the Cnidaria with the bilateral animals. It is a critically important point to take note that the diploblast monophyly tree shown in Figure 2E is also compatible with this view if the nonmetazoan outgroup is eliminated and the tree is rooted instead with the Porifera.

The work of Kim *et al.* (1999) produced what may be the most well-supported phylogeny of the primary metazoan taxa published to date. The authors performed an extensive examination of sequence evolution rates in the 18S rRNA gene, and utilized the Aguinaldo Taxon-pruning strategy (Aguinaldo *et al.* 1997) to produce a data set in which any specifically rate-induced artifacts were minimized. This may be an especially important issue for studies which span the diploblast/triploblast divergence in light of the apparent rate differences between these groups for the 18S gene (Cavalier-Smith *et al.* 1996) and possibly for others as well. The authors produced a maximum likelihood tree with much higher bootstrap values for the internal nodes in the diploblast radiation than any other study published to date, and once again confirmed the basal radiation of the Ctenophora to the Cnidaria. However, once again no resolution was possible within the Poriferan radiation and was still quite low for the link between the Porifera and the eumetazoans. Nonetheless the strategy employed in this paper provided a more strongly supported phylogeny for the bulk of the metazoan radiation than any other study before it.

Evolution of the Metazoan Mitochondrial Genome

While the protist phylum Choanoflagellata remains the best potential candidate for a true metazoan outgroup (Wainwright *et al.* 1993, Baldauf 1999, Van de Peer *et al.* 2000), at the moment there is no published mitochondrial genome organization data or sequence data available for this group. However the fungi are widely regarded as the sister taxon of the animals (Baldauf and Palmber 1993, Nikoh *et al.* 1994, Kumar and Rzhetsky 1996, Van de Peer *et al.* 2000). In any event there is little

doubt that the fungi and several protist phyla are derived from taxa which branch very close to the base of the metazoan radiation. As a consequence comparison of the structure and organization of the mitochondrial genome in these groups with that of the metazoa could be valuable in illuminating the kinds of processes which may have operated in producing the extant metazoan mtDNA.

Certainly the differences between animal mitochondrial genomes and those of protists and fungi are substantial. The mitochondrial genomes of the bilateral animals are well-characterized, with full mitochondrial DNA sequences available for 87 (as of Boore 1999) metazoan taxa, covering much of the range of the higher metazoan radiation. In very general terms the primary differences between fungal and animal mitochondrial genomes can be divided into size differences and gene content. In terms of size, fungal mitochondrial genomes are on average larger than those of metazoa, typically ranging from 30 to 80 kilobases (Clark-Walker 1992). While the triploblast mitochondrial genomes tend towards a median size of roughly 17 kilobases (Wolstenholme 1992), they do in some cases range up to in excess of 40 kilobases (Boore 1999). Much of the difference between fungal and metazoan mitochondrial DNA sizes is due to noncoding intergenic sequences, which are much more prevalent in fungal than in animal mtDNAs. As well, in those cases in which triploblast metazoan mitochondrial genomes extend up to atypically large sizes, this size difference is also due primarily to an increase in noncoding, often repetitive sequences which appear to have arisen due to duplications (Boore 2000).

Most of the remainder of the difference in size between fungal and triploblast metazoan mtDNAs is due to the loss (presumably the nuclear transfer) of mitochondrially encoded genes at some point early in the evolution of metazoa. Fungal mtDNAs possess a variety of protein-coding genes never found in those of the triploblast metazoa. The most common additional genes are those for ribosomal proteins, which can account for a substantial fraction of the non-repeat size difference between the fungi and metazoa (Clarke-Walker 1992). A variety of other genes are also found, ranging from unidentified open reading frames to additional components of the NADH dehydrogenase complex.

Finally, another component of fungal mitochondrial DNAs which sets them apart from those of the triploblast metazoa is the presence of introns within mitochondrial genes. As well, open reading frames are often found within the introns themselves. Both Type I and Type II introns are found in the fungi, and the issue of their origins, functional significance, mobility, and possibly horizontal transfer is quite complex. They have been shown to be unnecessary for the correct functioning of the *Saccharomyces cerevisiae* mitochondrial genome. Their presence is nonetheless an extremely common feature in the fungi.

As for the metazoa themselves there has been little ground for understanding the evolutionary processes involved in the establishment of their more conservative structure and content. Until recently there was no data from which to draw any inferences; however, this has changed with the publication of complete mitochondrial genome sequences for several anthozoans, as well as a significant number of partial sequences of mitochondrial genes from a wide array of cnidarian taxa. This lays the groundwork for an extensive analysis of the evolution

of the metazoan mitochondrial genome if the mitochondrial DNAs of other diploblast phyla can be characterized.

Bridge *et al.* (1992) described some peculiar properties of cnidarian mitochondrial genomes. Their work indicated that the Anthozoa, the most basal cnidarian class, possess circular mitochondrial genomes of an (at the time) undetermined size, but that the classes Cubozoa, Scyphozoa, and Hydrozoa all appear to have linear mitochondrial genomes, and this has been confirmed more recently (Pont-Kingdon *et al.* 2000) for the Hydrozoa. The size of the mitochondrial DNAs of the latter three classes were all relatively typical for metazoa, ranging from 14 to 18 kilobases, but several representatives of the genus *Hydra* showed an even more striking oddity in the form of two half-size linear mitochondrial genomes totaling 16 kilobases. The method of replication of such DNAs has not been determined.

The mitochondrial genome of the anthozoan *Metridium senile* (Pont-Kingdon *et al.* 1994, Beagley *et al.* 1998) was the first cnidarian for which full-genome sequence data was made available. The size was also typical for a metazoan mtDNA (17443 bases), but the genome was found to contain several features more strongly reminiscent of fungal mitochondrial genomes. The genetic code utilized in the Cnidaria appears to be that of the universal code with the single exception of TGA coding for tryptophan rather than a stop codon. This is somewhat interesting in that it clearly indicates that the multiple mitochondrial genetic code alterations found in the rest of the metazoa likely postdate the divergence of the Cnidaria from the rest of the metazoan line. However, this particular genetic code is not surprising in and of itself as it is widely found throughout both the fungi and the protozoa

(Elzanowski and Ostell 2000). It is almost certainly the primitive mitochondrial genetic code in the metazoan ancestor.

Most of the protein-coding genes found in the Cnidarian mitochondrial genomes are somewhat longer than those found in the higher metazoa (Pont-Kingdon *et al.* 1994). They are in fact closer in size to those found in representative protist and fungal lineages. In addition, the large and small-subunit ribosomal RNA genes are longer as well, and can be folded into secondary structures which are more closely allied with prokaryotic rRNA structure than are those of the triploblast metazoa (Pont-Kingdon *et al.* 1994, Beagley *et al.* 1998). Similar points of similarity were found between the structure of the initiator tRNA-fmet in *Metridium senile* with that of prokaryotes.

A unique feature of all cnidarian mitochondrial genomes for which large amounts of sequence data are available is the absence of tRNA genes apart from that of the initiator tRNA-fmet in the case of the Octocorallia, and this tRNA plus that for tryptophan in the case of hexacorals. Indeed, partial analysis of a hydrozoan mtDNA (Pont-Kingdon *et al.* 2000) appears to indicate the presence of only the latter two tRNAs, although the genome sequence is still incomplete. The loss of tRNAs must be a derived feature of the Cnidaria, in that this is not the case in either the fungal or protist outgroups to the metazoa, and in the triploblast metazoa themselves. A massive loss of tRNAs in the base of the metazoan radiation would have to be matched by a similarly extensive secondary gain of a complete set, which is very unlikely.

An apparent point of similarity between fungal, protist, and cnidarian mitochondrial genomes is the presence of two Type I introns in *Metridium senile* (Beagley *et al.* 1996). Both *cox1* and *nad5* are split by

these introns, and in the case of *nad5* the intron itself contains the full coding sequence for *nad1* and *nad3*. This is the first case in which introns have been found within a metazoan mitochondrial genome, and their presence may be a character of the primitive state of the mitochondrial genome in the metazoa, although the possibility of the presence of secondarily acquired characteristics in the cnidarian mitochondrial genome cannot be ignored. In particular no strong relationship was found between the sequences of these introns and those of fungal introns, and the particular site of insertion of the intron in *cox1* gene is not the same as that from the fungi or protists. Since the presence of these introns has specifically not been detected outside the order Actinaria this may in fact argue for the latter case.

The *cox1* intron in *Metridium senile* also codes for an unidentified open reading frame. Since the mitochondrial genomes of many fungi and protists encode a much larger number of protein-coding genes than do the metazoan mitochondrial genomes it might not be unexpected to find a higher number of genes encoded in those of the lower metazoa. This would be the case if rather than a single massive transfer event of genes from the mitochondrion to the nucleus had occurred, there had been a more gradual or at least episodic transfer. However it must be noted that this ORF has several signature sequences of site-specific nucleases which are commonly associated with Type I introns and service to facilitate transposition of the intron. Thus it is likely not a primitive component of even the cnidarian mitochondrial genome.

Apart from this ORF, found only in the actinarian hexacorals, the presence of an additional ORF of uncertain function has been found in the other anthozoan subclass, the Octocorallia (Pont-Kingdon *et al.* 1995,

Pont-Kingdon *et al.* 1998). The gene was first discovered in *Sarcophyton glaucum*, but has since been detected in *Renilla kolikeri* and *Muricea californica* (Beagley *et al.* 1995). The most likely interpretation of this particular gene is that it is a homolog of the bacterial *mutS* gene, a component of a DNA mismatch repair pathway. Phylogenetic analysis reveals that the gene is closely allied to the MSH1 gene of *Saccharomyces cerevisiae*, which itself is a participant in mismatch repair processes in the mitochondria (Chi and Kolodner 1994). The presence of this gene exclusively within the Octocorallia is a puzzle, in that it is unknown not only within the metazoa broadly but within the fungal and protist mitochondrial genomes as well. Although the nucleotide content of the gene is compatible with that of the rest of the mitochondrial genome of *Sarcophyton glaucum*, Pont-Kingdon *et al.* (1998) argue that the absence of this gene in the hexacorals, the higher metazoa, and in fungal and protist mitochondrial genomes is strong evidence that it has been acquired, possibly by nuclear transfer, some time after the divergence of the octocorallian line from that of the hexacorals. The distribution of this gene within the octocorals has not been surveyed, but the apparent low mitochondrial sequence evolution rate seen throughout the octocorals (France *et al.* 1996) suggests that it could be widespread.

What we are left with from the limited range of Cnidarian mitochondrial DNAs surveyed thus far is a picture of a genome which may indeed exhibit some features more closely linked to those of the protists and fungi than do the triploblast metazoa. Nonetheless the more striking features of these genomes, including a lack of most tRNAs, some extra open reading frames, and the presence of Type I introns are all evidently derived features of particular cnidarian taxonomic lines. In

the absence of these features, the cnidarian mitochondrial genome is still more closely allied with that of the triploblast metazoa than with the fungal or protist mtDNAs in terms of size, organization, and gene content.

That the Cnidaria may be rather unique in their mitochondrial genome evolution has been suggested by results from other laboratories. Romano and Palumbi (1997) found evidence which could be interpreted as suggestive of widespread and profound heteroplasmy in the mitochondrial genomes of scleractinian corals. Strong evidence for reticulate evolution within the genus *Acropora* (Hatta *et al.* 1999) has been found as well, suggesting that the evolutionary processes underlying their mitochondrial genome content could be more complex than is typical for most metazoans.

In terms of gene organization there appears to be a substantial amount of variation in the organization of at least anthozoan mitochondrial genomes. While *Renilla kolikeri* and *Sarcophyton glaucum* have identical gene arrangements and contents, another octocoral, *Acropora tenuis*, has been shown (van Oppen *et al.* 1999) to have a substantially different organization in spite of the close taxonomic relationship between the three, and even closer degree of nucleotide similarity (France *et al.* 1996). *Metridium senile*, the only hexacoral for which genome organization data is available, possesses an entirely different organization. Van Oppen *et al.* (1999) note that *Acropora* appears to possess longer intergenic sequences than do the other anthozoans characterized to date, and that these might fulfill the role in transcription which is thought to be played by tRNA genes in the higher metazoa (Clayton 1991). In addition, the presence of these longer

intergenic sequences could play a role in an increased frequency of genome rearrangements.

A body of work has repeatedly suggested that the actual rates of sequence evolution in the Cnidaria may be quite low relative to the very high rate seen in mammals, although little effort has been made to quantify the degree and commonality of this observation. Best and Thomas (1993) performed a brief study of mitochondrial *cox1* sequence evolution rate in the genus *Acropora* and found that the rate of sequence evolution at third-codon positions was very low compared to that found in mammals and was more similar to that of plants. France *et al.* (1996) while investigating the global phylogeny of the Anthozoa using mitochondrial 16S rRNA genes, made brief note of the low relative rates of sequence evolution found within the Cnidaria in relation to many other metazoan groups. The octocorals in particular were found to show extremely low rates of nucleotide sequence change, which is interesting in light of the fact that they are the only taxon known with certainty to possess the mitochondrially-encoded *mutS* homolog.

Romano and Palumbi (1996, 1997) also comment upon the apparent low rate of mitochondrial sequence evolution seen within the scleractinian corals (Hexacorallia), of which *Acropora* is a member. The rate is compared again to that of insects rather than the high mammalian rate, but a broad-scale comparison across the breadth of the metazoan radiation is not made. Van Oppen *et al.* (1999) take particular note of the low intrageneric rate of sequence evolution in the mitochondrial Cytochrome b gene, as well as the low intergeneric rate between *Acropora* and *Montipora*. Indeed, the authors do suggest that the presence of the *mutS* homolog may be a potential contributory factor

in this low rate, but a careful effort is not made to quantify the overall rate of sequence evolution within the context of the metazoan radiation.

Goals of This Study

The unique characters of the cnidarian mitochondrial genome cannot be understood in a vacuum, and their relationship to the full spectrum of the metazoan radiation will provide valuable insights into the evolutionary processes involved in producing the particularly conservative state of higher metazoan mitochondrial DNAs. In order to accomplish this we must understand the evolution of mitochondrial genomes in all of the basal metazoan phyla. The sponges will be particularly valuable in this as they are the most basal metazoan taxon, and are the inevitable outgroup to all other animals.

At present there is a gap in our understanding of the initial stages in the evolution of the metazoan mitochondrial genome as there is no data whatsoever describing the genome content and dynamics of nucleotide sequence evolution in the two additional primary metazoan lineages – the Porifera and Ctenophora. Towards this end I sequenced portions of the mitochondrial genomes of several demosponges and calcisponges. Analysis of this data will shed light upon several questions relevant to our understanding of metazoan evolution.

1) What is the ancestral state of the metazoan mitochondrial genome?

Characterization of the primary state of the metazoan mitochondrial DNA will allow inferences to be made concerning the process by which the conservative state of the triploblast phyla evolved. Since there is a substantial difference between the mtDNAs of protists and fungi and those of the higher metazoa in terms of both genome content and

organization, an understanding of the sponges is an important issue in determining the time scales and taxonomic range over which this difference evolved. In particular the data as it stands suggests that there may have been a single, massive transfer event of genes from the mitochondrion to the nucleus. Data from the Porifera could support this by showing a similarly conservative content, or refute it by showing a slightly or widely different content. Of course even if there is an identical gene content this merely drives the issue back one taxonomic step into the Choanoflagellates, but it serves nonetheless to frame the next reasonable set of questions.

2) Does sponge mtDNA content provide a coherent framework for understanding aberrant cnidarian mitochondrial genome characters?

That the Cnidaria possess a wider range of mitochondrial genome alterations than any other metazoan group seems certain. Examination of mtDNA data from just a few taxa has shown a variety of differences in both genome structure and content. What remains is to establish the polarity of these peculiar characters. Are they symplesiomorphic characters of the ancestral metazoan mitochondrial genome or are they strict apomorphic characters of subsets of the Cnidaria? There can be no substitute for poriferan mtDNA data to provide meaningful answers to these questions.

3) How is the high rate of mitochondrial DNA sequence evolution in the triploblast metazoans related to changes in the genome structure and content, and to the radiation of the metazoa as a whole?

The low rate of mitochondrial sequence evolution seen within the octocorallian cnidarians has been attributed, at least by suggestion, to

the presence of a possible mitochondrially-encoded error correction system gene. Whether the simple presence of this gene within the mitochondrion, rather than as a nuclear-imported gene product would be sufficient to drastically lower the rate of sequence evolution is unknown, and although there is a coherent argument to be made that this gene is itself a unique acquisition of the octocoral radiation, there nonetheless is no empirical basis for making this assessment. In the absence of information about mitochondrial DNA sequence evolution in the Porifera we cannot establish a time scale over which the apparent rate differences might have evolved, nor about the underlying processes.

4) Can mitochondrial DNA data from the triploblast metazoa be utilized to make meaningful inferences about the timing of metazoan origins?

This question is related to the previous one. If there are substantial mitochondrial DNA sequence evolution rate differences between the diploblast and triploblast taxa, then there will be a consistent bias towards underestimating the distance to the metazoan root using inferences made simply from mitochondrial sequence data. While many studies of this question have made no use of mitochondrial gene data, others have, and the influence of evolutionary dynamics upon the question must be addressed.

5) Are the mitochondrial and nuclear data congruent in their implications about the evolution of the metazoa?

Multiple nuclear gene phylogenies have placed the triploblast metazoa on an extended long branch relative to the diploblasts. The suggestion has been made that this may imply a period of accelerated sequence evolution in the triploblast ancestral taxa (Van de Peer 2000), but other interpretations are possible. The extent of this branch has varied

somewhat depending upon both the gene analyzed and the methodology used for the analysis. Does the mitochondrial data support this result, and to what degree? Since the dynamics of mitochondrial genome evolution are different from that of the nuclear genome due to factors such as haploidy and a lack of recombination, inferences drawn from this data may shed light upon the process of evolution of the higher metazoa.

[This page is intentionally blank.]

Chapter 2

Methods and Materials

Sources of Materials

Specimens of both live and preserved sponges and ctenophores were obtained from three primary locales – Howe Sound and Tzazuni Narrows Inlet in British Columbia, and the gulf coast of northern Florida. Table 1 provides information on the source locations and providers for each sample included in this study. This taxonomic classifications are listed in Table 2.

All samples were returned to the laboratory in sea water, and were immediately flash-frozen in liquid nitrogen and stored at –80 degrees Celsius. DNA extracts for immediate use were made with tissue taken prior to freezing, and after this point tissue samples were taken from the frozen tissue. Samples of *Grantia* sp. were obtained in isopropanol solution and were maintained at room temperature until use.

While efforts were made to clean obvious external and internal debris from organisms prior to freezing, it cannot be demonstrated that no contaminants of any sort were present. Indeed this is almost impossible to assure for sponges in any case, as they are by nature crowded with bacteria, algae, and a host of small marine organisms.

DNA Extraction

DNA extractions from most tissue samples were performed with the Isoquick Nucleic Acid Extraction Kit by Orca Research Inc. Approximately 200 mg of fresh or frozen sponge tissue (squeezed briefly to remove residual seawater) was placed into a 1.5 ml microcentrifuge tube and ground up in sample buffer with a heat-blunted 1.5 ml

TABLE 1
Sample Organisms

Species	Location		Provider
<i>Tetilla sp.</i>	Demospongiae	Howe Sound, BC	Seacology Inc.
<i>Cliona sp.</i>	Demospongiae	Florida	Gulf Specimen Marine Labs Inc.
<i>Microciona prolifera</i>	Demospongiae	Florida	Gulf Specimen Marine Labs Inc.
<i>Halichondria sp.</i>	Demospongiae	Florida	Gulf Specimen Marine Labs Inc.
<i>Unidentified demosponge</i>	Demospongiae	Howe Sound, BC	Seacology Inc.
<i>Clathrina sp.</i>	Calcarea	Howe Sound, BC	Seacology Inc.
<i>Grantia sp.</i>	Calcarea	Unknown	Northwest Biological Supply Co.
<i>Leucilla nuttingi</i>	Calcarea	Howe Sound, BC	Seacology Inc.
<i>Pleurobrachia sp.</i>	Ctenophora	Howe Sound, BC	Seacology Inc.
<i>Mnemiopsis sp.</i>	Ctenophora	Florida	Gulf Specimen Marine Labs Inc.
<i>Beroe sp.</i>	Ctenophora	Florida	Gulf Specimen Marine Labs Inc.

micropipettor tip. Following this 200 µl of the sample lysate was transferred into 200 µl of the kit lysis buffer and from there the kit protocol for a total nucleic acid (DNA and RNA) extraction was followed. Owing to the tendency to retrieve a smaller total DNA yield from the tissues of both sponges and ctenophores, a final suspension volume of between 30 and 50 µl of RNase-free water was utilized to dissolve the nucleic acid pellet rather than the 50 to 100 µl volume recommended in the kit protocol.

Restriction digestion of Isoquick-extracted genomic DNA samples proved difficult at best, and in order to investigate whether some co-extracted factor was interfering with digestion I performed CsCl purification of sponge total genomic DNA. As well, I wished to examine whether better PCR amplifications were obtained with CsCl-purified DNA rather than the dirtier Isoquick extractions.

In order to perform these extractions, a phenol/chloroform extraction of 10 grams of sponge tissue was made by grinding the tissue in liquid nitrogen and then placing it in 200 µg/ml Proteinase K buffer. The tissue extract was then mixed with one volume of phenol and centrifuged for 5 minutes at 10000 rpm, and the supernatant mixed with 1 volume of 24:1 chloroform:isoamyl alcohol and centrifuged again for 5 minutes at 10000 rpm. The supernatant was then precipitated in 95 % ethanol overnight at -20 degrees and the spun for 5 minutes at 10000 rpm. After a 70% ethanol rinse the DNA was desiccated and the resuspended in 200 l 1X Tris-EDTA ph 7.0 and added to a CsCl/H₂O solution with ethidium bromide and the refractive index adjusted to 1.386. The gradient was spun in a Beckman L8-80 ultracentrifuge for 36 hours at 55000 rpm in an TI-70 rotor and the genomic DNA band

Table 2 - Taxonomic Classification of all Species Utilized In This Study

Phylum Porifera			
Class Calcarea			
SubClass Calcarenea			
	Order Sycetida	Family Grantiidae	<i>Grantia sp.</i>
		Family Amphoriscidae	<i>Leucilla nuttingi</i>
SubClass Calcinea			
	Order Clathrinida	Family Clathrinidae	<i>Clathrina sp.</i>
Class Demospongiae			
SubClass Tetractinomorpha			
	Order Spirophorida	Family Tetillidae	<i>Tetilla sp.*</i>
	Order Hadromerida	Family Clonidae	<i>Cliona sp.</i>
SubClass Ceractinomorpha			
	Order Halichondriida	Family Halichondriidae	<i>Unidentified Demosponge</i>
	Order Poecilosclerida	Family Microcionidae	<i>Halichondria sp.</i>
Phylum Ctenophora			
Class Cyclocoele			
	Order Lobata	Family Bolinopsidae	<i>Microciona prolifera</i>
	Order Beroida	Family Beroidae	<i>Mnemiopsis sp.</i>
Class Typhlocoele			
	Order Cydippida	Family Pleurobrachiidae	<i>Beroe sp.</i>
			<i>Pleurobrachia sp.</i>

* Genus is also called *Cramiella*. All classifications according to Kozloff (1996) or as identified by GenBank.

extracted with a syringe. The ethidium bromide was extracted with CsCl-saturated isopropanol the final DNA extraction was stored at -20 degrees.

Primer Design

Initial design of primers for amplification of sponge mtDNA sequences was performed by alignment of representative metazoan sequences retrieved from GenBank. Sequences were excised from the GenBank files and translated with the appropriate mitochondrial genetic code in ESEE3 (Cabot and Beckenbach 1989). The compiled protein sequences were aligned using CLUSTALW (Thompson *et al.* 1994) and the nucleic acid alignments constructed directly from the amino acid sequence alignments using small QBASIC programs written for that purpose. Wherever possible the sequences utilized for alignment were biased towards a larger number of representative lower metazoan sequences under the assumption that better primers would be designed using sequences which are taxonomically closer to the base of the metazoan radiation. In some cases a small amount of cnidarian sequence was available and when this was the case a strong bias in the final primer design was made towards nucleotide content congruent with the cnidarian sequence itself.

Primers were designed in such a way as to provide a number of primers facing both into and out of the gene itself so as to provide candidate mitochondrial genomic sequences covering as much as possible of the genome. Degenerate primers were created whenever necessary. In all cases where candidate primers or primer regions had been located they were analyzed using NAR (Rychlik and Rhoads 1989) to

identify optimal regions for the location of PCR primers, for instance high melting temperature, low dimer formation probability and uniqueness of the primer sequence. Some primers were obtained directly from the literature and are detailed in Table 3 along with all primers designed in the study.

Once some sponge mitochondrial sequence had been obtained several additional sets of primers were designed in order to accommodate as much of the new sequence data as practicable, and to increase the chances of obtaining amplified products from previously recalcitrant DNA samples. These primers were designed in the same manner as described above, by manual addition and alignment of the poriferan mitochondrial sequence with the previous alignments. These primers are also detailed within Table 3.

PCR Amplification

Amplification of the initial candidate poriferan mtDNA bands was performed by “long PCR” with reagents from the GeneAmp XL PCR Kit by Perkin-Elmer. Amplification of trans-gene regions of the mitochondrial genome was attempted with a variety of mitochondrial primers. A single 2.6 kb band was amplified from *Tetilla* sp. using the primer 16S1 (Bridge et al. 1992) which was designed to amplify the 5' end of a 758-bp 16S rRNA band in *Metridium senile* (Cnidaria) and the primer CSC2R1, which was designed as a conserved Cytochrome Oxidase Subunit II primer from a range of metazoan *cox2* sequences including *Metridium senile* (Pont-Kingdon et al. 1994). The primer 16S2 (Bridge et al, 1992), also designed from cnidarian *rrnL* sequence, allowed the amplification of an approximately 750-bp *rrnL* band from whole genomic DNA of all of the

Table 3

Table of PCR primers used in this study. Standard IUPAC designations are used to denote degenerate bases in primers.

Notes:

^a Developed for this project, ^b(Liu 1993), ^c(Wei 1992), ^d(Bridge et al. 1992)

<i>rrnL</i>		
16S1	5'	-TCGACTGTTTACCAAAAACATAGC ^d
16S2	5'	-ACGGAATGAACTCAAATCATGTAAG ^d
16S3	5'	-GTATGAATGGCTTAACGA ^a
16S4	5'	-TAGGGATAACAGCGCAAT ^c
N	5'	-ATCCAACATCGAGGTCG ^b
<i>cox2</i>		
V	5'	-CGTTCCGGTTGATAACCTCATC ^b
C2F1	5'	-TGGAATTTGAYTCTTAYAT ^a
C2F2	5'	-CATTCTTTTATGCCWATTGT ^a
C2F3	5'	-AATAATGTTTCATATTA ACTAT ^a
C2F4	5'	-CAGTTAGGTTTTCAAGAT ^a
C2F5	5'	-CCATGGCAATTAGGTTTTCA ^a
C2F6	5'	-CAATTACCTTTTCAAGATGCTGC ^a
C2F7	5'	-GGCAGTTAGGTTTTCAAGAT ^a
C2F8	5'	-AATCATTCTTTTATGCCTAT
C2R1	5'	-TGATTAGCCCCRCARAT ^a
C2R2	5'	-TTAAAGCAAAC TTTTCCA ^a
C2R3	5'	-TCTACTACAATAGGCATA ^a
C2R4	5'	-CCGCATAWCTCAGARCA YTGKCC ^a
C2R5	5'	-AAAAAACTTGTYTGATTYAATC ^a
C2R6	5'	-ACAATAGGCATAAAAGAGTGATT ^a
<i>atp8</i>		
A8F1	5'	-GTCAGTATATATGAAAGT ^a
A8R1	5'	-TTAACATGGCAATAAAGA ^a
<i>atp6</i>		
A6F1	5'	-AACAGATGGCAATTTCTT ^a
A6F2	5'	-CTAACTCTTCAATGATGATG ^a
A6F3	5'	-TAGGGTTATTTCCTTATGTATT ^a
A6F4	5'	-TTAGATTGGCCGCTAATATA ^a
A6R2	5'	-CATCATCATTGAAGAGTTAG ^a
A6R3	5'	-AATACATAAGGAATAAACCTA ^a
<i>cox3</i>		
C3F1	5'	-AGCTGCTAGTTGGTATTGGCATT TTTGTAGA ^a
C3F2	5'	-TATCATTTAGTTAAACCAAG ^a
C3F3	5'	-GTAGTGTATTTTCATTATAGTC ^a
C3F4	5'	-GAGATGTTATTCGAGAAGC ^a
C3F5	5'	-TGGAGAGATGTAATTATAGA ^a
C3F6	5'	-GTTTGGTGGAGAGATGTAAT ^a
C3R1	5'	-AGGGCTGGGGTCTACTAAATGATAGGGATG ^a
C3R2	5'	-TAATACTCCATTGCTTGTA ^a
C3R3	5'	-TACCAACTAGCAGCTTCAA ^a
C3R4	5'	-CCATAAACTGAATCTGAAA ^a
C3R5	5'	-TTCAAAACCAAATGGTGAT ^a

species examined and from the purified 2.6-kb *Tetilla* sp. band.

Sequences of all DNA primers are given in Table 3.

In order to obtain the 2.6-kb *Tetilla* sp. band, the PCR reaction was performed on a Precision Scientific GTC2 DNA Thermal Cycler using a program of 94 degrees for 30 seconds, 52 degrees for 30 seconds, and 72 degrees for twelve minutes. Thirty-five amplification cycles were performed, the final cycle being followed by a final twelve-minute incubation at 72 degrees. Amplification of the 2.6 kb band using standard PCR mixes and cycles was not successful, but worked consistently with the long PCR protocol.

For most other PCR reactions standard cycles and buffer conditions were used. Typically BRL or Promega Taq enzyme was used in the buffer supplied by the company, with a magnesium concentration of 2.5 mM. Most amplifications were performed using a Precision Scientific GTC2 DNA Thermal Cycler. All of the amplifications were variations on a primary amplification program of 30 seconds denaturation at 95 degrees followed by a 15 second anneal cycle at the chosen annealing temperature, followed by a 30 second extension cycle at 72 degrees. All reaction programs ended with an extended 1-minute 72 degree cycles. A typical set of anneal cycles used for primary amplification from genomic DNA extract was 3 cycles at exactly the annealing temperature of the primer sequences, followed by 27 cycles at an annealing temperature of about 50 degrees. For reamplifications, on the other hand, 25 amplification cycles were typically used.

In terms of the amount of genomic DNA sample used, a fresh genomic DNA extraction was subjected to a serial dilution from 2X to 1024X, and known working primers (usually 16S1 and 16S2) were used

to find an optimal dilution factor for that extraction. This dilution factor was then used for all amplifications. For reamplifications from PCR reactions typically a 250 to 500-fold dilution of the original PCR product was subjected to a 25-cycle reamplification. Bands reamplified from excised agarose gel plugs were placed into 25 μ l sterile water overnight, and then 1 μ l of this elutant was utilized in a reamplification reaction.

DNA Cloning

In order to clone the initially amplified 2.5 kB *Tetilla* band, purified PCR product was subjected to a *Sau* 3AI digestion, blunt-ended and cloned into the *Sma* I site of dephosphorylated pUC18. Following this, the band was subjected to a partial *Sau* 3AI digestion and cloned into the *Bam* HI site of pUC18 and transformed by heat shock into competent *E.coli* DH5-alpha cells. Cells were plated onto LB/ampicillin/IPTG/Xgal plates. Random positive clones were picked and minipreps made with the GFX Micro Plasmid Prep Kit from Amersham Pharmacia. These clones were sequenced until a complete overlapping set had been obtained. Multiple overlapping sequence reads from separate clones were obtained for almost every portion of the sequence.

For the cloning of most other amplified bands in this study the TA Cloning Kit by Invitrogen Corp. was utilized. Fresh PCR product was ligated by means of the single adenine overhang produced by Taq polymerase on PCR products to the pCR2.1 vector according to the manufacturer's protocols and then transformed into competent INVaF' cells and plated onto LB/ampicillin/IPTG/Xgal plates. After incubation at 37 degrees for 12-16 hours white colonies were picked and minipreps made with the GFX Micro Plasmid Prep Kit from Amersham Pharmacia.

DNA Sequencing

The earliest samples sequenced in this study included cloned sequences for which minipreps of the pUC18 clones were sequenced directly using the M13 Forward primer and USB's Sequenase Version 2.0 kit on 5% polyacrylamide gels in 0.5X TBE.

However most DNA sequencing was performed using the Thermo-Sequenase Radiolabelled Terminator Cycle Sequencing Kit by USB Corporation. Purified PCR product or plasmid miniprep was mixed with primer, buffer, and ³³P-labelled dideoxynucleotides and subjected to 30 cycles of amplification on a Lab-Line Programmable Thermal Block II. The program used for cycle sequencing was a 95 degree denaturation for 30 seconds followed by a 55 degree anneal for 30 seconds then a 72 degree extension for 1 minute. Sequence reactions were run out on 5% polyacrylamide gels in 0.5X TBE.

Finally, a significant proportion of the sequence was obtained from cloned PCR products sent to the University of Calgary's Core DNA and Protein Services where they were subjected to cycle sequencing and sequenced directly on Perkin Elmer/Biosystems 377 DNA Sequencers. Typically 10 µl of purified plasmid miniprep premixed with primer was sent by courier to Calgary for sequencing.

Southern Blotting

DNA samples for Southern blotting were digested in a variety of restriction enzymes manufactured by BRL Inc, and were typically set up as 2-5 µl of genomic DNA extract combined with 2 µl of the appropriate BRL buffer, 2 µl of the restriction enzyme, and a volume of sterile water

sufficient to make up a 20 μ l total reaction volume. Digestions were performed at 37 degrees for two to four hours, and were then loaded directly into 0.6 % agarose/TAE gels in TAE buffer and allowed to run out overnight at approximately 25 mA.

After electrophoresis the gels were photographed on an ultraviolet light table, and the portion of the agarose gel containing the digested DNA was excised and prepared for capillary blotting. Prior to blotting the gel was placed in 1.5 M NaCl/0.5 M NaOH for 45 minutes, then rinsed with water and placed in 1.5 M NaCl/0.5 M Tris-HCl pH 7 for 45 minutes. Gels were then placed into a capillary blotting apparatus with a Amersham's Hybond-N blotting membrane and allowed to transfer overnight. After blotting the DNA was fixed to the membrane with a Stratagene UV Stratalinker 2400.

Probe labelling, hybridization, and detection was accomplished using the ECL Direct Nucleic Acid Labelling and Detection System kit by Amersham. Blots were prehybridized for 1 hour at 42 degrees, and then hybridized overnight at 42 degrees in a Robbins Scientific Hybridization Incubator Model 310 rotating hybridization oven. The chosen purified probe DNA sample was labelled with peroxidase enzyme according to the manufacturers instructions. Exposure of the film to the blot after washing was typically for an initial 5-minute period followed by an extended exposure gauged manually to give an optimal image.

Sequence Analysis

Assembly and editing of all sequence data in this study was carried out with the ESEE3 program by Eric Cabot (Cabot and Beckenbach 1989). Sequence alignments were generated with CLUSTALW

(Thompson *et al.* 1994) and the alignment files were reformatted into data sets useable by phylogenetic analysis software with a combination of BioEdit (Hall 1999), manual reformatting, and small QBASIC programs written for that purpose depending upon the needs of the particular program.

All neighbour-joining analyses were performed with either MEGA (Kumar 1993) or MEGA2 (Kumar 2000). MEGA2 was used to generate the phylogenetic tree figures in the manuscript, although TREEVIEW (Page 2000) was used when appropriate for small reformatting tasks. MEGA2 was also used to perform pairwise relative-rate tests. Two-cluster relative rate tests were performed with the PHYLTEST package (Kumar 1995), although use was also made of the LINTR package (Takezaki 1995) for the same test on occasion.

For maximum likelihood calculations the PAUP* package (Swofford 2000) was used most frequently owing to its flexibility, and all the nucleotide maximum likelihood analyses in this study were generated with it. Occasional use was also made of PHYLIP Version 3.5 (Felsenstein 1999) for nucleotide maximum likelihood analysis. For amino acid sequence data the program PROTML (Adachi and Hasegawa 1994) was used for generate phylogenetic trees.

Analysis of codon usage patterns was made either with MEGA, or with the GCUA package (McInnerny 1998). Much basic manipulation of tables of results in numerical format was performed with Excel 2000 (Microsoft). Final versions of most charts were prepared with Macromedia Freehand Version 9 (Macromedia Inc) or with Sigmaplot Version 4 (SPSS Inc).

Chapter 3

Isolation and Characterization of a 3.5 kb Demosponge mitochondrial DNA Fragment Amplification and Sequencing of the Mitochondrial Band

Gross characterization of the sequence

The full sequence of the 3545-bp demosponge mtDNA band (Figure 3) reveals the presence of four open reading frames at positions 997-1737, 1825-2015, 2016-2750, and 2848-3545. Analysis of the first ORF reveals very high sequence similarity, at both the nucleotide and predicted amino acid sequence level, to the previously published sequences of mitochondrial Cytochrome Oxidase Subunit II (*cox2*) from *Metridium senile* (Pont-Kingdon et al. 1994) and from a wide range of other metazoan and non-metazoan taxa. The third ORF, at positions 2016- 2550, reveals sequence similarity to mitochondrial ATPase subunit 6 (*atp6*) from a variety of metazoan and non-metazoan taxa. The second ORF cannot be easily identified on the basis of DNA sequence. The fourth ORF exhibits unambiguous similarity to Cytochrome Oxidase Subunit III (*cox3*). The DNA sequence from base 1 to approximately base 844 exhibits a high degree of similarity to the 3' end of the mitochondrial large ribosomal RNA (*rml*) subunit in a wide variety of taxa. The sequence from bases 1751 to 1823 demonstrates similarity to *E. coli* tRNA-Lys and a variety of other prokaryotic tRNAs, while the sequence from bases 2759 to 2830 shows high sequence similarity to a tRNA gene as well, with an anticodon for arginine.

Verification of Amplified DNA Source

Our immediate goal was to assess whether the sequence represented true poriferan mitochondrial DNA or DNA amplified from a symbiotic or contaminating organism. Many sponges are known to harbor dense intracellular populations of zooxanthellae, zoochlorellae, and cyanobacteria (Simpson 1984). Additionally, since the DNA extraction was made from whole tissue the co-extraction of DNA from small marine organisms inhabiting the internal canals and choanocyte chambers in the sponge, or from larvae of marine organisms including echinoderms, amphipods, annelids, and copepods was a significant possibility.

The full amino acid sequences of the *cox2* genes from five metazoan phyla along with one fungal *cox2* sequence and the *cox2* sequences of a red algae and a green algae were aligned. The inferred amino acid sequence of the candidate poriferan *cox2* band was aligned with these sequences as well, using the cnidarian mt genetic code for translation (Pont-Kingdon *et al.* 1994) as it appeared to provide a translation consistent with the positions of conserved amino acids in the other *cox2* genes (see discussion below). Alignments were performed first with CLUSTALV (Higgins 1994) and then inspected by eye to optimize the alignment. Alignments of the DNA sequences of these genes were then made using the amino acid alignments.

Results of neighbour-joining analysis of both the amino acid and DNA alignments for the *cox2* gene are shown in Figure 4. The phylogenies recovered from both the nucleotide and amino acid data sets are consistent with one another, and show very high bootstrap support

(>95% in all cases) for placement of the candidate poriferan sequence as a sister taxon to the one published cnidarian *cox2* sequence, that of the anthozoan *Metridium senile*. This placement is supported as well by maximum parsimony analyses on both the nucleotide and amino acid data, although at only 77% and 65% of bootstrap replicates with DNA and amino acid parsimony respectively. In both cases however the sequence is supported as metazoan at >98% of bootstrap replicates. Metazoan origin is also supported by DNA maximum likelihood analysis.

Figures 5 and 6 show the same analysis applied to the full ATPase Subunit 6 nucleotide sequence and translated amino acid sequences. The *atp6* results are generally consistent with those produced by the *cox2* data, but the *cox3* data shows some significant departures from the other two genes, particularly within the triploblast data set. In addition, the precise internal branching order of both the diploblast and triploblast taxa is labile between data sets. However, this is not relevant here for a simple determination of the overall phylogenetic affinity of the amplified DNA sequence.

In order to ensure that the mitochondrial sequence was repeatably obtained from this species in general, rather than from the one individual, the sequence from base 1654 to base 2286 was amplified from three other individuals of the same species. This region includes the 3' end of the *cox2* gene, a putative tRNA gene, and the 5' end of the putative ATPase subunit 8 gene. In all cases a band with identical sequence was produced, although there was some evidence of heteroplasmy at the 3' end of *cox2* in two individuals.

Since both of the primers utilized in amplifying this sequence were developed partially from cnidarian mitochondrial sequence data, the

results of this analysis leave open the possibility that the product could represent an amplified contaminating cnidarian mitochondrial sequence. In order to address this issue, a portion of the mitochondrial large ribosomal RNA subunit gene was taken from the amplified candidate sequence, and also amplified from a fresh-water calcisponge (*Grantia* sp.) and a marine calcisponge (*Leucilla nuttingi*). We reasoned that it would be unlikely that a common cnidarian contaminating species would be present in all three of these animals, and further that comparison of this sequence with the same region from representatives of most cnidarian classes should place sponge sequence basally to the cnidarian radiation.

Partial mitochondrial large ribosomal RNA subunit sequences were retrieved from Genbank. These sequences were aligned with partial mitochondrial lrRNA sequences obtained from *Tetilla* sp., *Grantia* sp., and *Leucilla nuttingi* with the 16S1 and 16S2 primers, using CLUSTALV with a fixed gap weight penalty of 25 and a floating gap penalty of 25. Regions of unambiguous sequence alignment were selected for analysis, corresponding approximately to the three high-identity sequence blocks of France *et al.* (1996) for the same cnidarian sequences.

Alignments were analyzed by neighbour-joining using MEGA (Kumar *et al.* 1993) with both complete deletion of gaps and pairwise deletion, using the Kimura 2-parameter correction. Five hundred bootstrap replications were performed, and the sponge sequences were found to represent a radiation basal to the Anthozoa, likely the most basally derived cnidarian class (Bridge *et al.* 1995) at 99% of bootstrap replications (Figure 7). Bootstrap values for branchings within the Cnidaria are generally very low, due largely to the fairly small length of sequence analyzed, and the position of the echinoderm sequence is

Figure 3

Nucleotide sequence of the 3545-bp *Tetilla* sp (Demospongiae) band. The sequence contains three protein-coding genes: Cytochrome C Oxidase Subunit II (*cox2*); ATPase Subunit 8 (*atp8*); ATPase Subunit 6 (*atp6*); and the 5' portion of Cytochrome Oxidase Subunit III (*cox3*). In addition, it contains the 3' end of the large ribosomal subunit RNA gene (*rrnL*), and two tRNA genes with anticodons for lysine and arginine. The sequence is available in Genbank with the accession number AF035265. The exact 3' end of the *rrnL* gene is uncertain, but likely downstream of the two conserved sequence blocks CSB#1 and CSB#2, and upstream of the candidate polycistronic transcript cleavage site. This region is indicated with a string of "=" symbols. The anticodons of the tRNA genes are indicated with three asterisks. The sequence appears to be the sole sense strand for all genes in this sequence, as all genes are encoded 5' to 3' on the sequence shown. Presumptive amino acid translations for all protein-coding genes are shown. Primer sequence is not shown.


```

      L R L L E V D N R L I V P I Q T Q V R V L V T A A D V
1441 ttaaggctggtggaagtggataataggttaattgtgcctatacaaacacaggtgcgagtgtagtaactgcggcggtgt
-----

      L H A F A V P S L G V K V D A V P G R L N Q T S F F
1521 ttacatgcatttgctgtgccatcttttaggggtgaaggtgatgccgtgccgggacgattgaatcagacaagttttttc
-----

      L K R P G V F Y G Q C S E L C G A N H S F M P I V V E
1601 ttaaaagacccggagtgatgttgacaatgttctgagttatgcggggcaaatcattctttatgcctattgtagtagag
-----

      G V S L E K Y V S W V A T Q S E E A *
1681 ggcgtatcgtagaaaagtagtatcttgagtagcaacacagtcctgaggaggttaagtttttatttttaagtgtgttaa
-----|-----

                                     M P Q L E
1761 ctcatgtgtagagtagtagacttttaactgtgaagtagttggttcgagtcacaaccacatcagtgccacagttagag
--- tRNA-Lys -----***-----|-----

      A V T F L C Q Y I W K L V I L F V L F S I L V N V I L
1841 cggttactttttgtgtcagtatatatgaaagttggaattttgtttgtcttattctcaactctgtaaagtgtatattg
- ATP-8 -----

      P R L Q W Q I V T R N Q V N S I E M K K E R I K L E T
1921 cccagattacaatgacagatcggtactagaaatcaagttattctatagaaatgaagaaagaaagaattaaattggagac
-----

      I L I I * M F A A Y F D Q F N V V K L I T M Q V F V G
2001 aattttaataataataatgtttgcagcttattttgatcaatttaattgttgaagtttaataacaatgcaagttttgtagg
-----||----- ATP-6 -----

      D W L L I F T N S S M M M V I A V I I S W L L S E G
2081 ggattgattgttgatttttactaactcttcaatgatgatggtgatagccgttattatatcttgattactatcagaggaa
-----

      K R L I P N R W Q F L I E S A Y I N I H S V V Q E N L
2161 agagattaatacctaacagatggcaattttcttattgagtcctgcatatataaacattcattcggtggttcaagagaattta
-----

      G K A G Q K F F P F V L C L F I F I A M L N V L G L F
2241 ggtaaggcgggccaaaagtttttccctttgtgttatgtctttttatctttattgccatgttaaactgttttagggttatt
-----

      P Y V F V P T A Q I V I T L G L S V S I I I A V T L
2321 tccttatgtattttgtaccaactgcgcaaatagtaataactttgggtctttctgtatcaataataatagctgtaacattat
-----

      L G V L T F K L K F F S I L M P G G V P L V L A P F L
2401 taggggttttaacatttaagttaaaatttttagtatattaatgccaggcggggtccattagttggcgccgtttctt
-----

      V V I E T A S Y L I K A I S L G V R L A A N I S A G H
2481 gttgtaatagagacagcgagttatttaataaaggctatttcgttaggagttagattggccgctaataatctgctgggca
-----

      L L F A I L S G F A F N M L S S G M I I S S L V P M
2561 tttattatttgctattttatcaggatttgcggttaacatgttatcaagtgggatgataattcaagtttagtttcttatgt
-----

      L I M V F I T L L E I M V A V I Q A Y V F C L L T T I
2641 taattatggtatttataacgttattagagataatgggtggcgtaattcaggcttatgtgtttgtttattaacaactatt
-----

      Y L G D T I A L H *
2721 tatttgggggatactatagcatttcattaagttaagatctacccatagctcaattgggtgagcatttaccttctaagtaa
-----|----- tRNA-Arg -----***-----

```

```

                M M Q Q V Y H P Y H L
2801 gaggttgtaggttcaagtcctattgggtaataaaaaattaagagagctatgatgcaacaagtttatcatccntatcattta
-----|-----|--- cox3 (5' end) --

V K P S P G P Y I G G C G A F L T T V G A V V Y F H Y
2881 gttaaaccaagcccnggaccatatatagggggttggtgcttttttaacaacggttggggcggtagtgtattttcatta
-----

S P A W V L S F G L V I I V V T M V V W W R D V I R
2961 tagtcctgcttgagttttaagtttgggttagtaataatagtagtaacaatggtggttgatgacgagatgttattagag
-----

E A T Y Q G H H T L V V K Q G L K Y G M I L F I L S E
3041 aagcaacatatcaaggtcatcacattagttgtaaagcaggggctaaaatatggaatgattttattttatcggag
-----

V C L F F S F F W A F F H S S L V L ? V E I G A V W P
3121 gtatgtttatttttttcttttttttgagcttttttcatagtagtttagttctancggtagaaaataggcgagtttgacc
-----

P R G G W P V N P F S V P L L N T V V L L S S G A T
3201 gcctaggggggggtgacccgtaaacccgttttcggttcctttgttgaatactgttgttttgctaagttcaggggcaactg
-----

V T W T H H A I I S G K R K E A I I G L V F T V F L G
3281 ttacttgaacgcatcatgcaataattagtggttaagagaaaagaggctattatagggttagttttacagtttttttaggg
-----

I L F T G L Q A I ? Y ? ? ? P F T I P D S V Y G S T F
3361 attttgtttacaggggttacaagccattgnntatnannagnctccttttacaattccngattcagttttatgggtcaacttt
-----

F L T T G A T G ? H V L I G S S F L L V C L V R L I
3441 ctttttaacaactggggccacggggnacatgttttaataggtagttctttttattagtagtttagtgcgacttatat
-----

Y Y Q F T R H H
3521 actatcaattcacgagacatcatca
----->

```

Figure 4

Neighbour-joining trees of aligned *cox2* DNA and amino acid sequences. Number show percentages of 500 bootstrap replications which support that branch. A. Neighbour-joining tree derived from DNA sequence alignments using codon positions 1 and 2 only, and the Kimura 2-parameter correction using only transversions and complete deletion of sites with gaps. B. Neighbour-joining tree derived from amino acid sequence data using Poisson distances and complete deletion of sites with gaps.

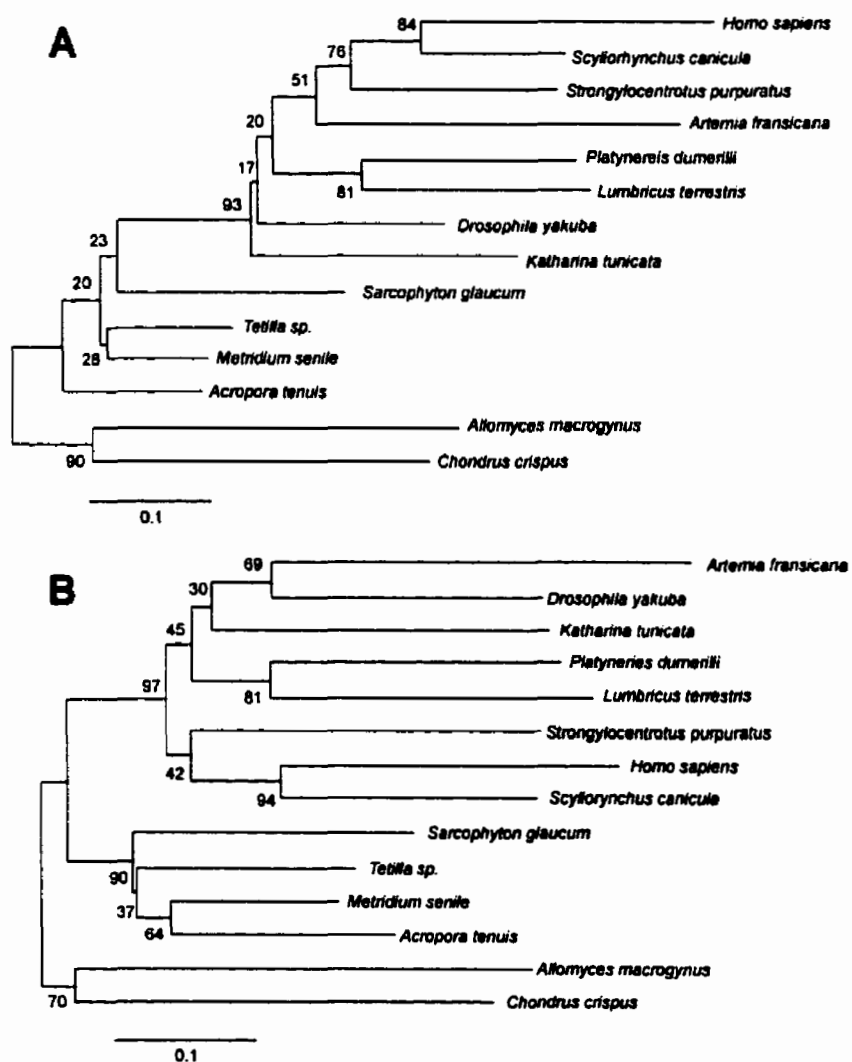


Figure 5

Neighbour-joining trees of aligned ATPase Subunit 6 DNA and amino acid sequences. A. Neighbour-joining tree derived from DNA sequence alignments using codon positions 1 and 2 only, and the Kimura 2-parameter correction using only transversions and complete deletion of sites with gaps.

B. Neighbour-joining tree derived from amino acid sequence data using Poisson distances and complete deletion of sites with gaps.

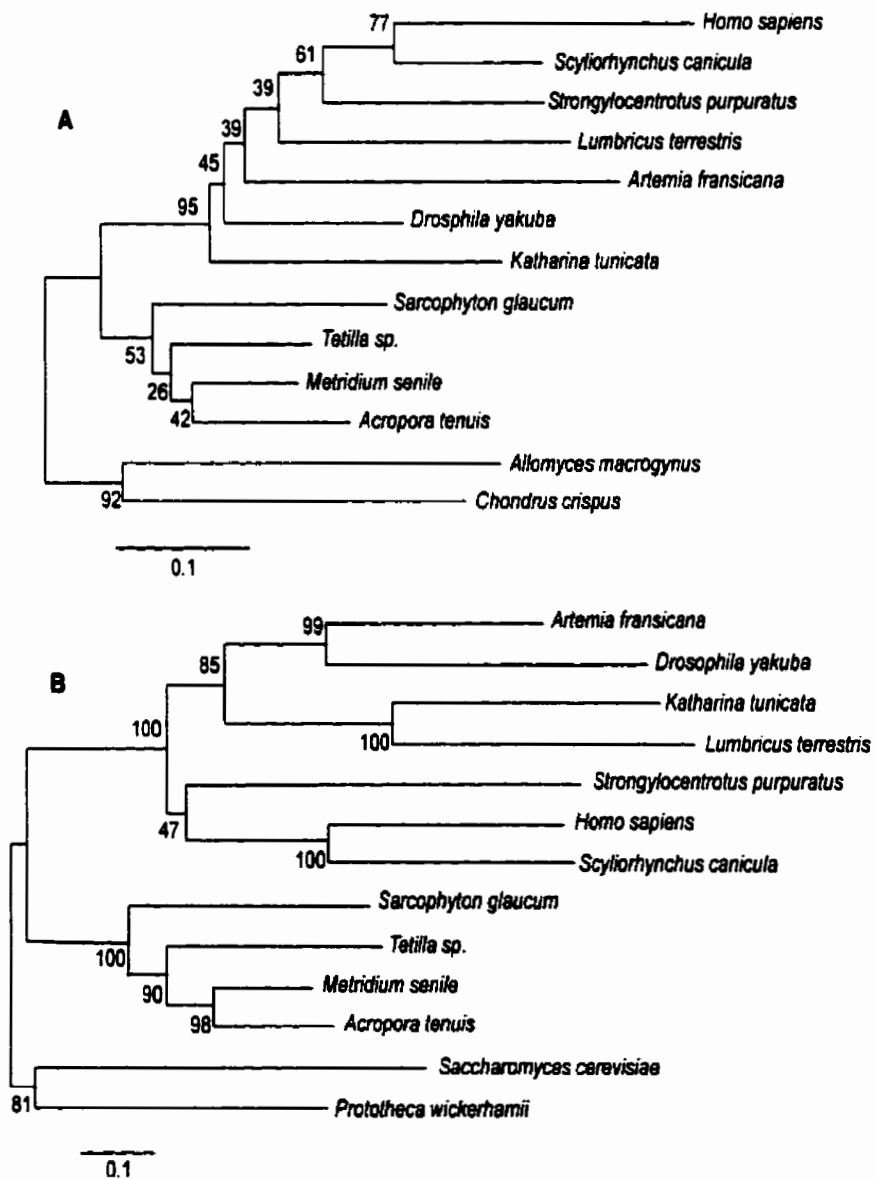


Figure 6

Neighbour-joining trees of aligned *cox3* DNA and amino acid sequences. Numbers show percentages of 500 bootstrap replications which support that branch. A Neighbour-joining tree derived from DNA sequence alignments using codon positions 1 and 2 only, and the Kimura 2-parameter correction using only transversions and complete deletion of sites with gaps. B. Neighbour-joining tree derived from amino acid sequence data using Poisson distances and complete deletion of sites with gaps.

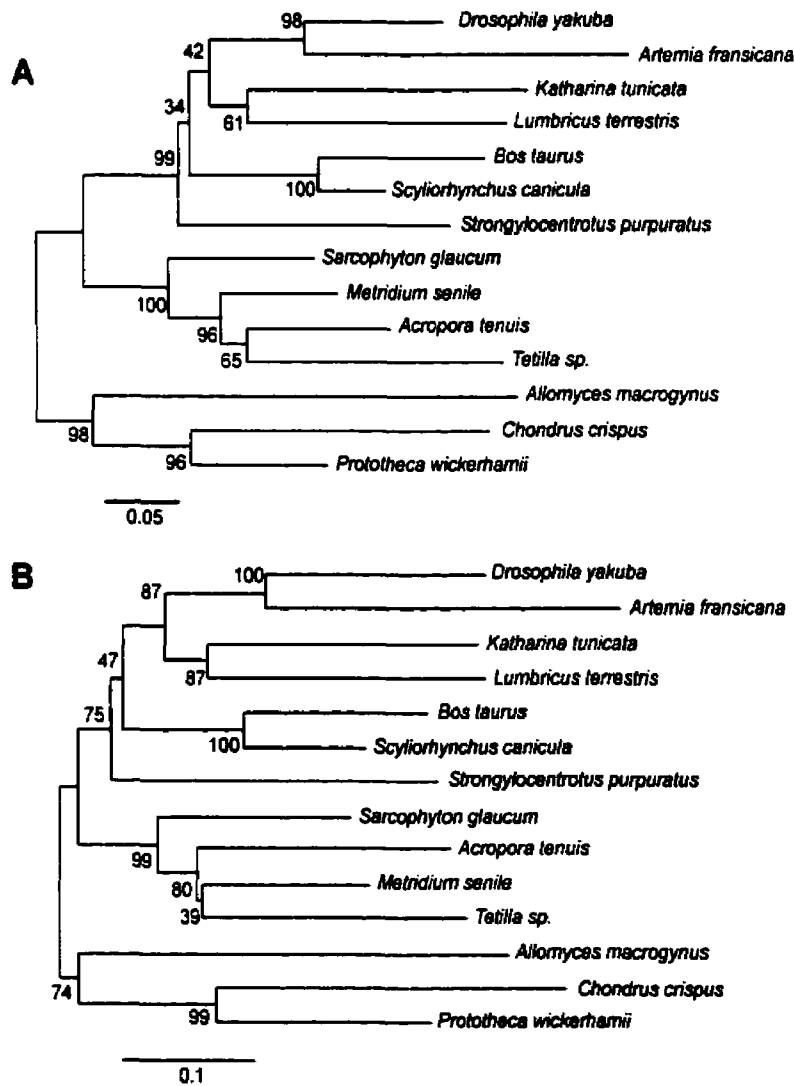


Figure 7

Neighbour-joining tree derived from short (~750 bp) Cnidarian, Echinoderm, and candidate Poriferan mitochondrial 16S rRNA sequences using the Kimura 2-parameter correction, transversions only, and complete deletion of sites with gaps. Use of other distance correction parameters was found to have no significant effect upon either branch order or branch length. Only the three highest-confidence blocks of conserved sequence were utilized to produce this tree. Numbers show percentages of 500 bootstraps which show support for that branch. The placement of the Echinoderm sequence is aberrant, and is likely a result of a combination of much higher relative rate of sequence evolution and lower confidence sequence alignment.

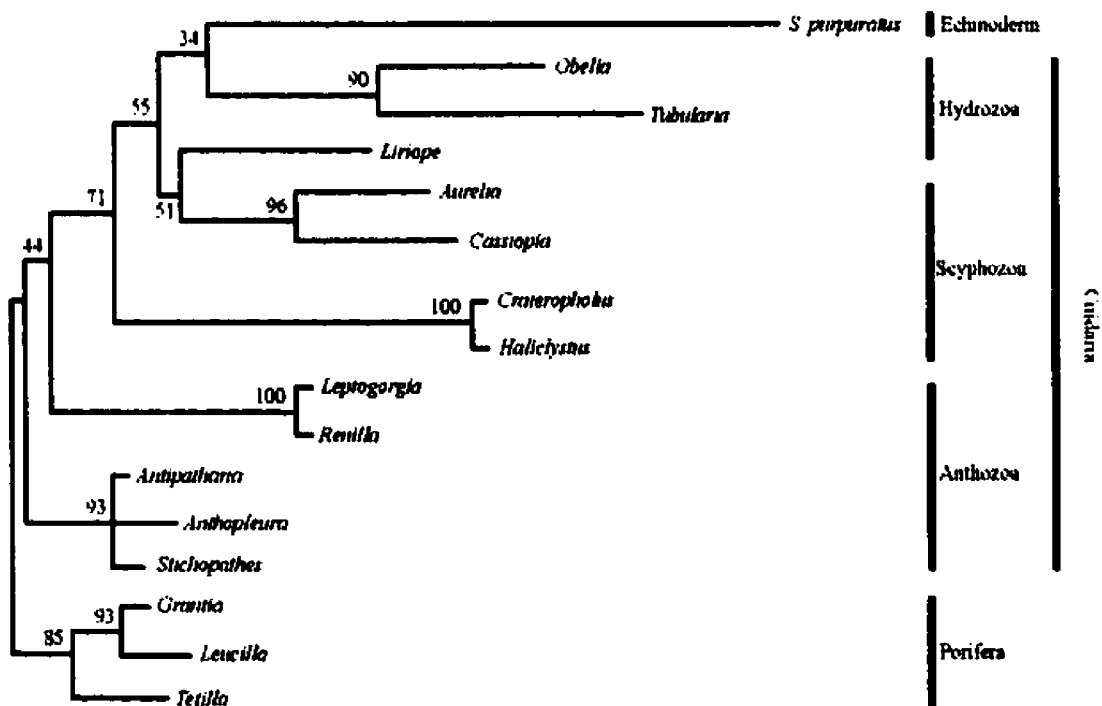
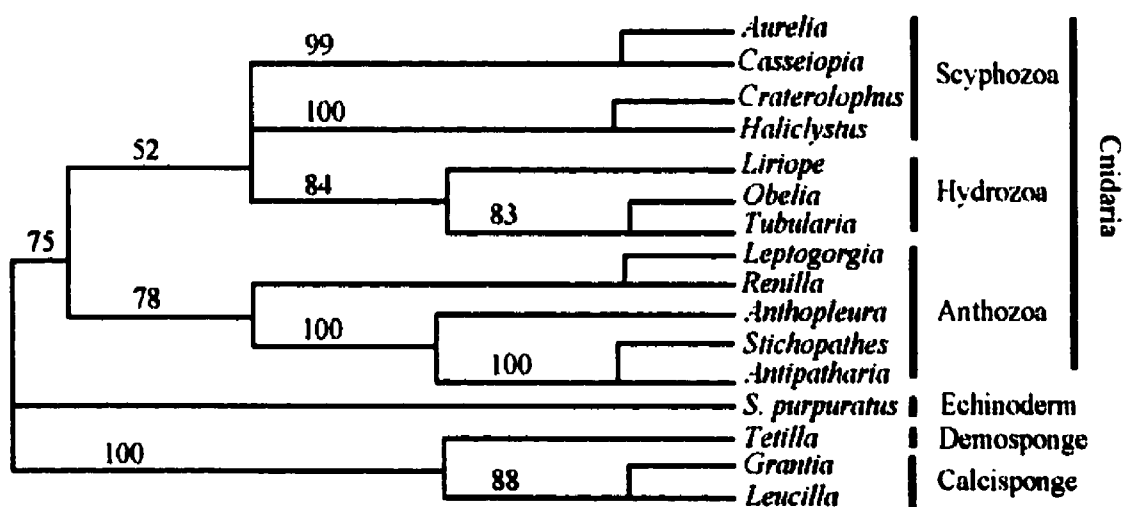


Figure 8

Bootstrap 50% majority-rule consensus tree derived from Cnidarian, Echinoderm, and candidate Poriferan mitochondrial *rnl* sequences by maximum parsimony. The full (~750 bp) sequence alignments were utilized to produce this tree. Number show percentage support for that branch out of 100 bootstrap replications.



erroneous due to its much higher rate of sequence change. Inclusion of the more variable regions of the *rml* sequences in this analysis further lowers the bootstrap values within the Cnidaria, but maintains the relative position of the Porifera although the branch length leading to the poriferan sequences increases significantly. For all analyses, essentially identical trees were obtained with other distance correction methods including gamma distances.

The full *rml* data set, including the more highly variable regions, was analyzed by maximum parsimony using PAUP 3.1.1 (Swofford 1993). Using the heuristic search option, two most parsimonious trees were found, both of which grouped the Porifera separately from the Cnidaria. In neither case was the placement of the poriferans as a separate radiation from the cnidarians affected, and in both cases the cnidarians, echinoderms, and poriferans represented an unresolved polytomy on the tree. A bootstrap 50% majority rule consensus tree based upon 100 replications placed the Porifera as a separate radiation to the Cnidaria at 100% of replications (Figure 8).

In order to investigate the effect of alignment upon these results a series of alignments was generated with CLUSTALV by varying the fixed and floating gap penalties (10/10, 10/25, 25/10, 25/25, 50/10, and 50/25). These complete alignments were analyzed by neighbour-joining with MEGA using the Kimura 2-parameter correction and complete deletion of sites with gaps. In all cases the Porifera were found to cluster as a separate radiation from the Cnidaria, although the exact placement of the cnidarian classes with respect to one another varied somewhat, as did the placement of the echinoderm sequence.

Protein-Coding Genes

The ORF from bases 997 to 1737 codes for Cytochrome C Oxidase Subunit II. Comparison with other mitochondrial *cox2* genes from both metazoan and non-metazoan taxa (Figure 9) reveals that the poriferan sequence shares a common 5' insertion with Cnidaria relative to the rest of the metazoa, although the sponge insertion is 2 amino acids shorter than the cnidarian sequence. The presence of this 5' sequence appears to be a shared feature with both the fungal and algal *cox2* genes, suggesting that a deletion of some 17 amino acids occurred following the divergence of the triploblast lineage from the diploblasts. The metazoa are distinguished from the fungal and algal sequences however, by the common presence of a deletion at the 5' end of the *cox2* sequence, corresponding to a 6 amino acid deletion relative to the fungi and the algae. Both diploblast genes are longer than the triploblast mitochondrial *cox2* gene (Table 4).

The ORF at bases 2016 to 2550 shows a high degree of similarity to the 5' 178 amino acids of ATPase Subunit 6 (Figure 8) from a wide range of taxa, and the ORF at bases 1825-2015 may code for ATPase Subunit 8, based solely upon the hydropathy profile of its inferred amino acid sequence (Figure 12). If this is the case then the gene appears to overlap the 5' end of *atp6* by one nucleotide. This situation is known from many other metazoan phyla (Wolstenholme 1992). Figure 10 shows the results of amino acid sequence alignments with this gene. The metazoan genes are again longer than their protist orthologs, and are distinguished by two 5' deletions.

Figure 9

Alignment of the full demosponge *cox2* with representative *cox2* genes from both metazoan and non-metazoan taxa. Note both the 5' indel in common between the triploblast metazoan phyla, and the smaller indel which sets the metazoan sequences as a whole apart from the non-metazoan sequences. Amino acid residues identical with the *Tetilla* sp. sequence are indicated with a dot, and gaps introduced for alignment are indicated with a dash.

<i>Tetilla</i>	MEKFA-LSSL---LIVLR---DAPEPW-QLGFQDAASPMEEIIFFH-DQIMFILTIIIVTIVLWLLIRA	60
<i>Metridium</i>	.TNL--..NNW---..INQFGY-.L....-..L....H.....-..V....I..I.T....IVK.	22
<i>Sarcophyton</i>	.KT---PINA---..LEASLLR-.A..F..S.....L..L..-N.....I..I.T....I..G	25
<i>Acropora</i>	.LN---NFFY---IVNVCCKK-.I..S.-H....E..A.....V.....L.V....IV....VE.	26
<i>Hydra</i>	.KNLIF.I.FFTC.LFNNFIYK.I..IN-..S..ES..SCSNML.....NT..YMI..LVL.G...FSI	42
<i>Strongylocentrotus</i>	.GT-----A.F.L...S..L...LTY...-YALIV..L.TIL.FYG.VSL	25
<i>Homo</i>	.AH-----AA.V.L...T..II..L.T...-HALI..IFL.CFL..YA.FLT	22
<i>Scyliorhinus</i>	.AH-----PS.....I..L.H...HTLI.VFL.S.L..YITAI	26
<i>Katharina</i>	.AF-----R.W....G...II.QL...-HAIL..I..ISLLSYGAVSL	25
<i>Artemia</i>	VSQ-----F...L.NGN..LI.QL...-HALLVVL.TSL.GFE.AAL	19
<i>Drosophila</i>	.ST-----AN..L..R...LI.QL...-HALL..V..TVL.GY..IFIL	24
<i>Lumbricus</i>	.PN-----G.VI....S..LQLVS...-HALLV..LVL.V..GYA.LAL	26
<i>Allomyces</i>	.KLLN-FIY-----N..H.Q...-I...G.TTTFYG.VDL..-N.I.Y.I.VIIG.T.IQTSV	35
<i>Chondrus</i>	.NNIL-NFY-----PAVITT-VA.N..-I....P.T.I..G..NL.Y..-L..FICV.SVF.S.M.G.T	36
<i>Tetilla</i>	LTLKHYKY-----LFEGLIEIWTLPACVLIFIAFPSKLKLYLMDEVIDPALTIIKAIGHQWYSYE	124
<i>Metridium</i>	.SG.A.HR-----VD...L....I...II..L.....M.....	35
<i>Sarcophyton</i>	.GQA.HR-----I..VT...V..II..II.V.....VE.GV.V.....	44
<i>Acropora</i>	.NG.Y.DRD-----ID...L..V...I..VI.V...S.....VS.....	43
<i>Hydra</i>	IIN.N.N.F-----L.NNF..V...I..II..I..I..F...SI..NVE.S..V..IV.....	66
<i>Strongylocentrotus</i>	.VSSNTNRF-----F...QEL.T...VI..LI..L..L..Q.....K..F...F....R...	49
<i>Homo</i>	.T.LTNTN-----ISDAQE..TV..IL..II.VL..L..RI..IT..N..S..S....T...	55
<i>Scyliorhinus</i>	VST.LTN-----ILDSQE..V..IL..II..I..L..RI..I..IN..H....M.....	48
<i>Katharina</i>	INNSFLSRS-----TL.RQE...V..IL..V...L...Q...L..LEE...VV....R...	52
<i>Artemia</i>	FSN.FLHR-----LD.QA..TV..VI..II.VA..L..IR...I..IHN...VT.....	57
<i>Drosophila</i>	FENNYVNR-----LH.Q...IL..II..L...L..R...L..INE.SV..L..R.....	51
<i>Lumbricus</i>	ILN.QVNR-----II.AQTV.T...IL..LI..LVL..L..RI..IT...RQ.SI.V..T....R...	59
<i>Allomyces</i>	MLDSYGNSDKHVYKYSHH...V...IT..LI.VA...F...V...V..V..Y...G.N	68
<i>Chondrus</i>	.WHFEQNQNKIPSS-.TH....M...VT..FI.LI..V..FS...A...I..S...I..TL.....	67
<i>Tetilla</i>	YSDY-----GKTIEFDSYMIPTSDLNKGDRLLEVDNRLIVPIQTQVRVLVTAADVLAFAVPSLGVK	188
<i>Metridium</i>	...-Q.E.L....V...E...F.....V...N.H...G...S...A.A...	48
<i>Sarcophyton</i>	...Q-TTLGEDS.L....QP.....MV...D.Y...G...S...A...	63
<i>Acropora</i>	...QE-----E.L....V...S..F.....V...N.H...I..G...S...A..I.	58
<i>Hydra</i>	.NF-----SPN.....T...Q..F...T..D.VL.VN.KIKLI.SS...I.CWT.....	93
<i>Strongylocentrotus</i>	.T.FK-----DL...V...VSF.NP.....VL.M.NFI...SS...SW....T...	71
<i>Homo</i>	.T..G-----GLI.N..IL.PLF.EP....D...VVL.EAPI.III.SQ...SW...T..L.	85
<i>Scyliorhinus</i>	.T..E-----DLG...I.Q.Q..TP.QF...T.H.IV...ESPI...S.E...W...A...	72
<i>Katharina</i>	..FL-----NL...SLE..EE..Y.....H.RV..MK.K...SW.....	71
<i>Artemia</i>	..LN-----D.Q...I..SNE.ST.IY..D...RQC..IKAI.LMI.SDA...SW....I.	86
<i>Drosophila</i>	..FN-----N...I...NE.ITDGF...D...VVL..NS..I..I...I.SWT..A...	75
<i>Lumbricus</i>	.T.FL-----NV..I...L...LP..Y.....MV...LEI..I..I..I.SWT..A...	81
<i>Allomyces</i>	.G...-NEDD.S...F.V..DE.EM.QF.....VV..VD.R..IV.....I.S...I.	91
<i>Chondrus</i>	...-LNEDD.S.NY...EE..EI.QF.....MVI..N.HI..I.....SW...FRS.	94
<i>Tetilla</i>	VDVAPGRLNQTSFFLKRPGVFYGCSELGCGANHSFMPVVEGVSLKLYVSVATQSEEA	247
<i>Metridium</i>	M..I.....G..I...I.....I.....I.A...D..IN..I.G.D.-	62
<i>Sarcophyton</i>	M.....G.LA..G.L.....I.A..MD..I..LSSLCADL	82
<i>Acropora</i>	T.....AGI..I...T...I.....I.A...DR.IN.MLSL.N.-	76
<i>Hydra</i>	...I.....K.F.....L.....SIYS..Q..FIN.-SINKT--	115
<i>Strongylocentrotus</i>	M.....T..AA.T.....I.....R...I.S.PENTFEN..TQYL..-	91
<i>Homo</i>	T..I.....T.TAT...Y.....I.....L.LIP.KIFEIGP-V-F-TL	108
<i>Scyliorhinus</i>	I.....A.IIS...Y.....I.....I...A.P..HFET.SSLIL...	91
<i>Katharina</i>	A.....LR..ANY.....I.....I...L.V.DRSSFIK.IIFNG.A-	94
<i>Artemia</i>	M..D.....SRLLVNI.....I..SG...f...I.A.GERDFLK.LEL.IS--	113
<i>Drosophila</i>	.GT.....N..IN..L.....I.....R.I...I.S.FVNYFIK.ISRN-NS-	99
<i>Lumbricus</i>	IG.TTTQ.....I.....A..AINTKSFIR..S-NFK-P	103
<i>Allomyces</i>	I..I...I..A..LID.E..Y.....G...AI.A...DR..A..DA.L-S-	111
<i>Chondrus</i>	C..I.....L.I..E..Y.....I..I..G.....AT..PN...ISNKLN.-	113

Figure 10

Alignment of the full demospone ATPase Subunit 6 gene with representative *atp6* genes from both metazoan and non-metazoan taxa. Note the indel beginning at *Tetilla* sp. position 72 which sets the metazoan sequences as a whole apart from the non-metazoan sequences. Amino acid residues identical with the *Tetilla* sp. sequence are indicated with a dot, and gaps introduced for alignment are indicated with a dash.

<i>Tetilla</i>	MF-AAYFDQFN-VVKLMTMQV-----FVGDWLLIFTNSSMMVMVAVINSWLLSEGKSLMPNSWQFLIESA	63
<i>Metridium</i>	.G.....K-----VVD-----IAI.....ML..AVALM.LK.NR.I..R..AVM..I	25
<i>Acropora</i>	.S-G.....K-----MVA-----IAL.....ILV.VVVL..FK.VQ.I.K...S...LI	25
<i>Sarcophyton</i>	.A..SF.....-VVRM-----MG.....MM.....I.LFS.MLMGGCSYK...T.L.AMQ.IL	31
<i>Strongylocentrotus</i>	.T..SILG..FP-----ET.FFIPMNVS.AFCLSWLVEIYPV.WA-.SRF.SIWLGF	42
<i>Homo</i>	.N-ENL.AS.IA-----PT.LGLPAVLIILFPPLLIPTSKYL-IN.RLIITQQWL	44
<i>Scyliorhinus</i>	.I-MSF.....LS-----PSFLGIPLIALAISIPWLMFPTTNRWLN.RLLT.QAWF	43
<i>Artemia</i>	.M-.SL.SV.DP-----TSSFLSNWLSML.PLLFMVMSFW.I.SRP...AK.V	36
<i>Katharina</i>	.M-MDI.SS.DD-----NNFTL.SF.YLIW.LGLAPVLISQNMFWFSKSRFYSM.LFF	45
<i>Lumbricus</i>	.M-PDI.SS.DP-----YMENTL.PLN.LFL.TNTAILMIQSSEFWLNARTSAFKSPV	47
<i>Drosophila</i>	.M-TNL.SV.DP-----SAIF.L.LNWLSTFLGLLMIPISTYW...SRYNIFWN.I	40
<i>Saccharomyces</i>	..ITSPL..ELNNYFGFYLFNYHFD.SNFGFYLGSLALIAISL.MMNLTYPGS.AKMV.QKFGMAM.AI	61
<i>Prototheca</i>	.IST-PL..TMSMMPLHIGNYY--SFTNSSLYMFLA.GTAGLLHFVFT-QN.GF.V.SR..S.V.MI	54
<i>Tetilla</i>	YMNIHSVVQ---ENLGKAGQKFFPFVLCLEFIAMNLVGL-FPYVFVPTA-QMVTMLGLSVSMMMAVT	127
<i>Metridium</i>	.DHF.GL.K---D.S.PQ---Y....FT....MVF..I.....TV.V-H..V....F.IVIG..	52
<i>Acropora</i>	.EHF.G..K---D..SE.LSY..LIVS..F.MVF.....-....T.V-HI.V....F..I.G..	52
<i>Sarcophyton</i>	.NHFLGL.K---DSTANK.TQY.ALIMTM.T...G..L...-....T...-HIMI.F...L.IL...	65
<i>Strongylocentrotus</i>	RS..LEMIF---QKTSPTAPWAGLMAGV.VL.LLV....FP..A.QSPSN.SL.YS.GEPLW..MK	91
<i>Homo</i>	IKLTSKQMM---TMHNTK.RTWSLMLVS.I...TT.L...-L.HS.T..T..LS.N.AMAIPWAGTV	91
<i>Scyliorhinus</i>	INRFYQLM---QPMNLG.H.WAILFTA.ML.LITI.L...-L..T.T..T..LSELNMAFALPLWLT	93
<i>Artemia</i>	L.GLNREMS---LLM.P.SFGANIL.IA..L..LFN.FI...-...I.TA.S-HLAV..S.A.PLWISFI	80
<i>Katharina</i>	KNFMGGQ.M---RAT..NISG.INM.ISI.LLLILV.LS..-L...SLSS-HLSFAFCFGLPFWLSLI	96
<i>Lumbricus</i>	NDT.FTQLS---RTSTTHLKGSLTSLTI.FMLV.I.LM..-I..M.STSS-HL.F..S.GFPIWLSLM	98
<i>Drosophila</i>	LLTL.KEFK---TL..PS.HNGST.IFISLFSILF.NFMG-LFPYIFTSTSHLTL..S.ALPLWLCFM	96
<i>Saccharomyces</i>	.FTMLNL.ENQIHSSKTVS..SY...IWS..VL.LFS.L.S.-I..GYAT..-LIF....I..LIGA..	104
<i>Prototheca</i>	.EFVS.LM.EQI---.AK.S.Y..L.FT..V.LLFT.LM.M-I..S.TA.S-HLAV.F...L.LFI.M.	93
<i>Tetilla</i>	LLGVLTFKLKFFSMLMPGGVPLVLAPFLVVMETASYLMKAISLGVSLAANMSAGHLLFAILSGFAFNML	196
<i>Metridium</i>	.G.LWK..WN.L.I...A.A..A...L..LI..V..ISS.....R.....A..G....	74
<i>Acropora</i>	.A.FWS..GD..VF..S.A..G...L..L..V.FISS.....R.....L.....A..G....	74
<i>Sarcophyton</i>	M..MTQYRW.N.A..M..S.A..L...L..S..VM...SR.....I.....V.....M.AS.G.A.I-	92
<i>Strongylocentrotus</i>	I..FYLAFKSRL.H.V.Q.T.SA.M.LM.W...L.LFAQPMA..LR...LT....IFL..TAMWLLS-	133
<i>Homo</i>	IM.FRSKIKNALAHFL.Q.T.TP.I.M..II..I.L.IQFMA.A.R.T..IT....MHLIGSATLA.S-	137
<i>Scyliorhinus</i>	.I.MFNQPTIALGH.L.E.T.TP.V.V.III..I.LFIRPLA...R.T..LT....MQLIATA..VL-	136
<i>Artemia</i>	.YTWIKETTNALAH.V.L.T.AP.M..M.L..II.NMIRP.T.S.R...I...I..TA.S-HLAV..S.A.	123
<i>Katharina</i>	FS.D.YVS-EVV.H.L.S.A.GI.N...LV..V.ISVRP.T.S.R...I...I..LGLIGSYLSVG..	136
<i>Lumbricus</i>	ISTFAHSPK.STAHFL.D.A.DW.N...LI..T.VFVRPLT.SFR.....IVLSLMGIYCAAAW-	145
<i>Drosophila</i>	.Y.WINHTQHM.AH.V.Q.T.AI.M..M.CI..I.NIIRPGT.A.R.T...I...LTL.GNTGPS.S-	138
<i>Saccharomyces</i>	I..LQQH.A.V.GLFL.S.T.TP.I.L..LI.FV..IASGL...IR.G..IM...TMS..G.LI.TFMG	145
<i>Prototheca</i>	IM.FQVHG.H...F.L.K.A..I...L..LV..SFS.....R.F..M...T.VK.....WT..S	122
<i>Tetilla</i>	-----SSGMMISSLVPLIMVMTLLEMMVAVIQAYVFCLLTTIYLGDTMALH	244
<i>Metridium</i>	-TTAGVEN-IFPVILIM-----S...AA.....S.....A..IV... 95	
<i>Acropora</i>	-----VASGPVG-VF.L.M.....VA.....A.....A..IV... 91	
<i>Sarcophyton</i>	-SKGMLVL.LAPMLVMI-----IA..L.....M.IN.SLN... 116	
<i>Strongylocentrotus</i>	---SSLMI.VPILIIIFILLF-----V..G..C.....TA.IHF..QQNI--- 161	
<i>Homo</i>	--TINLP.TLIIFTI..ILLT-----I..IA..L.....T..VSL..H.NT--- 164	
<i>Scyliorhinus</i>	--TMPTVALLTSIV.FLLT-----I..VA..M.....V..LSL..QENV--- 165	
<i>Artemia</i>	---LYVTSIVVF.QIIL-----M..FS..I..S...MT.M.L.ASE----- 148	
<i>Katharina</i>	-FCYPILVL.LLLFVQ.FYF-----F.IG.SL...I..S..VSL.SD.HPY--- 168	
<i>Lumbricus</i>	-FSS-VS.TALL.LTAIGYI-----F.VAICL...I...LSL.SD.HAH--- 176	
<i>Drosophila</i>	-YLL-----VTFLVQAIAL-----V..SA.TM..S...AV.S.LSSEVN--- 167	
<i>Saccharomyces</i>	L-----NLITF..IGFL.ITVL.AIS...FGI.F.....AI..CGFIN.SLN... 176	
<i>Prototheca</i>	V-----GGMLSVA.ML.FAVVFALMG..IG..CL.....TI.VC...N.AIN... 153	

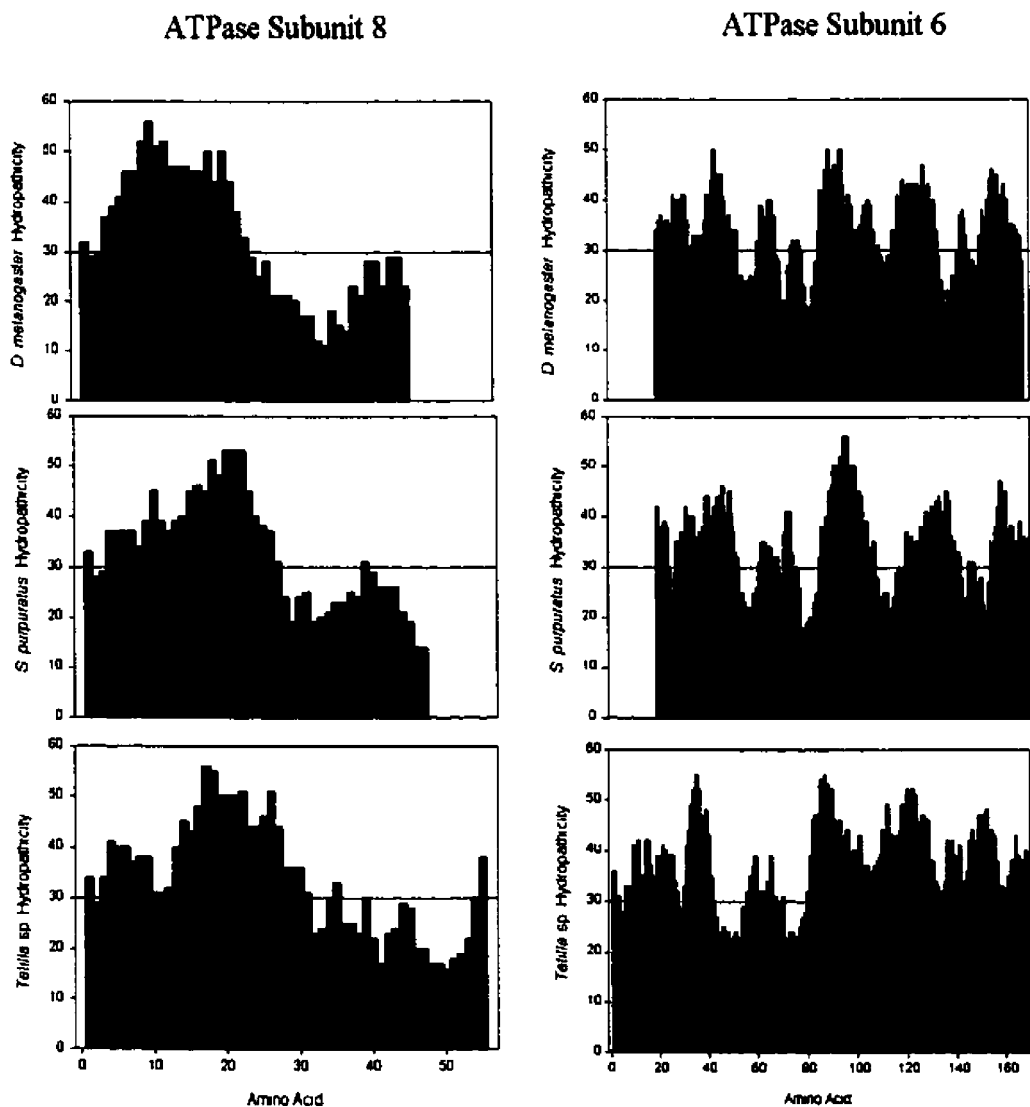
Figure 11

Alignment of the partial demosponge *cox3* with representative *cox3* sequences from both metazoan and non-metazoan taxa. Amino acid residues identical with the *Tetilla* sp. sequence are indicated with a dot, and gaps introduced for alignment are indicated with a dash.

<i>Tetilla</i>	M-----MQQVYHPYHLVKPSGPYIGGCGAFLTTVGAVVY-FHY--SPAWVLSFGLVLIIVTMVWVW	59
<i>Metridium</i>	-----KST.....D...W...V.A...LFI...S.....QT...LM.AITLSL...I...	22
<i>Sarcophyton</i>	-----SKT.....D...W...SM...L.I...S.L.....YI.PMYL.VIT..IL...F...	23
<i>Acropora</i>	V-----RTVLC.....E...W.CL.AG...FI...S.M.-...GLVQIMYL.VLVV..II...F...	31
<i>Bovine</i>	-----THQT.A...M.N...W.LT.ALS.L.M.S.LTMW...F--NSMTL.MI...TTNML...YQ...	35
<i>Purpuratus</i>	-----AIQ.....DQ...W.LD.AFSGGLMM.S.N.LW...T--QKTNLTIV.FLLITN...N...	36
<i>Yakuba</i>	-----STHSN..F...DY...W.LT.AI...MT.VS.M.KW...Q--YDISLFLN.I.TIL.VYQ...	37
<i>Artemia</i>	-----N.LN.....NI...W.LAT.M...AM.S.L.KW...S--FDSMLFFT.INNNHICFDQ...	35
<i>Katharina</i>	-----IRN.F...EF...W.LV.SM...CL...LAAW...G--FSLFLVWV.VIL.LL...Q...	32
<i>Lumbricus</i>	-----IRQ.F...EY...W.LTSSI...TLAI.LASW...N--HGFLC.TLAAFL.I.S.IQ...	36
<i>Dogfish</i>	-----AHQA...M.D...W.LT.ATA.L.M.S.LAIW...F--HSLLL.YL.TLLLL...IQ...	34
<i>Allomyces</i>	.TKKLINQFKSFQV.....E...W.L-ASV.C.M.TLGG.MK..GFAAGDMGTPL..ILVLAS.TL...	46
<i>Crispus</i>	.T-TLSQISKSVQR...F...D...W.F-VASLCAFSCAIGG.MYM.AYVNGSFTI..ISFFCLLV.FT...	51
<i>Prototheca</i>	.R-----K..F...D...W.L-LASF.AFLSTTGG.MYM.AYSGGGLL.AI.QLSLFLV.Y...	39
<i>Tetilla</i>	RDVIREATYQGHHTLVVKQGLKYGMILFILSEVCLFFSFFFAFFHSSLLXXXIEIGAVWPVPRGGWPVNPFS	128
<i>Metridium</i>	F..L..M.....L.....LF.....IAPNI..L.....Q.IN.L...	38
<i>Sarcophyton</i>	..I...F...QI...R..R.....T...F..A.....SPTI..L.....V.IEVL.D...	43
<i>Acropora</i>	Q.....S.F..F.S.....I...L.....LF.....APA..L.V...Q.VN.L.S...	50
<i>Bovine</i>	S.F.....PA.QK..R.....I...LF.TG...Y...APT.P.L.GC...T.IH.L.LE	61
<i>Purpuratus</i>	..I..K.NF..S..AI.NK.MR.....T...F..A.....APS...VA...S.IT.L...L	60
<i>Yakuba</i>	..S..G...L.YA.TI..RW.....LF.V.....SPAI..L.S...M.IISF...Q	61
<i>Artemia</i>	..S...F..M.STK.GF..RW.....I...FF.V...G.....SPNI.V..L...SEVESF--L	63
<i>Katharina</i>	G.F..Y...R.SR..RW.....T...LF..A...Y...APCL...SC...Q.VA.L...Q	56
<i>Lumbricus</i>	..V..G..M...SL.TT..RW.....T...NF.LA.....SPTP...CS...T.IH.L...E	60
<i>Dogfish</i>	..I...G.F...PP.QK..R.....V...FF.LG...Y...APT.P.L.GC...T.IN.LD..E	61
<i>Allomyces</i>	KT...Y.I.TT.VV...V..IL...T.....APS..L.ST...V.IE.L...E	69
<i>Crispus</i>	S.FE...GI.Q...RF.V...I..ILF..A.....APT.I...SI...K.INVL.WE	78
<i>Prototheca</i>	G.....NP.QV..R...L...IMF..A.....APT.I...K.IQVL.WE	61
<i>Tetilla</i>	VPLLNTVLLSSGATVWTWTHAIISGKRKEAIIGLVFTVFLGILFTGLQAIXXXXXPETIPDSVYGSTFF	194
<i>Metridium</i>	A.....A...L...KT...N...TI..V..LI.....ME.YEA..A.S.....	56
<i>Sarcophyton</i>	AI.....A...V...L...QS.TC..I..VQ..A...ME.YEA...S.....	62
<i>Acropora</i>	A.....A...A...E...L...SL...V...G.E.YEA...S.....	63
<i>Bovine</i>	S...A..VSI..A..SLME.D..HMLQA.FI.IT..VY..L...SE.YEA...S.G.....	91
<i>Purpuratus</i>	G.....V.LS.S.S.LA.N.T.S.QA.FL..A..SY..A...WE.IDA...A.....	85
<i>Yakuba</i>	I...AI..A..V...A..SLMESNHSQTTQ..F...L...Y..I...YE.IEA...A.....Y	90
<i>Artemia</i>	SA...SI..A..V...A..LMENNFOCCLQ..L...L..LY.SF..GLE.MEAS...A..I.....	96
<i>Katharina</i>	A...A..VS...A..SL.D.DQGG.N.S.LT...I..AY..F...GE.LETS...A..C.....	84
<i>Lumbricus</i>	A...A..V...A..SLM...ID.TQA..IL..C..AY..F...GE.MAA...S.A...T...	86
<i>Dogfish</i>	A...A..V...A..GLME.N...QA.TL.II..VY..A...ME.YEA...A.G...T...	87
<i>Allomyces</i>	..T...II..T..C.I.VS.AK...D.GATMTY..IL..IT.AMM.L...WVE.VNA...A.....	101
<i>Crispus</i>	I.F...LI..L..C...C..SLV.NL.NQSVLS.FL.IV.A.V..TF...YE.SMAD.RLS.GI...Y	116
<i>Prototheca</i>	I.F...II..T...S...A...LA.N...QG..S.AI..L.AVI...F..LE.LEA...S.GI...Y	90
<i>Tetilla</i>	LTTGATG-HVLIGSFLVLVCLVRLIYYQFTRHH	226
<i>Metridium</i>	VA..FH.M..I..TT..A...A..V.H.....	68
<i>Sarcophyton</i>	MA..FH.F..I..TI..T...A...G.HY...	75
<i>Acropora</i>	MA..FH.F.....TT..II.....KLN...SRQ	78
<i>Bovine</i>	VA..FH.L..I...T...I..FF.QLKEH..SN.	108
<i>Purpuratus</i>	VA..FH.LQ.I..TT..M...F.TAGRH..ST..	103
<i>Yakuba</i>	MA..FH.V.....TT.....L.HLNNH.SKN.	106
<i>Artemia</i>	.A..FH.L...TI..MI..IL.HAKCY.S-Q.	113
<i>Katharina</i>	VA..FH.F...V..L...T.W.NFSCH.SSS.	101
<i>Lumbricus</i>	VA..FH.L.....AI..A.TWSHH.SAG.	102
<i>Dogfish</i>	VA..FH.L..I...T...A...L.QVL.H..SE.	102
<i>Allomyces</i>	VA..FH.T..M..TI..T.S.N.ITS.HL.SG.	119
<i>Crispus</i>	MA..FH.F...V.TIS.A...I..LQ..L.QQ.	132
<i>Prototheca</i>	.A..FH.F...I..TI..IA...F..VLHH.SKS.	107

Figure 12

Kyte-Doolittle hydropathy plot of presumptive *atp8* and *atp6* sequences from *Tetilla* sp. compared with hydropathies of the same genes from *Drosophila melanogaster* and *Strongylocentrotus purpuratus*. All hydropathies were calculated using the GREASE program from the FASTA package (Pearson and Lipman, 1988). Only the 5' 170 amino acids of *atp6* are shown.



The ORF at bases 2848 to 3545 represents the 5' 697 nucleotides of the Cytochrome Oxidase Subunit III gene. This identification is very clear based upon overall nucleotide and amino acid similarity. Figure 11 shows the strong amino acid sequence conservation present in this gene. The metazoan genes are shorter than the fungal and brown algal genes via a 5' deletion, but appear to be very highly conserved overall.

Codon Usage, Nucleotide Bias, Genetic Code

The frequencies of nucleotides in the third codon position of the three protein-coding genes is proportional to their relative usage in the full mtDNA segment (Table 5). For third codon positions, codons ending in A or T constitute 75.9% of those used, compared to 66.7% of first codon positions and 58.5% of second codon positions. This suggests an overall stronger AT bias in third codon positions relative to AT content of the sequence as a whole (66.7%). Of the six codons not found to be used in this segment of the mitochondrial genome, five of them end in either G or C, the exception being CGT (arginine).

All of the codons whose specificity most frequently changes are found in *cox2*, and comparison of the *cox2* nucleotide and amino acid sequences from *Tetilla* sp. with a variety of other taxa allows a reasonable estimate of their translations to be made. In the case of *Metridium senile* (Pont-Kingdon et al. 1994) evidence regarding the assignment of the TGA codon to tryptophan rather than a stop codon was found to be inconclusive, but the authors felt that their translations were consistent with TGA being assigned to tryptophan. Further sequence analysis

Table 4

Comparison of mitochondrial protein-coding gene sequences at the amino acid level. All values were derived from the literature, or in the case of the *Tetilla* sp. sequence were calculated from the translated amino acid sequence.

Note: ^a Pont-Kingdon *et al.* 1994, ^b Jacobs *et al.* 1988, ^c Okimoto *et al.* 1992, ^d de Bruijn 1983, ^e Boore and Brown 1994, ^f Boore and Brown 1995, ^g Perez *et al.* 1994, ^h Paquin and Lang 1996, ⁱ Leblanc *et al.* 1995, ^j Wolff *et al.* 1994, ^k Beagley *et al.* 1998., ^l NF=not found.

	<i>atp8</i> Length (amino acids)	<i>atp6</i> Length (amino acids)	<i>cox2</i> Length (amino acids)	% <i>cox2</i> Amino Acid Identity With <i>Tetilla</i> sp.	% <i>cox2</i> Amino Acid Identity With <i>M. senile</i>	% <i>cox2</i> Amino Acid Identity With <i>D. melanogaster</i>	% <i>cox2</i> Amino Acid Identity with <i>P. wickerhamii</i>
<i>Tetilla</i> sp.	63	244	247	-	74.4	53.0	59.3
<i>Metridium senile</i> ^a	72 ^k	230	248	74.4	-	55.2	55.4
<i>S. purpuratus</i> ^b	55	230	229	57.1	57.3	61.6	46.3
<i>C. elegans</i> ^c	NF ^l	199	231	37.1	38.6	43.2	39.8
<i>D. melanogaster</i> ^d	53	224	228	53.0	55.2	-	47.1
<i>K. tunicata</i> ^e	53	231	229	56.3	53.6	62.4	48.8
<i>Lumbricus terrestris</i> ^f	53	232	228	50.6	49.2	59.6	46.3
<i>Artemia franciscana</i> ^g	53	220	228	47.4	49.2	61.8	43.6
<i>Allomyces macrogynus</i> ^h	47	262	245	54.5	55.7	45.5	54.8
<i>Chondrus crispus</i> ⁱ	NF ^l	253	254	54.9	53.9	47.8	64.7
<i>P. wickerhamii</i> ^j	NF ^l	250	258	59.3	55.4	47.1	-

Table 5

Total nucleotide composition and nucleotide composition by codon position of *Tetilla* sp. *cox2*, *cox3* (partial), *atp8*, and *atp6* genes compared with other metazoans. All calculations were made with MEGA (Kumar 1993).

Species	%T	%A	%G	%C
<i>Tetilla sp</i>	38.4	28.3	20.8	12.5
<i>M. senile</i>	26.1	26.1	19.1	17.6
<i>S. purpuratus</i>	26.9	26.9	17.4	24.0
<i>L. terrestris</i>	27.8	27.8	15.4	25.3
<i>D. melanogaster</i>	32.9	32.9	12.2	13.6
<i>H. sapiens</i>	29.0	29.0	13.1	32.1

Species	%T1	%A1	%G1	%C1
<i>Tetilla sp</i>	30.8	27.7	28.9	12.6
<i>M. senile</i>	28.4	27.6	29.3	14.7
<i>S. purpuratus</i>	26.5	25.5	25.1	22.9
<i>L. terrestris</i>	25.6	27.4	23.7	23.3
<i>D. melanogaster</i>	35.7	29.0	21.2	14.1
<i>H. sapiens</i>	20.1	29.7	20.5	29.7

Species	%T2	%A2	%G2	%C2
<i>Tetilla sp</i>	44.3	21.3	13.6	20.7
<i>M. senile</i>	41.6	22.0	12.9	23.4
<i>S. purpuratus</i>	40.4	20.6	15.5	23.5
<i>L. terrestris</i>	39.7	19.1	14.5	26.7
<i>D. melanogaster</i>	41.8	22.9	13.9	21.4
<i>H. sapiens</i>	39.9	21.6	11.2	27.3

Species	%T3	%A3	%G3	%C3
<i>Tetilla sp</i>	40.0	35.9	20.1	4.1
<i>M. senile</i>	41.6	28.6	15.0	14.9
<i>S. purpuratus</i>	28.1	34.4	11.8	25.7
<i>L. terrestris</i>	29.2	36.8	7.9	26.0
<i>D. melanogaster</i>	46.4	46.7	1.6	5.2
<i>H. sapiens</i>	17.7	35.5	7.4	39.3

Table 6

Codon composition and relative synonymous codon usage values for the full protein-coding sequence of the *Tetilla* band, derived from all protein-coding genes. The concatenated coding regions include a total of 788 codons. RSCU values were calculated with the General Codon Usage Analysis Version 1.0 package (McInerney 1998).

AA	Codon	N	RSCU	AA	Codon	N	RSCU
Phe	UUU	54	(1.86)	Ser	UCU	17	(2.08)
	UUC	4	(0.14)		UCC	1	(0.12)
Leu	UUA	71	(4.06)		UCA	10	(1.22)
	UUG	19	(1.09)		UCG	6	(0.73)
Tyr	UAU	27	(1.86)	Cys	UGU	8	(1.78)
	UAC	2	(0.14)		UGC	1	(0.22)
ter	UAA	3	(0.00)	Trp	UGA	15	(0.00)
ter	UAG	0	(0.00)	Trp	UGG	3	(1.00)
Leu	CUU	8	(0.46)	Pro	CCU	13	(1.68)
	CUC	0	(0.00)		CCC	3	(0.39)
	CUA	6	(0.34)		CCA	11	(1.42)
	CUG	1	(0.06)		CCG	4	(0.52)
His	CAU	19	(2.00)	Arg	CGU	0	(0.00)
	CAC	0	(0.00)		CGC	0	(0.00)
Gln	CAA	18	(1.33)		CGA	4	(1.26)
	CAG	9	(0.67)		CGG	0	(0.00)
Ile	AUU	35	(1.36)	Thr	ACU	20	(1.82)
	AUC	2	(0.08)		ACC	0	(0.00)
	AUA	40	(1.56)		ACA	19	(1.73)
Met	AUG	23	(1.00)		ACG	5	(0.45)
Asn	AAU	11	(1.29)	Ser	AGU	14	(1.71)
	AAC	6	(0.71)		AGC	1	(0.12)
Lys	AAA	12	(0.92)	Arg	AGA	10	(3.16)
	AAG	14	(1.08)		AGG	5	(1.58)
Val	GUU	39	(1.81)	Ala	GCU	20	(1.67)
	GUC	1	(0.05)		GCC	6	(0.50)
	GUA	30	(1.40)		GCA	11	(0.92)
	GUG	16	(0.74)		GCG	11	(0.92)
Asp	GAU	16	(1.88)	Gly	GGU	11	(0.96)
	GAC	1	(0.12)		GGC	4	(0.35)
Glu	GAA	9	(0.64)		GGA	11	(0.96)
	GAG	19	(1.36)		GGG	20	(1.74)

has supported this interpretation (CT Beagley, pers. com.). In the case of *Tetilla* sp., a TGA codon occurs four times, and in three of those cases the tryptophan residue at that position is universally conserved in the *cox2* sequences of taxa ranging from *Drosophila melanogaster* to *Prototheca wickerhamii*. In each of these cases the *Metridium senile cox2* sequence contains a TGG codon for tryptophan at that position. This is also the case for four conserved codons in the *cox3* sequence. The TGA codon then is likely to be translated as tryptophan in *Tetilla* sp.

The mitochondrial genetic codes of higher metazoan phyla commonly utilize methionine as the translation for ATA codons, but the translation of this codon in turbellarian, platyhelminths and echinoderms is isoleucine, that of the universal genetic code. In *Metridium senile* ATA codons also translate to isoleucine (Pont-Kingdon et al. 1994). In *Tetilla* sp., there are nine ATA codons; in the *cox2* nucleotide sequences of *Strongylocentrotus purpuratus*, *Katharina tunicata*, *Artemia franciscana*, *Drosophila melanogaster*, and *Lumbricus terrestris*, there are a total of 35 codons corresponding to seven of these positions. Isoleucine is found at eleven of these positions in the amino acid sequence compared to methionine which is only found six times. Two of these positions are well conserved for isoleucine and show no methionine codons in any of the other sequences, even when fungal (*Allomyces macrogynus*), red algal (*Chondrus crispus*), and green algal (*Prototheca wickerhamii*) *cox2* sequences are included in the analysis. A similar case can be observed in *cox3*, in which there are eight more ATA codons, one of which is nearly universally conserved as an isoleucine position across all alignments. It seems reasonable to infer that the

mitochondrial ATA codon in *Tetilla* sp. codes for isoleucine, the universal genetic code translation.

The codon AAA has been found to maintain its standard genetic code assignment of lysine in all metazoan phyla with the exception of the echinoderms and plathelminths, in which it evidently codes for asparagine. Cnidarians appear to maintain the standard genetic code assignment (Wolstenholme 1992). In the case of the sponge *cox2* gene, six AAA codons are found, one of which corresponds to the position of an AAG-encoded lysine residue in *Metridium senile*, and AAA-encoded lysine residues in *Artemia* and *Lumbricus*, and two of which correspond to presumptive AAA-encoded lysine residues in *Metridium senile*. In the *cox3* gene one more conserved *Tetilla*/*Metridium* AAA position is found, along with two more weakly conserved lysine codons. Although there are no strongly conserved lysines at any of these positions, the results are consistent with AAA codons specifying lysine in *Tetilla* sp.

The two codons AGA and AGG are the most frequently reassigned in the mitochondrial genetic code of metazoans. While in the universal code they specify arginine, they are variously reassigned throughout the higher metazoan lineages. The *Metridium senile* mitochondrial genetic code was found (Pont-Kingdon et al. 1994) to utilize the standard genetic code specificity of arginine. In the case of the *Tetilla* sp. mitochondrial *cox2* gene, there is only a single AGA codon, which corresponds to an AGA codon in *Metridium senile*. This codon position codes for an arginine residue in *Strongylocentrotus purpuratus*, and in the fungal and algal lineages as well. The *cox3* gene has a further two AGA codons, one of which occupies a universally conserved arginine position, and the other a moderately well-conserved arginine position. The usage is consistent with

AGA specifying arginine. Three examples of the AGG codon are found in *Tetilla* sp. *cox2*, two of which correspond to very highly conserved arginine residues, with perfect conservation in all of the species utilized in the alignments. These two positions are coded for in *Metridium senile* by AGA codons.

Four initiation codons are found in this mitochondrial sequence: for *cox2*, *atp8*, *atp6*, and *cox3*. In three cases an ATG initiation codon is found, whereas the ATP8 gene appears to begin with a GTG initiation codon. The use of GTG instead of ATN as a mitochondrial initiation codon is infrequent in metazoa. It is known to occur in chordates and echinoderms, but not in arthropods, nematodes, or cnidarians (Wolstenholme 1992).

The mitochondrial termination codons of most metazoan phyla have been found to include TA or T partial termination codons to which 3' adenines are added by polyadenylation of the RNA transcript (Ojala *et al.* 1981). The mitochondrial genome of *Metridium senile* showed no such partial stop codons (Wolstenholme 1992). The mitochondrial sequence characterized here for *Tetilla* sp. includes the 3' ends of only three protein-coding genes (*cox2*, *atp6*, and *atp8*), and all appear to end in a complete TAA termination codon (Figure 3). The TAA terminator of *atp8* overlaps with the putative start (ATG) of *atp6*, and would require alternate processing of a polycistronic transcript to translate both genes.

The tRNA-Lys and tRNA-K Genes

The sequence from bases 1751 to 1823 appears to code for a tRNA which has a UUU anticodon for lysine (Figure 13). This immediately sets the poriferan sequence apart from those cnidarian mitochondrial

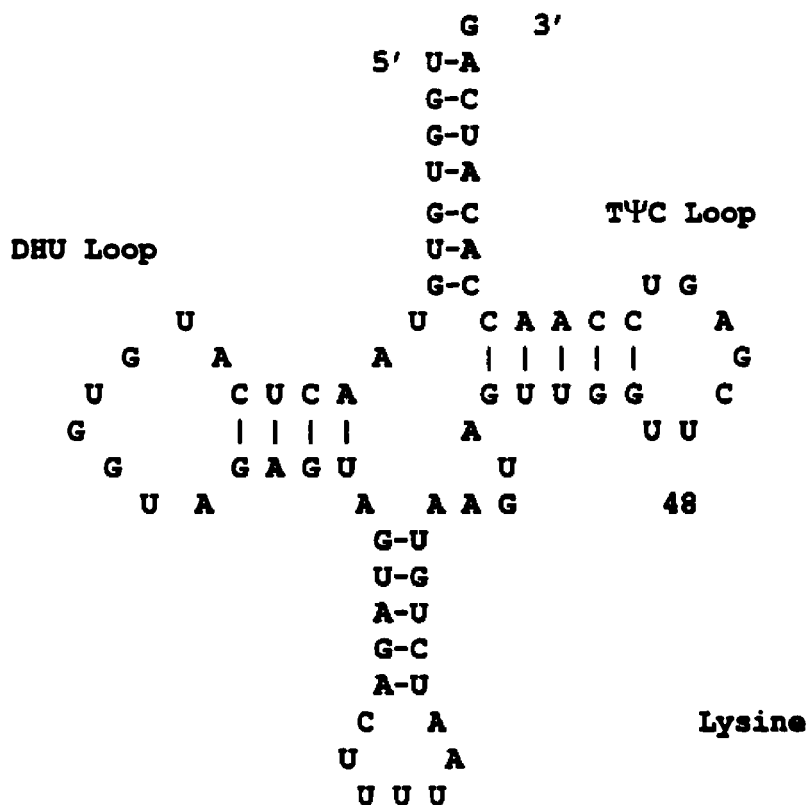
DNAs for which some description exists in the literature: *Metridium senile*, *Hydra attenuata*, and *Sarcophyton glaucum* (Wolstenholme 1992; Pont-Kingdon et al. 1994), which appear to contain only two tRNA genes, for tryptophan and methionine. The octocoral *Renilla kolikeri* contains only a single tRNA, for f-Met (Beagely et al. 1995). Whether or not any other tRNA genes exist in this mitochondrial genome is unknown, but this result makes it likely that the reduction in the number of mitochondrially encoded tRNA genes may be a derived cnidarian feature.

Structurally the tRNA gene is almost completely concordant with the general structure of prokaryotic and nuclear cytoplasmic tRNAs (Rich and RajBhandary 1976). Wolstenholme (1992) lists 24 invariant and semi-invariant tRNA bases; the *Tetilla* sp. tRNA-K gene has 22 of them: T8, R9, G10, Y11, A14, G18, G19, A21, R24, Y25, R26, Y32, T33, R37, G53, T54, T55, C56, R57, A58, Y60, and C61. It possesses both the DHU and T Ψ C arms, including the full consensus sequence of the T Ψ C loop (UUCRANY). The presence of this full consensus sequence in a metazoan mitochondrial tRNA-Lys^{UUU} is uncommon but not unknown (e.g., hominoids). As well, the presence of an A rather than a Y at position 48 appears to be a character exclusive to metazoan tRNA-Lys^{UUU} sequences (Sprinzl et al. 1996). As in all other metazoan mitochondrial tRNAs, there is no encoded 3' terminal CCA sequence. This trinucleotide is most likely added post-transcriptionally (Wolstenholme 1992).

Relative synonymous codon usage values for the protein-coding genes in this genome (Table 6) show a slight preference for the AAG codon over the AAA codon. This is odd in light of the overall stronger preference for A or T-terminal codons in these sequences, however no further tRNAs for lysine may be required in this genome as the 3'

FIGURE 13

Presumptive secondary structure for the RNA encoded by the tRNA-Lys gene in *Tetilla* sp. The folded tRNA contains the full sequence of the DHU arm and TΨC arm, as well as the full consensus sequence of the TΨC loop (GTTCRANY).



terminal uridines could make a wobble pair with either of these two codons. Bases 2759 to 2830 also encode an apparent tRNA (Figure 14). The anticodon sequence is UCU, and thus is the anticodon for arginine. The distribution of relative synonymous codon usages (Table 6) shows an almost 2:1 preference for the AGA codon over the AGG codon in specifying arginine positions. Of the other four possible synonymous codons, only CGA is utilized at all. Overall the match of this tRNA to the canonical tRNA sequence is even better than for the lysine tRNA. Twenty-three of twenty-four canonical bases are represented (Wolstenholme, 1992).

The lrRNA Gene

The sequence from bases 1 to 844 is similar to the 3' end of the mitochondrial lrRNA gene from a wide range of metazoan taxa. It is not possible with the sequence data alone to determine the precise location of the 3' end of the gene or make any inferences regarding its length relative to other metazoans. Two small conserved sequence blocks (Figure 3) exist which are present at the 3' ends of *rrnL* in a variety of taxa ranging from ciliate protists to *Drosophila melanogaster*. This suggests that the 3' end of *rrnL* could exist nearer to these blocks than bordering the 5' end of *cox2*. As well, there is a 143-base sequence from bases 846 to 988 which can be folded into a three-armed structure similar to sequences which may direct cleavage of polycistronic transcripts in other lower metazoan phyla (C.T. Beagley, pers. com., Clary and Wolstenholme 1985; Okimoto et al. 1992). This sequence is 8 bases downstream from the sequence from bases 792 to 837, which has the capacity to fold into a 46-base hairpin. Thus it seems likely that the

region between bases 787 and 845 contains the true 3' end of the lrRNA gene.

Discussion

1) The Porifera cluster unequivocally within the metazoa.

Numerous lines of evidence (Robitzki et al. 1989, Gamulin et al. 1997) including nuclear ribosomal and protein-coding gene phylogenies (Wainwright 1993, Van de Peer and Wachter 1997, Kobayashi 1996, Borchellini 1998) provide evidence that the Porifera must be regarded as falling within the metazoan radiation. It might be argued that the sampling of nonmetazoan taxa was limited, however repeated studies have demonstrated the close basal relationships between the metazoan and the fungi, so inclusion of fungal lineages along with algal is likely sufficient to bracket the placement of the Porifera with the rest of the metazoa. The metazoa as a whole, including the Porifera, constitute of monophyletic radiation distinct from the protists. The results of this study with mitochondrial sequence data from both the demosponges and calcisponges demonstrate this without ambiguity.

2) The Porifera exhibit the closest phylogenetic affinity with the metazoan phylum Cnidaria.

Both the partial lrRNA gene sequence and the DNA and amino acid sequences of *cox2* show much higher pairwise similarity to cnidarian mitochondrial DNA sequences than any other metazoan phyla to each other. This result is somewhat surprising given the presumed very ancient divergence times between these phyla (greater than ca. 543 Myr), and between each and the other metazoan phyla examined (also greater than ca. 543 Myr) (Conway-Morris 1993). The results suggest that there

has been a higher level of mitochondrial sequence conservation between these two diploblast phyla than there has been in the triploblast phyla. *Metridium senile* does not have a mitochondrially-encoded candidate DNA mismatch repair enzyme, and the question remains open as to whether the low rate of sequence evolution is due to a mismatch repair system encoded in the nuclear genome, or due to other factors, for instance constancy of temperature or environment, or other selective constraints.

Comparison of poriferan mitochondrial sequence evolution rates against both octocoral and hexacoral mitochondrial sequence evolution rate will be worthwhile in light of this question. If the rate of mitochondrial sequence evolution in the diploblast phyla is as consistently low relative to the triploblasts as the results of this study suggest, however, then the mitochondrial sequence data for Porifera and Cnidaria may be of no use in interpreting the true relationship of the diploblasts to the triploblasts. Nonetheless it may be useful in inferring the relationship of the diploblast phyla to one another. Further work on rate variation within diploblast mitochondrial genomes will determine the extent to which it will be useful for this end.

Chapter 4

Variability in Mitochondrial Genome Organization Within the Diploblast Radiation

Introduction

The organization of metazoan mitochondrial genomes is variable, with tRNA genes showing the most mobility and the protein-coding genes somewhat less (Wolstenholme 1991, Boore 1999). In general there tends to be more commonality in the arrangement of mitochondrial genes in more closely related taxa, but this is by no means without exception. In fact some phyla show a very high degree of variability (e.g., the Nematoda) whereas some (for instance the Chordata) possess large sets of common gene arrangements.

That these gene arrangements are not purely random and that there exists a strong taxonomic link between gene arrangements has been shown on a purely algorithmic basis by Sankoff *et al.* (1992) who used the organization of full mitochondrial genomes to generate a global phylogeny of the metazoa which was largely in keeping with standard views. Although the basis of their methods of phylogeny inference was a fairly complex algorithm to minimize the number of intermediary steps between different organizations, the theory underlying their methods is conceptually simple. Since rearrangements of the mitochondrial genome appear to occur with great rarity from a purely temporal standpoint, common arrangements are likely shared derived characters and can be utilized to infer the evolutionary history of a group of organisms.

What mitochondrial genome organization and content might demonstrate about the root of the metazoan radiation is not clear, however, due to a lack of available data. In fact the full organizations of

only two Cnidarian mitochondrial genomes have been determined to date (Beagley *et al.* 1995, Beagley *et al.* 1998) and partial organizations for only three others (Beaton *et al.* 1998, van Oppen *et al.* 1999, Pont-Kingdon *et al.* 2000). These genomes have shown some striking differences from those of the higher metazoans, and some commonalities as well. The presence of mitochondrial introns (Beagley *et al.* 1996), and additional genes (Pont-Kingdon *et al.* 1998) as well as both highly divergent (in the case of *Metridium senile*) and nondivergent (in the case of *Sarcophyton glaucum* and *Renilla kolikeri*) genome organizations leaves the question of the ancestral status of the metazoan mitochondrial genome very much open. The lack of data from the other two primary metazoan lineages, the Porifera and Ctenophora, is an important gap in our understanding of the evolution of mitochondrial evolution, particularly given the differences between the mitochondrial genomes of metazoa and those of the protists from which they arose (Cummings 1992). Will the sponges demonstrate an intermediary organization, content, and structure between that of the Cnidaria and that of the higher metazoans? Is the compact organization of the metazoan mitochondrial genome a primary feature of the kingdom and if so is this relevant to our understanding of the origins of the metazoa itself? What does this organization say about the relationships between rate of sequence evolution and rate of genome rearrangement in the metazoa? Is there a correlation (van Oppen *et al.* 1998) between length of intergenic sequences and frequency of rearrangement? Since the sponges are the most basally derived metazoan lineage, characterization of the organization and evolution of their mitochondrial genomes is a necessary step towards understanding the parameters of mitochondrial genome

evolution in the kingdom as a whole, and towards placing any novel characters into a meaningful context against the history of animal life.

Methods

PCR amplification using combinations of primers facing outward from *rnl*, *cox2*, *atp6*, and *cox3* was attempted for a number of different sponge species in order to amplify large contiguous portions of the mitochondrial genome. All PCR was performed with primers either obtained from the literature, in the case of the *rnl* primers, or designed in our laboratory. PCR amplification of all bands was by combinations of primers facing outwards from genes, in hopes of amplifying across gene junctions. In fact for most of the samples the same trans-gene regions were amplified, usually with a small combination of primers. Typically a region from *rnl* to the 3' end of *cox2* was amplified, as was a region from *cox2* to *atp6* or *cox3*. A region from *atp6* to *cox2* was also amplified in most cases. In all cases a standard set of PCR cycles was used, a 15 second 94 degree denaturation followed by a 30 second 42 degree anneal then a 30 second 72 degree extension. Thirty-five cycles were used on primary amplifications from source DNA, and twenty-five cycles for reamplifications from gel plugs or diluted PCR products.

In order to best align the intergenic segments of the mitochondrial bands for making comparisons of their structures and functions, the protein-coding genes and the *rnl* genes were first aligned independently of the larger genomic segments. These alignments across the metazoan radiation allowed for each protein-coding gene to be imported into a file with the fully characterized 3.5 kb *Tetilla* sp. band and the uncharacterized full mitochondrial segment for that species. Regions

between the genes were then characterized by making alignments with and across all known DNA sequences. As a result we obtained fully characterized DNA sequences for all species.

Genome Size

Figure 15 shows a southern blot of restriction-digested *Tetilla* sp. genomic DNA extract probed with labeled 16S1-C2R1 2.5kb amplified mitochondrial segment. Lane 6 shows the results of digestion with the CAGCTG-specific restriction enzyme *Pvu* II. Two primary bands are visible in this lane. There is strong hybridization to an approximately 16.5 kb band, as well as a much higher molecular weight band of >30 kb. We interpret the high molecular-weight band as the relaxed circular form of the mitochondrial DNA molecule, likely nicked in the purification process. In the case of this lane incomplete digestion of the mitochondrial genome has left a significant proportion of the sample in closed circular form. Migration of nicked circular mitochondrial genomic DNA is slower than that of the linearized form (Bridge *et al.* 1992) due to slower migration of the non-supercoiled circular form through agarose gels (Johnson and Grossman 1977). For the *Pvu* II digestion the most reasonable interpretation of the bands seen in the Southern blot is that there is one *Pvu* II site in the *Tetilla* sp. mitochondrial genome, so a linearized mitochondrial band is being produced, albeit with a significant amount of incomplete digestion.

In addition, the combination of the higher molecular weight band and the 16.5 kb one is found in Southern blots utilizing undigested genomic DNA as well (not shown), which we interpret as a combination of nicked and broken mitochondrial DNA molecules. As well, digestion

Figure 15

Southern blot of restriction-digested whole CsCl-purified *Tetilla* sp. genomic DNA extraction probed with a 2.5 kb amplified *Tetilla* sp. mitochondrial band. Only one single *Hind* III site is present in the fully sequenced portion of the mitochondrial genome.

Key-

A: Lambda/*Hind* III

B: *Tetilla* sp DNA *Pvu* II/*Hind* III

C: *Tetilla* sp DNA *Sal* I/*Hind* III

D: *Tetilla* sp DNA *Sal* I

E: *Tetilla* sp DNA *Pvu* II/*Sal* I

F: *Tetilla* sp DNA *Pvu* II

G: *Tetilla* sp DNA *Hind* III

H: Lambda/*Hind* III



of the genomic DNA extraction with many other restriction enzymes generate either the exclusive combination of the high molecular weight and 16.5 kb band, or a band or set of bands equal to or smaller than 16.5 kb. If the apparently larger band were representative of the true size of the poriferan mitochondrial genome then we would expect to see bands of an intermediate size range produced from partial digestion products. In fact these were never seen in Southern blots made from digestions with any set of enzymes, despite the presence of many partial digestion bands of less than 16.5 kb. Thus we interpret the true size of the poriferan mitochondrial genome to be approximately 16500 base pairs, although a reasonable error range of at least one kilobase should be allowed given the crude manner of measuring band size.

While some fungal mitochondrial genomes (Paquin and Lang 1996), as well as that of the choanoflagellates (OGMP 2000), and ciliates (Cummings 1992) exist as 45 to 76 kb covalently closed circles, the median size of the metazoan mitochondrial genome is very close to 16.5 kilobases. It is interesting that this sponge mtDNA molecule is congruent with the typical size of both deuterostome and protostome mitochondrial DNAs (Wolstenholme 1992) rather than being closer to the somewhat larger Cnidarian genome sizes of 18-20 kilobases and tends to support the proposition that cnidarian mtDNAs are possessed of a number of highly derived features (Wolstenholme 1992, Beagley *et al.* 1999).

Restriction Mapping

Figure 15 also shows the results of restriction digestions of the *Tetilla* sp. genomic DNA extraction with several other sets of enzymes. These results are sufficient to construct a preliminary restriction map of

the entire mitochondrial genome and to place the 3.5 kb sequenced mitochondrial band (approximately 21% of the entire genome) into context with the map as a whole. Figure 16 shows a restriction map assembled from the measured fragment sizes on this Southern blot, anchored arbitrarily at the *Hind* III site immediately downstream of the full 3.5 kb sequenced section. There are discrepancies of a few hundred nucleotides between the measured and predicted fragment sizes in the map, but this is likely attributable to the relatively high error level on the fragment size estimates. The exact positions of the restriction sites on the map, with the exception of the *Hind* III site included within the sequenced region, must be regarded as approximations.

Genome Organization

The organizations of the orthologous segments of the mitochondrial genomes of the four demosponges and one calcisponge sequenced in this study are shown in Figure 17. The most obvious feature of these genomes is the commonality of the protein-coding gene organization *cox2* – *atp8* – *atp6* – *cox3*, the only possible exception being in the *Cliona* sequence. It is worth noting that the absence of sequence information in this section of the *Cliona* mt genome is due to repeated inability to amplify trans-gene regions which included *cox2*, and not necessarily due to a direct absence of the gene in this position. Similarly, the absence of the 16S-*cox2* connection in the *Chalina* sequence is supported by the easy amplification of the 16S gene segment and the trans-gene segments across the *cox2-atp8/6* and *atp8/6-cox3* boundaries, but complete lack of success in amplifying across the *rrnL-cox2* boundary using primers which had worked successfully in other combinations.

Figure 16

Restriction map derived from restriction digestion data in Figure 15.

Predicted and actual fragment sizes based upon this map are detailed.

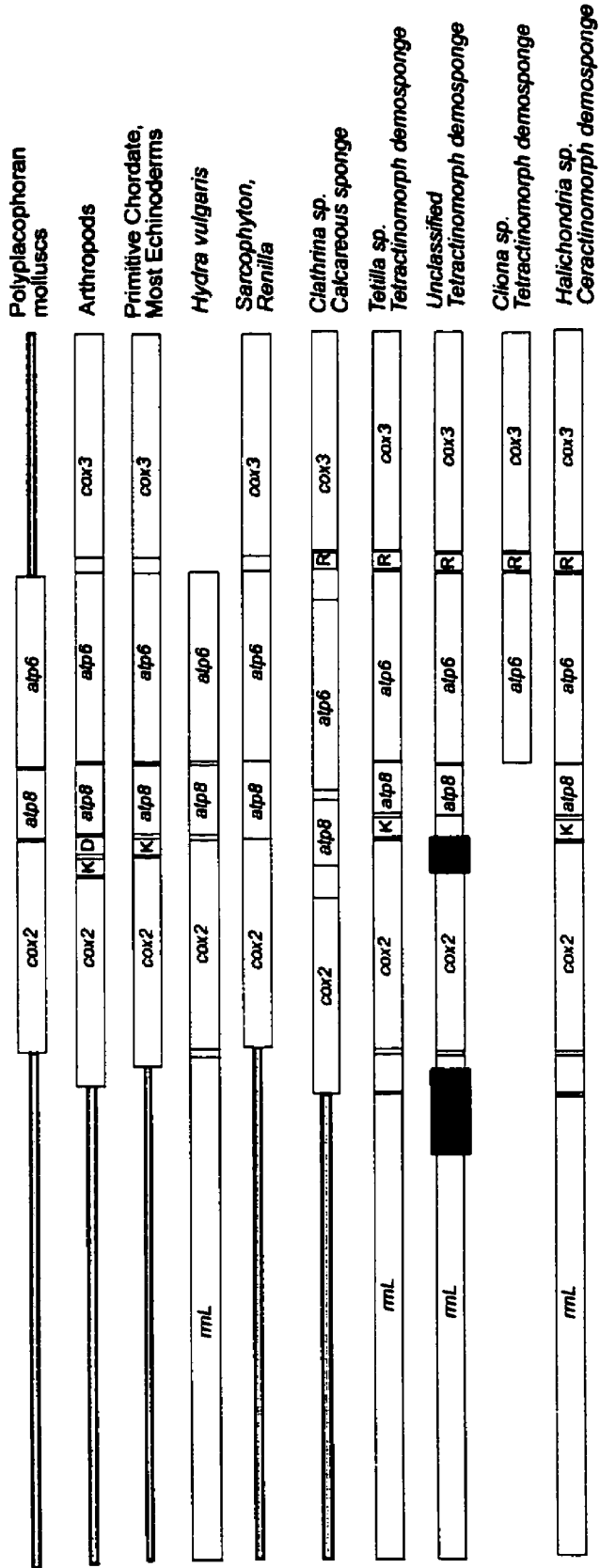
The total size of the mitochondrial genome has been somewhat arbitrarily set at 16 500 base pairs, which is only an approximation.

In fact this particular organization is likely the most widely conserved within the metazoa. It is probably the primitive chordate arrangement, and is common throughout the echinodermata as well (Boore 1999). In addition it is very common within all arthropod classes, and is found in two of three characterized mitochondrial genomes of the anthozoan Cnidaria. It may also be present in the Hydrozoa (Pont-Kingdon *et al.* 2000) in that the arrangement *cox2* – *atp8* – *atp6* is found for *Hydra attenuata*. In fact this arrangement is most specifically lacking in the Mollusca (although *Katharina tunicata* has this arrangement modified only by a single transposition of the ATPase Subunit 6 gene), the Annelida, and the Nematoda (Boore 1999). Given the wide distribution of this arrangement within the Porifera there seem to be strong grounds for regarding it as the primitive metazoan arrangement, and that the different organizations found widely throughout the Mollusca, Annelida, and Nematoda are all derived conditions.

The most striking difference between the poriferan mtDNAs and those of the Cnidaria is the presence of tRNA genes for arginine and lysine. All of the cnidarian mitochondrial genomes characterized to date lack the majority of tRNA genes, retaining only those for tryptophan and methionine (Beagley *et al.* 1995, Beaton *et al.* 1998). This appears to confirm the conjecture (Boore 1999) that the loss of tRNA genes is a derived feature of the phylum Cnidaria. There are some notable similarities between the poriferan arrangement and that of the octocorralian taxa *Renilla kolikeri* and *Sarcophyton glaucum* in that the same arrangement of these four protein-coding genes is found. The hexacorralian *Metridium senile* shows no definite organizational relationship with that of the Porifera save the proximity of ATPase

Figure 17

Relative organizations of orthologous sections of the mitochondrial genomes of four demosponge and one calcisponge species compared with organizations from a variety of other metazoans. All the information presented here has been derived directly from the sequence data. Gaps in sequence data continuity are indicated with black boxes, however these sections are known to be contiguous with their neighbours owing to PCR amplifications across them. Thin lines indicate alternate gene organizations in those taxa.



subunits 8 and 6. The partial gene order published for *Acropora tenuis* (van Oppen *et al.* 1999), another hexacorallian, shows proximity of *cox2* and *cox3* but in opposite transcriptional orientations.

In terms of gene arrangement the presence of a lysine tRNA between *cox2* and *atp8* is also a very common feature throughout the metazoa, being found widely within the Arthropoda (although typically in association with an asparagine tRNA), in the Chordata, and in the Echinodermata (Boore 1999). The Mollusca, Annelida, and Nematoda again appear not to retain this primitive arrangement. Little real phylogenetic information can be derived from these arrangements, except to provide a framework within which to analyze apparent synapomorphies.

tRNA Gene Sequences

Table 7 shown structural alignments of all of the tRNA genes present in the sequenced portions of the mitochondrial genomes described here. All tRNAs were characterized firstly by application of the tRNAscan-SE software package (Lowe and Eddy 1997) and then examined by hand for optimal foldings. In total the tRNA genes conform to the consensus sequence of conserved base positions for mitochondrial tRNAs (Wolstenholme 1991). Even the calcareous sponge *Clathrina's* tRNA-Arg shows no strong variation from the poriferan pattern.

Noncoding Sequences

Table 8 shows a compilation of most of the noncoding sequences present in the sequenced portions of the mitochondrial genomes of the sponges. Two noncoding sequences, the sequences between the large

Table 7

Aligned poriferan tRNA genes from all sequenced mitochondrial genome segments. All tRNAs have been folded with the tRNAscan-SE algorithm (Lowe and Eddy 1997) with preferences set to the mitochondrial and chloroplast tRNA set and with relaxed folding constraints.

	acceptor stem		stem	D-loop	stem		stem	anticodon loop	stem	variable loop	stem	T-Phi-C loop	stem	acceptor stem	
<i>Halichondria</i> tRNA-Lys	TGGTGTA	AA	ACTC	ATGCGGAA	GAGT	A	ATAAG	CTTTTAA	CTTAT	AAGCA	GTTGG	TTCGACT	CCAGC	TACACCA	G
<i>Tetilla</i> tRNA-Lys	TGGTGTG	TA	ACTC	ATGTGGTA	GAGT	A	GTAGA	CTTTTAA	TCTGT	AAGTA	GTTGG	TTCGAGT	CCAAC	CACATCA	G
<i>Halichondria</i> tRNA-Arg	AAACACG	TA	GCTC	AAATGGTA	GAGC	A	TTTAC	CTTCTAA	GTAAG	AGGTT	GTAGG	TTCGAG	TCTAT	CGGGTTT	A
<i>Clathrina</i> tRNA-Arg	AAACGCA	TA	GCTC	AATTGGTA	GAGC	A	TTTAC	CTTCTAA	GTAAG	AGGTT	GTAGG	TTCAAGT	CCTAC	TGGGTTT	A
<i>Tetilla</i> tRNA-Arg	CTACCCA	TA	GCTC	AATTGGGT	GAGC	A	TTTAC	CTTCTAA	GTAAG	AGGTT	GTAGG	TTCAAGT	CCTAT	TGGGTAA	T
<i>Demo2</i> tRNA-Arg	ATGCCCG	TA	GCTC	AATGGTA	GAGC	A	TTTAC	CTTCTAA	GTAAG	GGGGT	GTGGG	TTCAAGT	CCTAC	GGGGGAT	G
<i>Cliona</i> tRNA-Arg	TGACTTA	TA	GCTC	AATTTGGGA	GAGC	A	TTTAC	CTTCTAA	GTAAG	AGGTT	GTAGG	TTCAAGT	CCTAC	TGGGTTA	A

ribosomal RNA subunit and *cox2* for *Tetilla* sp. and *Halichondria* sp. were not included because no firm basis exists for establishing the true end point of the rRNA gene itself. These two sequences are nonetheless interesting in that they are likely the longest found in these demosponge mitochondrial genome segments, and are capable of folding into complex stem/loop structures (Figure 18). Also shown in Figure 18 is a candidate folded structure for the presumptive origin of replication for the *Sarcophyton glaucum* mitochondrial genome (Beaton *et al.* 1998), which happens to lie at the upstream end of the *cox2-atp8-atp6-cox3* segment in the *S. glaucum* mitochondrial genome and is the largest noncoding sequence in this genome. There is nonetheless no basis for identifying these regions in *Tetilla* and *Halichondria* as candidate origins of replication. Although they both fold into complex stem/loop structures, a complete mitochondrial genome sequence from each species would be required in order to establish likely candidate regions. Furthermore, the octocoral *Renilla kolikeri* shares the same *cox2-atp8-atp6-cox3* organization (Beagley *et al.* 1995) with no evidence that the smaller 96-bp noncoding sequence in this region is the origin of replication, and indeed there are several larger intergenic sequences in that species. Indeed, while the *Tetilla* and *Sarcophyton* sequences have an elevated %A+T level relative to the protein-coding gene sequences in their respective genomes, a feature typical of replication or transcription origin sites in both metazoan and protist mitochondrial DNAs (Wolff *et al.* 1994, Beaton *et al.* 1998), the *Halichondria* sequence itself has a significantly lower %(A+T).

The remaining noncoding intergenic sequences are all listed in Table 8. In three of four cases the calcareous sponge *Clathrina* has an

Table 8

Compilation of intergenic noncoding sequences for all sponge mitochondrial genome sequences. Only the *Tetilla* sp. and *Halichondria* sp. intergenic regions between *rrnL* and *cox2* are not included, owing to the difficulty in determining the exact endpoint of the rRNA gene.

Genes	Sequence	Length
	-----	-----
<i>cox2 - Lys</i>		
- Tetilla sp.	gtttttatttta	12
- Halichondria sp	tcggtacgcattaaagtaacgacg	24
<i>Lys - atp8</i>		
- Tetilla sp.	t	1
- Halichondria sp.	a	1
<i>cox2 - atp8</i>		
- Clathrina	cgtatggattaatgaatcaaatcttaataataataaaatcggttacgcaatctt atgattgcacgattttaccatgttacgaagtaagtaacaatggaagtgg gaaaaactc	111
<i>atp8 - atp6</i>		
- Tetilla sp.	none	0
- Demol	accccatggaagagtttagggat	23
- Halichondria sp	agttg	5
- Clathrina	ttttaagtcctatgttttagtaattaagagaattaagtg	38
<i>atp6 - Arg</i>		
- Tetilla sp.	gttaa	5
- Cliona sp.	aaatcgattgaattattaaagtaattcaatcggtttatttataacttgatttt aaaaaaacaagaaa	63
- Demol	gcggccatcacgcctgggtg	18
- Halichondria sp	t	1
- Clathrina	aatattacgccattgggttaataaccataataaagttatcgaagtaaaagatcg taccttggttaagataacttaccacttggtaagtattctttactacgagattg attcaacctaattggtcgaatagcaatataaaattcgt	142
<i>Arg - cox3</i>		
- Tetilla sp.	ctattgggtataaaaaataagagagct	28
- Cliona sp.	gattgagtct	10
- Demol	agagagaataa	11
- Halichondria sp	t	1
- Clathrina	taacaattaatca	13

Figure 18

RNA translations of noncoding intergenic mitochondrial sequences immediately upstream of *cox2* for *Tetilla* sp., *Halichondria* sp., and the cnidarian *Sarcophyton glaucum* folded with the Mfold Version 3.0 server (Zuker *et al.* 2000) using default parameters altered only with a temperature setting of 18 degrees (assumed to be more reflective of the true environmental conditions of the sponge sequences). In all cases several candidate foldings were produced for each sequence, and the single folding with the lowest delta G value is shown. A. *Tetilla* sp. noncoding region. B. *Halichondria* sp. noncoding region. C. *Sarcophyton glaucum* presumptive origin of replication.

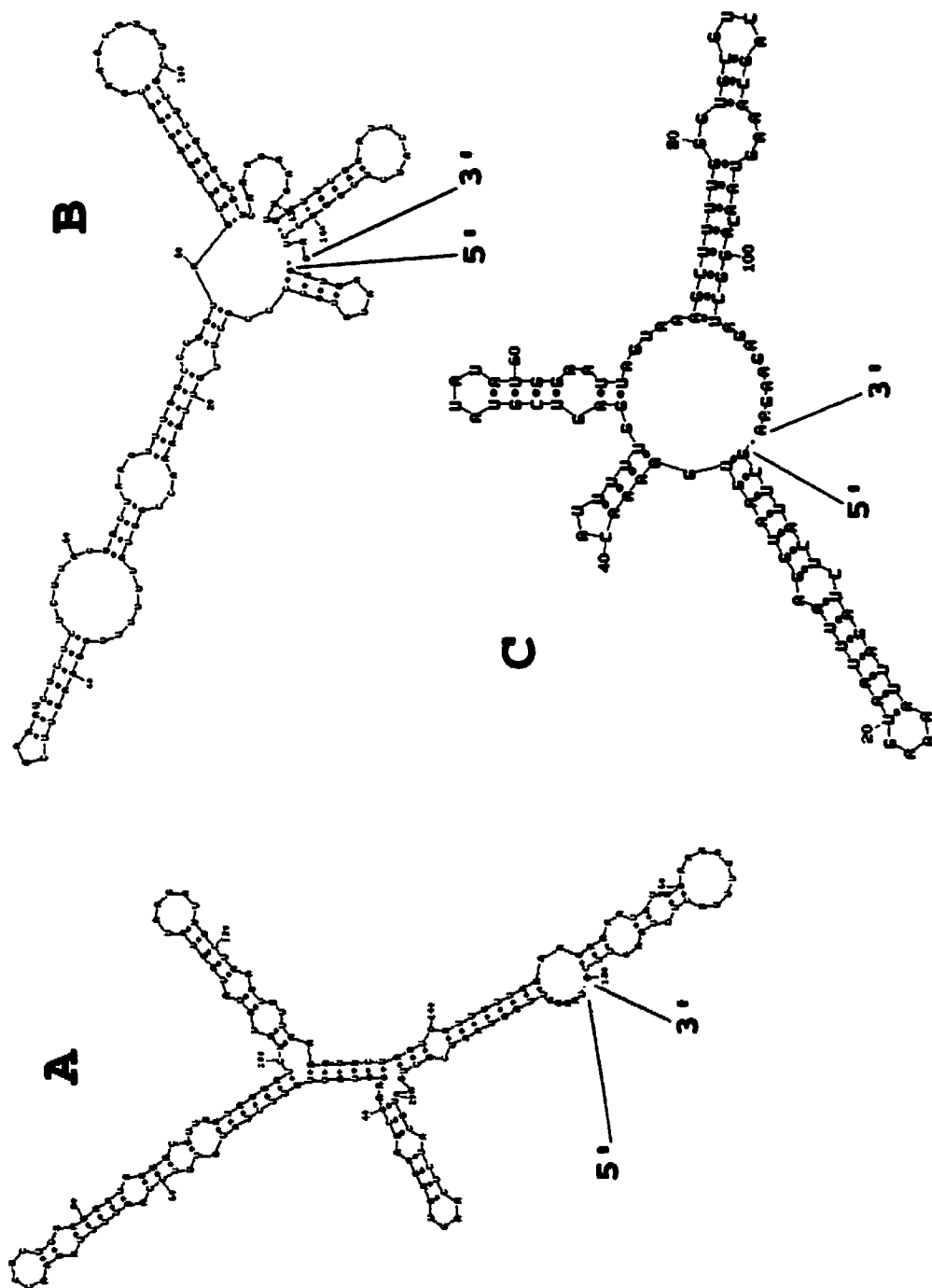
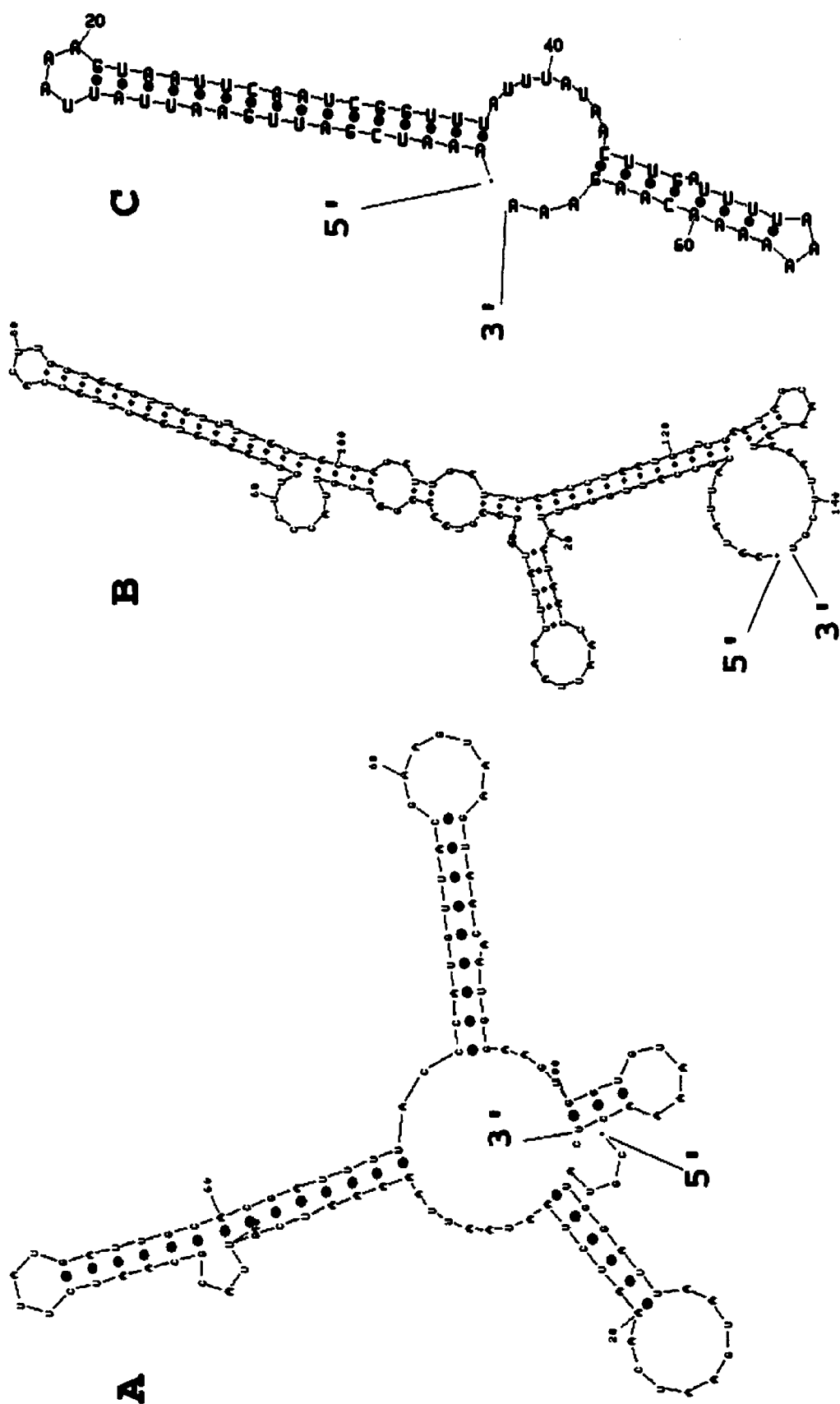


Figure 19

RNA translations of noncoding intergenic mitochondrial sequences of *Clathrina* sp. and *Cliona* sp. folded with the Mfold Version 3.0 server (Zuker *et al.* 2000) using default parameters altered only with a temperature setting of 18 degrees (assumed to be more reflective of the true environmental conditions of the sponge sequences). In all cases several candidate foldings were produced for each sequence, and the single folding with the lowest delta G value is shown. A. *Clathrina* sp. noncoding region between *cox2* and *atp8*. B. *Clathrina* sp. noncoding region between *atp6* and the arginine tRNA gene. C. *Cliona* sp. noncoding region between *atp6* and the arginine tRNA gene.



intergenic sequence almost or more than twice as long as any of the corresponding demosponge sequences. In two of these cases the sequences are quite long (111 and 142 nucleotides respectively) and can also be folded into complex stem/loop structures similar to those of noncoding intergenic regions which have been implicated in transcriptional processing (Okimoto *et al.* 1992, Boore and Brown 1994, Beagley *et al.* 1998). These sequences are nonetheless not unusually large for intergenic sequences found in the Cnidaria (Beagley *et al.* 1995), the higher metazoans (Beaton *et al.* 1998, Boore 1999), and in protists (Cummings 1992, Wolff *et al.* 1994). The shorter *Cliona* intergenic sequence between *atp6* and the tRNA-Arg can also be folded into a complex stem-loop structure as shown in Figure 19. Since the actual role of these stem/loop structures remains largely speculative, little else can be said of them at this time.

Discussion

1. *The size of the Tetilla sp. mitochondrial genome is approximately 16500 base pairs.*

Despite the inherent level of error in measuring mitochondrial genome size from Southern blots, this value is within the normal size range of most metazoan mitochondrial genomes. This suggests that the transfer of genes from the mitochondrial to the nuclear genome which occurred at some point in the evolution of the metazoa must at least predate the full

diversification of the Demospongiae and Calcarea. The poriferan class Hexactinellida at present remains uncharacterized, so we cannot say that this transfer occurred prior to the origin of the phylum Porifera itself.

2. There is a very low level of variability in the organization of the Porifera examined here compared with that seen within the Anthozoan subclasses Octocorallia and Hexacorralia.

That there is variation in mitochondrial genome organization within the Porifera is likely shown by the results of the *Clathrina* sp. amplifications, in which no *rrnL-cox2* bands were amplified even when known good primers were used for amplification under favorable conditions. Within the Demospongiae the *Cliona* sp. sample also appeared to show a similar situation, although less definitively as the *cox2* amplifications for *Cliona* were typically very difficult.

3. The organization of tRNAs and their structure is not atypical for metazoan sequences.

This result is significant in its implications for the phylum Cnidaria. Since all of the mitochondrial genomes examined thus far from the anthozoan Cnidaria share a nearly complete lack of tRNAs, necessarily implying import from the nucleus. There has clearly been a significant event in the evolution of the Anthozoa at least; either a second massive transfer of genes from the mitochondrion to the nucleus or a generalized loss of tRNAs and import of strictly nuclear-derived copies. That this event occurred only in the evolution of the Cnidaria is clearly the case.

4. The relatively short size of the noncoding intergenic segments of the sponge mitochondrial genomes is concordant with the range of intergenic segment variation seen in the rest of the metazoa.

The Demospongiae in particular show no extremely long intergenic sequences in this section of the genome, and appear to follow the pattern common in the higher metazoans of having little or no intergenic spacer DNA. Some of the longer of these noncoding elements have a high T/A content, which is in accord with intergenic spacer sequences in both the metazoa and the protists. Portions of these segments can be folded into complex stem/loop structures which have been repeatedly implicated in transcript processing. All of these factors do not strongly set the metazoan mitochondrial DNAs apart from those of the higher metazoan radiation.

Chapter 5

Mitochondrial Sequence Evolution Rate Variation Across the Diploblast Radiation I: *rrnL*

INTRODUCTION

One of the more notable results of the initial characterization of mtDNA genes from Poriferans was a high degree of sequence similarity to published cnidarian mtDNA sequences. Amplification of contaminating cnidarian sequences was ruled out, and thus the sponges examined do appear to show an unusually low level of sequence divergence relative to the Cnidaria than do the sponges or Cnidaria to any of the triploblast phyla, or indeed to that between many individual triploblast phyla. However, both a wider representation of taxa within the diploblast radiation and a more careful approach to quantifying the relative levels of sequence divergence within and between the diploblasts and triploblasts is called for in order to provide a confident estimate of the significance of this observation.

Bases 1 to approximately 810 of the *Tetilla* sp. sequence represent the 3' end of the *rrnL* gene, and were utilized to demonstrate that the amplified sponge sequences clustered together in a radiation basal to representative sequences of all cnidarian classes, and apart from fungal and triploblast sequences (Watkins and Beckenbach 1999). However no attempt was made at the time to utilize this segment of the gene for serious quantification of the rates of sequence evolution in the diploblast phyla. It was noted that the divergence level within and between the

diploblast phyla was in this case similarly low compared with the extremely long branches which led to the triploblast phyla.

This particular segment of the *rml* gene, corresponding roughly to positions 1366 to 1965 of the *Metridium senile* lrRNA sequence (Pont-Kingdon *et al.* 1994) is particularly well-characterized throughout the phylum Cnidaria. Attempts to study inter- and intra-specific relationships within the Cnidaria have provided a wealth of sequence data for this gene, most of which are readily available in Genbank. As a consequence comparison of rates of *rml* sequence evolution in the Cnidaria with the rest of the diploblast radiation could be very thorough, if a sufficiently broad sample of taxa from the sponges and ctenophores were obtained. Since *rml* fragments had already been obtained from one demosponge and two calcareous sponges, a broader sampling of the poriferan radiation should be achievable.

Bridge *et al.* (1992) developed a set of *rml* primers which were found to have broad specificity in amplifying the same region within this gene within the Cnidaria and Ctenophora. These primers were in fact the same one used, as published, to amplify sponge *rml* fragments as well. They were here utilized to amplify all of the sequences in this study.

Previous analyses of *rml* sequences in the phylum Cnidaria have broadly suggested low rates of sequence evolution within the phylum at a variety of taxonomic levels. France *et al.* (1996) surveyed a wide range of species within the class Anthozoa, and noted an extremely low level of sequence divergence both at the interspecific and intraspecific levels, with by far the lowest level of divergence within the Octocorallia. Much of the sequence divergence that could be observed was concentrated

within variable length sections of the *rrnL* region they sequenced. As well, their results supported those of Bridge *et al.* (1995) in finding evidence for a strict two-subclass organization to the Anthozoa. Romano and Palumbi (1996,1997) surveyed the same region of *rrnL* in the scleractinian corals) and found a very high degree of sequence conservation. When including less conserved regions of the *rrnL* sequence, however, they found a total divergence rate compatible with that of insect mitochondrial DNA, although still much lower than the mammalian rate.

Since the discovery of a homolog of the bacterial *mutS* DNA mismatch repair gene in the mitochondrial genome of the octocorallians *Renilla kolikeri* (Beagely *et al.* 1995), *Sarcophyon glaucum* (Pont-Kingdon *et al.* 1995), and *Muricea californica* (Beagley *et al.* 1995) it has become a commonplace when confronted with a low overall sequence divergence rate in cnidarian mitochondrial genes to reference these reports and speculate that there may be a connection between the two. However, to date no study has placed the presence or absence of this gene within the mitochondrial genome into an overall context with the rate of sequence evolution in any given subset of cnidarian taxa. Indeed it has been argued that the presence of this gene in the Octocorallia is likely due to gene transfer from the nucleus to the mitochondrion and that its presence is very unlikely in any other metazoans (Pont-Kingdon *et al.* 1998).

Alignments pose an ongoing problem in attempting to make phylogenetic inferences from rRNA molecules in general because of the difficulty in assigning truly orthologous relationships to individual sites. Factors such as slippage during replication may lead to duplications and

deletions. In addition, selection pressure for maintenance of opposite sides of stem structures (Hillis and Dixon 1991) results in lack of independence of sites. It has recently been argued (Hancock and Vogler, 2000) that in fact attempts to generate truly structure-based alignments in highly variable regions may themselves be misleading owing to the tendency for replicational slippage combined with structural constraints to produce numerous short identical or nearly identical sequences. For this reason the approach taken in this study to align the *rmL* sequences was that utilized thus far in the literature. A sequence-weighted progressive multiple alignment approach as implemented in the CLUSTALW algorithm described by Thompson *et al.* (1994) was used. Independent alignments of my sequences with published cnidarian data have in fact generated conserved sequence regions which are strikingly similar to those already published (France *et al.* 1996, Watkins and Beckenbach 1999), and which can be matched to conserved stem-loop structures on the *M. senile* *rmL* model (Beagley *et al.* 1998).

Despite the very large representation of cnidarian *rmL* genes published, to date no significant effort has been expended to elucidate their overall implications about the rate of sequence evolution at the base of the metazoan tree. In part this has been due to a lack of corresponding sequence from other diploblast phyla, and to the difficulty in making reliable inferences about sequence alignment relative to the triploblast phyla or to reasonable outgroups. In fact, alignment with outgroup taxa is even more problematic than to the triploblasts.

In addition, this further complicates the task of making reasonable relative-rate estimates. Tests of the efficacy of relative rate tests (Muse and Weir 1992, Robinson 1998) have demonstrated the importance of

using the closest possible basal taxon to the two sets of taxa whose rates are being compared in order to provide realistic assessments of their relative rates. Very long outgroup branches, as are seen in the case of the *rrnL* data, can lead to erroneous rejection of significant rate differences.

In this study we utilize a large subset of the published *rrnL* sequences for all cnidarian classes, combined with an extensive subset of triploblast sequences. These sequences are aligned with amplified *rrnL* sequences from multiple poriferan and one ctenophoran species. This analysis represents the first complete sample of mitochondrial DNA sequences from all major phyla in the diploblast radiation. The problematic phylum Placozoa was not included owing to lack of easy sample availability, but debate currently exists regarding the proper placement of this phylum (Nielsen 1996, Abouheif *et al.* 1998). The best current estimates ally the Placozoa closely with, if not within, the cnidarian radiation (Bridge *et al.* 1995, Abouheif *et al.* 1998), and thus their absence is not felt to be a significant detriment to the implications of the current study.

Amplification and Sequencing of Mitochondrial rrnL

Amplification of mitochondrial *rrnL* sequences from Porifera and Ctenophora were all performed using the 16S1 and 16S2 primers (Table 3). Reaction conditions were 3 cycles at an annealing temperature of 48 degrees, followed by 27 cycles at 50 degrees, all with a 72 degree extension for 30 seconds. Typically a primary band of the appropriate size (500 to 700 bases) was obtained, with a number of secondary bands of varying sizes. Purification of the single mitochondrial band was

performed by taking a plug sample of an agarose gel under low-intensity ultraviolet light illumination, placing the plug in 20 microliters of water overnight at room temperature, and reamplifying for 25 cycles at an annealing temperature of 50 degrees with a 30 second 72 degree extension.

The only exception to this was the Ctenophore *Pleurobrachia bacheii*, which presented two closely-spaced bands on the agarose gel near 650 bp. Excision and reamplification of gel plugs on these two sequences produced two separate sequences, both of which were apparently homologous to mitochondrial 16S rDNA. These two bands were labelled PleuroSubA and PleuroSubB. Phylogenetic analysis of these sequences when compared with cnidarian 16S sequence and the same sequence from another Ctenophore, *Mnemiopsis leyidii*, placed both *P. bacheii* sequences into a single clade with the cnidarian class hydrozoa and the *M. leyidii* sequence. The alignment of Hydrozoan and Ctenophoran sequences may simply represent long-branch attraction between Hydrozoa and Ctenophora, as they are typically the two longest-branched clades in the diploblasts, and repeated phylogenetic analyses of nuclear genes (Wainwright *et al.* 1993, Kobayashi *et al.* 1996, Collins *et al.* 1998, Borchellini *et al.* 1998, Kim *et al.* 1999) have not supported a sister-group relationship between the Hydrozoa and Ctenophora as was proposed by Hyman (1940).

Insufficient informative variation exists to resolve the internal nodes in this relationship, but nonetheless both sequences appear to be compatible with a ctenophoran origin. It is possible that one of the two bands simply represents a contaminating hydrozoan sequence in the Ctenophore sample, or that one of them represents the amplification of a

nuclear copy of a mitochondrial sequence. Interestingly, a similar situation has also been observed in the scleractinian corals. Romano and Palumbi (1997) observed the amplification of multiple mitochondrial 16S bands in scleractinian corals and found strongly inconsistent results in taxonomic relationships inferred from the two sets of sequences. The authors speculate that this could be due to heteroplasmy due to interspecific mating, which has been demonstrated in some scleractinian species (Hatta *et al.* 1999), but strangely do not discuss the possibility of simple nuclear insertions of mitochondrial DNA sequences. In any case this result is quite interesting in light of our finding of a similar situation in a ctenophoran mitochondrial 16S amplification. This result was pursued no further in this study, however, and both the PleuroSubA and PleuroSubB bands were excluded from the study to provide an unbiased examination of the data, as there is currently no way to distinguish which represents the best sample of ctenophoran mtDNA. Most likely, one or both represent nuclear inclusions of mitochondrial DNA sequences. Indeed, further amplifications from *Pleurobrachia* using the same primers have yielded more apparent 16S sequences (J. Stewart, personal communication). This result increases confidence that nuclear copies of mitochondrial sequences have been amplified, and in the absence of any contiguous multi-gene sequence, it will be impossible to decide which represents the true *Pleurobrachia* mitochondrial sequence. I decided not to include the *Pleurobrachia* sequences in my analyses. In the case of the *Mnemiopsis* sequence, only a single 16S band was amplified.

Amplification of demosponge and calcisponge *rrnL* sequences was considerably less problematic. Secondary bands were occasionally

amplified, but several reamplifications and sequencings of these bands presented no significant matches to *rml* sequences. Five demosponge and three calcareous sponge *rml* sequences were amplified, representing two of the three sponge classes.

Sequence Alignment

Cnidarian and triploblast mitochondrial 16S rDNA sequence data were obtained from Genbank. Since the exact portions and lengths of sequences available in Genbank varied depending upon the original study in which they were obtained, some editing was necessary to bring all the sequence data into a single aligned data set which covered a purely orthologous section of the gene. In the case of full genome sequences the 3' half of *rml* was excised and quickly aligned with the *Tetilla* sp. sequence. The section of the gene surrounding the orthologous *Tetilla* sequence was then excised from the sequence and saved for a more thorough group alignment in a main file in FASTA format. In the case of published partial *rml* sequences the Genbank entry was simply excised from the file and saved to the group sequence FASTA format file for multiple sequence alignment in CLUSTALW. Once a full alignment had been performed, sections at the 5' and 3' ends of the sequences which were not represented in all taxa were excised from the final file.

The multiple sequence alignment was performed with CLUSTALW (Thompson *et al.* 1994) with the default pairwise and multiple alignment parameters. Multiple alignment with a range of gap creation and extension parameter changes were not found to significantly alter the alignment of the two regions homologous to the Conserved Sequence

Blocks 1 and 2 of France *et al.* (1996). Changing the alignment parameters altered the positioning of bases in the poorly conserved regions of the gene, but these were not included in the analysis. The full final alignment of the *rml* from both diploblasts and triploblasts is shown in Figure 20. This alignment consists of 352 nucleotides, roughly half of the sequence length typically amplified from any given diploblast phylum. This is a very short segment of *rml* however the alignment itself is extremely good in this region. Inclusion of less stringent portions of the sequence alignment was not felt to be justified, owing to the need to select only truly orthologous sites in the gene itself. In fact this criterion is absolutely critical if meaningful inferences are to be made with regard to relative rates between the diploblast and triploblast lineages.

The resulting sequence alignment, while short, is nonetheless quite good as can be seen in Figure 20. The level of sequence conservation in the three combined subsections of *rml* which are congruent with the conserved sequence blocks of France *et al.* (1996) is very high, although noticeably less so in the triploblast phyla. This is an immediate area concern since relative rate estimates for nucleotide sequence evolution must necessarily depend upon the alignment and characterization of orthologous sites. In the best of circumstances this is difficult to achieve with certainty in rRNA alignments, and thus the results we will present must be read in context with the results for protein-coding genes presented in Chapter 6.

The primary problem in making alignments in this set of taxa can be seen in Table 9. Alignment of just the triploblast sequences results in an unambiguously alignable 450-nucleotide region, of which 155 sites are

informative about relationships within the triploblast taxa. The overall level of similarity of the triploblast sequences with one another and the diploblasts with one another means that inclusion of both the triploblast and diploblast sequences in the alignment reduces the unambiguously alignable region to 352 nucleotides, of which 152 are informative for triploblast-triploblast relationships (not a significant reduction) and 166 are informative about relationships within the diploblasts. Alignments made with just the diploblast sequences included results in a 437 nucleotide alignment of which fully 227 sites are informative about relationships within the diploblasts. Furthermore, alignments made with just the diploblast taxa included universally produce two well-conserved and one semi-conserved sequence blocks in the sequences which fall almost exactly on the nucleotide boundaries of the Conserved Sequence Blocks which France *et al.* (1996) described for alignments of the same gene within the Anthozoa.

For this reason further study was made of the *rmL* sequence alignments using only the diploblast data set, with a length of 437 nucleotides. Sequence evolution rates within the diploblast phyla were examined in the same manner as for the triploblast/diploblast alignments, in order to characterize the relative rates of mitochondrial DNA sequence evolution within and between the diploblast phyla using this longer data set. These results will later be compared with those obtained for longer sections of protein-coding genes, albeit from a more limited set of taxa.

FIGURE 20

Mitochondrial *rnl* alignments across diploblast and triploblast radiations. Alignments of amplified *rnl* segments were made with CLUSTALW set to default alignment options, then further aligned by eye to minimize number of gaps. The resulting data file was then edited to remove sections with large numbers of gaps. Bases are shaded which possess a nucleotide concordant with at least 60% of the other bases in that column. The resulting data file is concordant with the conserved sequence block of the 16S alignment of France (1996). Species utilized in this data set, and their corresponding Genbank accession numbers are as follows: *Hydra vulgaris* (AF100773), *Liriope tetraphylla* (U19377), *Tubularia indivisa* (U19379), *Stichopathes spiessi* (U40286), *Cerianthus borealis* (U40288), *Anthopleura elegantissima* (U40292), *Sarcophyon glaucum* (AF064823), *Corynactis californica* (U40293), *Favia fragum* (U40295), *Protodendron* sp. (U40296), *Alcyonium* sp. (U40297), *Paragorgia* sp. (U40299), *Chrysogorgia chryseis* (U40306), *Narella bowersi* (U39786), *Muricea fruticosa* (U40302), *Metridium senile* (NC_000933), *Renilla muelleri* (U19372), *Aurelia aurita* (U19373), *Casseiopia* sp. (U19374), *Tetilla* sp. (AF035265), *Halichondria* sp. (AF362003), *Cliona* sp. (AF362004), *Microciona prolifera* (AF362005), *Grantia comoxensis* (AF035267), *Leucilla nuttingi* (AF035266), *Clathrina* sp. (AF362006), *Uncharacterized demosponge* (AF362007), *Mnemiopsis* sp. (AF362008), *Strongylocentrotus purpuratus* (X12631), *Felis catus* (NC_001700), *Dobsonia moluccensis* (AF179290), *Homo sapiens* (J01415), *Oncorhynchus mykiss* (L29771), *Scyliorhinus canicula* (NC_001950), *Lumbricus terrestris* (LTU24570), *Platynereis dumerilii* (NC_000931), *Artemia franciscana* (X69067), *Drosophila yakuba* (X03240).

	10	20	30	40	50	60	70	80	90	100	110	120
Hydra	AT					AG	GA	TTA	ATC	AT	ATTTT	A
Liriope						AG	GA	TTAA	ATC	AT	ATTTT	A
Tubularia						AG	GA	TTAA	ATC	AT	ATTTT	A
Stichopath						AG	GA	TTAA	ATC	AT	ATTTT	A
Cerianthus						AG	GA	TTAA	ATC	AT	ATTTT	A
Anthopleur						AG	GA	TTAA	ATC	AT	ATTTT	A
Sarcophyto						AG	GA	TTAA	ATC	AT	ATTTT	A
Corynaetis						AG	GA	TTAA	ATC	AT	ATTTT	A
Favia						AG	GA	TTAA	ATC	AT	ATTTT	A
Protodendr						AG	GA	TTAA	ATC	AT	ATTTT	A
Alcyonium						AG	GA	TTAA	ATC	AT	ATTTT	A
Paragorgia						AG	GA	TTAA	ATC	AT	ATTTT	A
Chrysogorg						AG	GA	TTAA	ATC	AT	ATTTT	A
Narella						AG	GA	TTAA	ATC	AT	ATTTT	A
Muricea						AG	GA	TTAA	ATC	AT	ATTTT	A
Metridium						AG	GA	TTAA	ATC	AT	ATTTT	A
Renilla						AG	GA	TTAA	ATC	AT	ATTTT	A
Aurelia						AG	GA	TTAA	ATC	AT	ATTTT	A
Cassiopeia						AG	GA	TTAA	ATC	AT	ATTTT	A
Tetilla						AG	GA	TTAA	ATC	AT	ATTTT	A
Halichondr						AG	GA	TTAA	ATC	AT	ATTTT	A
Cliona						AG	GA	TTAA	ATC	AT	ATTTT	A
Microciona						AG	GA	TTAA	ATC	AT	ATTTT	A
Gracilia						AG	GA	TTAA	ATC	AT	ATTTT	A
Leucilla						AG	GA	TTAA	ATC	AT	ATTTT	A
Clathrina						AG	GA	TTAA	ATC	AT	ATTTT	A
Demosponge*						AG	GA	TTAA	ATC	AT	ATTTT	A
Mnemiopsis						AG	GA	TTAA	ATC	AT	ATTTT	A
purpuratus						AG	GA	TTAA	ATC	AT	ATTTT	A
felis catus						AG	GA	TTAA	ATC	AT	ATTTT	A
dobsonia						AG	GA	TTAA	ATC	AT	ATTTT	A
homo sapien						AG	GA	TTAA	ATC	AT	ATTTT	A
Oncorhynch						AG	GA	TTAA	ATC	AT	ATTTT	A
scyllorhin						AG	GA	TTAA	ATC	AT	ATTTT	A
lumbicus						AG	GA	TTAA	ATC	AT	ATTTT	A
platynerai						AG	GA	TTAA	ATC	AT	ATTTT	A
artemis						AG	GA	TTAA	ATC	AT	ATTTT	A
drosophila						AG	GA	TTAA	ATC	AT	ATTTT	A

Hydra	130	140	150	160	170	180	190	200	210	220	230	240
Liriope	130	140	150	160	170	180	190	200	210	220	230	240
Tubularia	130	140	150	160	170	180	190	200	210	220	230	240
Stichopath	130	140	150	160	170	180	190	200	210	220	230	240
Carianthus	130	140	150	160	170	180	190	200	210	220	230	240
Anthopleur	130	140	150	160	170	180	190	200	210	220	230	240
Sarcophyto	130	140	150	160	170	180	190	200	210	220	230	240
Corynactis	130	140	150	160	170	180	190	200	210	220	230	240
Pavia	130	140	150	160	170	180	190	200	210	220	230	240
Protodendr	130	140	150	160	170	180	190	200	210	220	230	240
Alcyonium	130	140	150	160	170	180	190	200	210	220	230	240
Paragorgia	130	140	150	160	170	180	190	200	210	220	230	240
Chrysogorg	130	140	150	160	170	180	190	200	210	220	230	240
Murelia	130	140	150	160	170	180	190	200	210	220	230	240
Muricea	130	140	150	160	170	180	190	200	210	220	230	240
Metridium	130	140	150	160	170	180	190	200	210	220	230	240
Renilla	130	140	150	160	170	180	190	200	210	220	230	240
Aurelia	130	140	150	160	170	180	190	200	210	220	230	240
Cassiopeia	130	140	150	160	170	180	190	200	210	220	230	240
Tetilla	130	140	150	160	170	180	190	200	210	220	230	240
Halichondr	130	140	150	160	170	180	190	200	210	220	230	240
Cliona	130	140	150	160	170	180	190	200	210	220	230	240
Microciona	130	140	150	160	170	180	190	200	210	220	230	240
Grantia	130	140	150	160	170	180	190	200	210	220	230	240
Leucilla	130	140	150	160	170	180	190	200	210	220	230	240
Clathrina	130	140	150	160	170	180	190	200	210	220	230	240
Demosponge*	130	140	150	160	170	180	190	200	210	220	230	240
Purpuratus	130	140	150	160	170	180	190	200	210	220	230	240
Felis catus	130	140	150	160	170	180	190	200	210	220	230	240
Dobsonia	130	140	150	160	170	180	190	200	210	220	230	240
Homo sapien	130	140	150	160	170	180	190	200	210	220	230	240
Oncorhynch	130	140	150	160	170	180	190	200	210	220	230	240
Scyllorhin	130	140	150	160	170	180	190	200	210	220	230	240
Ictalurus	130	140	150	160	170	180	190	200	210	220	230	240
Platyneres	130	140	150	160	170	180	190	200	210	220	230	240
Artemia	130	140	150	160	170	180	190	200	210	220	230	240
Drosophila	130	140	150	160	170	180	190	200	210	220	230	240

TABLE 9

Number of parsimoniously informative sites in mitochondrial *rrnL* alignments made with various combinations of taxa. All calculations were made with the data sets which have been edited to remove all ambiguous alignment sites. Calculations of the numbers of parsimoniously informative sites were made with MEGA (Kumar 1993). DD= Diploblast-to-diploblast comparisons. DT = Diploblast-to-triploblast comparisons. TT = Triploblast-to-triploblast comparisons

Triploblast *rrnL* Alignments **450 nt**

TT Informative Sites 155 sites

Diploblast/Triploblast *rrnL* Alignments **352 nt**

DT Informative Sites 222 sites

DD Informative Sites 166 sites

TT Informative Sites 152 sites

Diploblast *rrnL* Alignments **437 nt**

DD Informative Sites 232 sites

1. *rrnL* Sequence Evolution Across the Metazoan Radiation

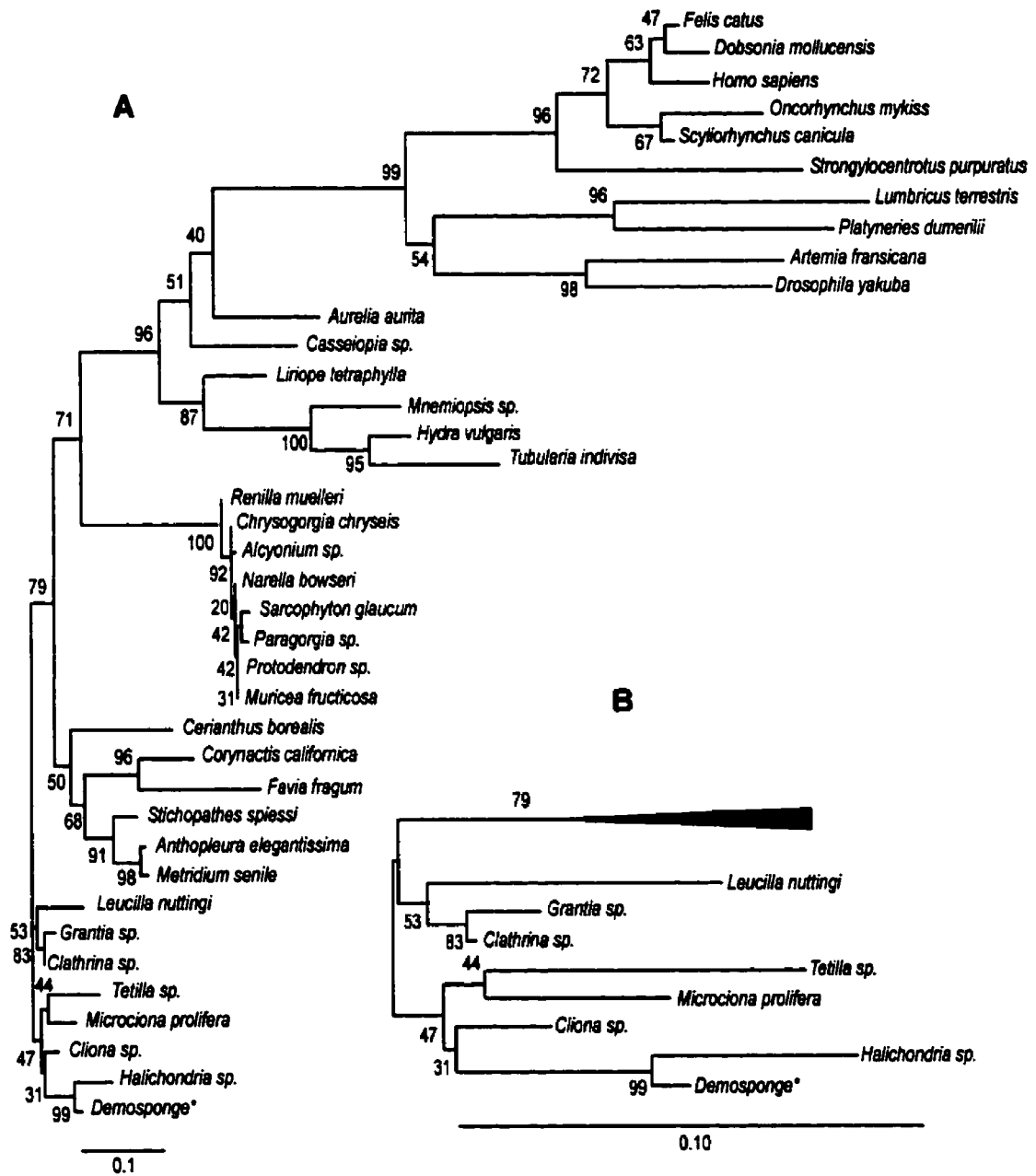
Construction of Phylogenetic Trees

Figure 21 shows the results of 500 neighbour-joining bootstrap replications on the 352-nucleotide combined diploblast and triploblast data set. The single most notable feature of this tree is the extremely poor rooting of the deep internal branches within and between the metazoan phyla. In particular the deepest rootings within the Cnidaria and between the Cnidaria and triploblast radiation are obscured to such a degree that there is no clear distinction between the taxa themselves. As well, the clustering of the phylum Ctenophora with the cnidarian class Hydrozoa is apparent, which harkens back to the ideas of Hyman (1940) which proposed that the Ctenophora ally most closely with the trachyline hydrozoans. This result, however, is incongruent with the bulk of molecular work which has been performed to elucidate the taxonomic position of this phylum (Wainwright *et al.* 1993, Borchellini *et al.* 1998). We must therefore regard the clustering of the Hydrozoa (which are by far the longest-branched cnidarians) and the Ctenophora (in which sequences are also quite divergent) as a potential case of long-branch attraction.

The triploblast radiation exists on a long branch which clusters with the Ctenophora, Scyphozoa, and Hydrozoa (which themselves all exist on relatively long branches). The Porifera, on the other hand, appear to exist on an extremely short set of internal branches. This clustering of rRNA sequences in a fashion which correlates directly with branch length distribution has been noted previously in connection with attempts to use nuclear 18S rRNA sequences to infer the global eucaryotic phylogeny (Stiller and Hall 1999) and is interpreted as

FIGURE 21

The results of 500 neighbour-joining bootstrap replicates on the combined diploblast and triploblast mt *rml* data set, performed with MEGA2 (Kumar 2000). Kimura 2-parameter gamma distances were utilized, with a gamma parameter of 0.64, determined empirically with PAUP* (Swofford 2000), and with pairwise deletion of gaps for all sequence pairs. The tree has been arbitrarily rooted with the Demospongiae. A. The full tree of the metazoan radiation as inferred by neighbour joining. Numbers above the nodes represent the percentage of bootstrap replicates supporting that node. B. Expanded view of the Porifera in the neighbour-joining tree, with the rest of the metazoan radiation collapsed.



indicative of branch-length artifacts in that context. That there is little phylogenetic information present to allow any confidence in the inferred phylogenies is illustrated by the bootstrap support levels, which fall near or below 50% for many deep internal nodes.

A more informative approach to characterizing the relative rates of sequence evolution in the metazoa using this data set is to constrain the inferred phylogeny to one which has broad support in the literature. A constrained maximum likelihood tree was therefore constructed using the full data set. The constraint backbone tree is shown in Figure 22. This tree represents a consensus established by a variety of studies, and for which significant support exists at least in the 18S ribosomal RNA subunit sequence data (Wainwright *et al.* 1993, Cavalier-Smith *et al.* 1996, Collins 1998, Kim *et al.* 1999), and which is also generally supported by both paleontological (Conway-Morris 1998) and morphological (Nielsen *et al.* 1996) data. Mapping of the relative levels of sequence divergence onto this well-supported tree may provide some insight into the true relative levels of sequence divergence between taxa in the data set. Most of the relative rate tests to be performed are nonetheless completely independent of this tree, as they depend upon straightforward pairwise comparisons of sequence divergence relative to an outgroup.

The constrained maximum likelihood tree is shown in Figure 23. Many features of the constraint tree are already supported by the simple neighbour-joining tree (Figure 21). In particular the relative rooting between the calcisponges, demosponges, and cnidarians is the same. The only features significantly different between the two trees are the rooting of the Ctenophora and of the triploblast radiation, as well as the

placement of these lineages into a clear basal context with respect to one another. Since both the Ctenophora and the triploblasts appear on long branches, and apparently misroot with the Hydrozoa and Scyphozoa, constraining the inferred tree to that of Figure 22 only serves to emphasize the extent of their relative levels of sequence divergence. The Ctenophora, it should be noted, exhibit approximately the same high level of sequence divergence whether they are rooted in the manner of Figure 24, or are placed as a sister taxon to the Cnidaria (Willmer 1990), or indeed are rooted as a sister taxon of the triploblasts as has been suggested by some studies based upon higher-level characters (Nielsen *et al.* 1996).

Constraint of the inferred tree to a taxonomically reasonable framework results in a dramatic decrease in the lengths of the internal branches between nodes in the basal portion of the tree. This must be regarded at least partially as a function of the limited size of the data set available for analysis with this gene. An insufficient number of informative characters exist which allow reliable resolution of the deepest internal nodes. On the other hand, it is likely at least partially due to the nature of the data itself. The existence of multiple long branches leading to the major metazoan phyla, combined with a short set of internal branches separating them would lead directly to the misrooting of nonmetazoan outgroups typically seen in attempts to reconstruct the basal metazoan phylogeny.

Other portions of the constrained tree are congruent with previous work. The specific portions of the anthozoan radiation addressed in the study of France *et al.* (1996) which utilized much the same data set, are congruent with the results of that earlier work, although the exact

FIGURE 22

The constrained backbone tree used for maximum likelihood calculations. This tree has been constructed as a consensus of current understanding of basal metazoan relationships. Alteration of the rooting of the Ctenophora to a place between the Cnidaria and the triploblast radiation has no significant effect upon the results of the tree searches.

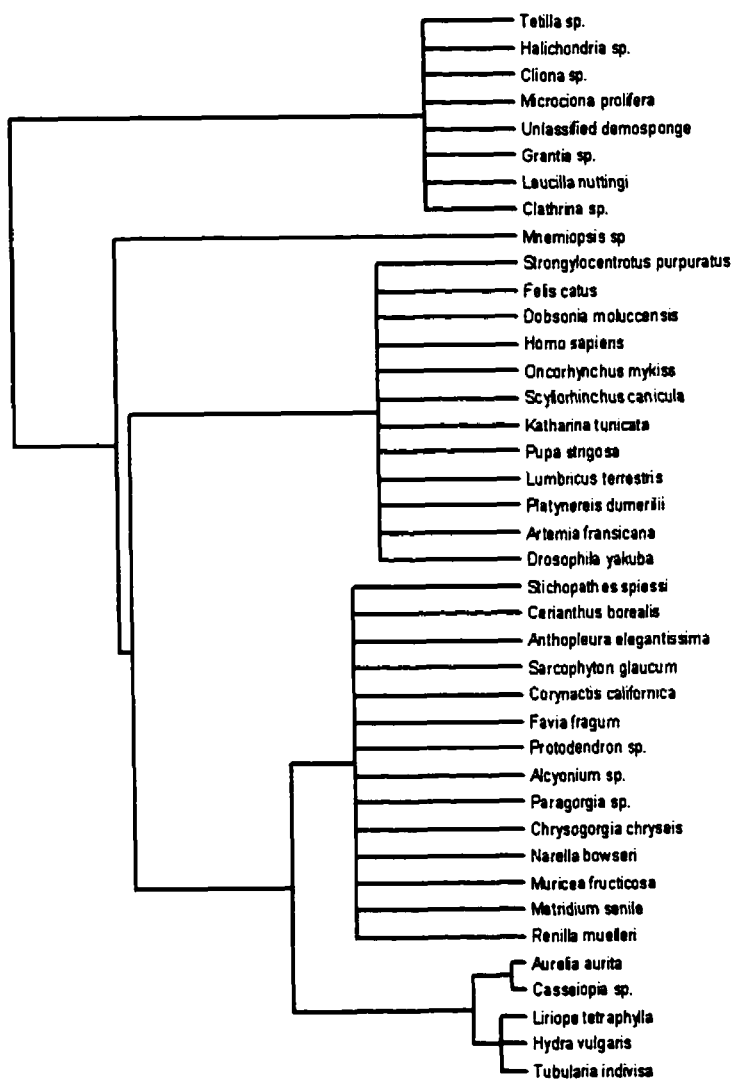
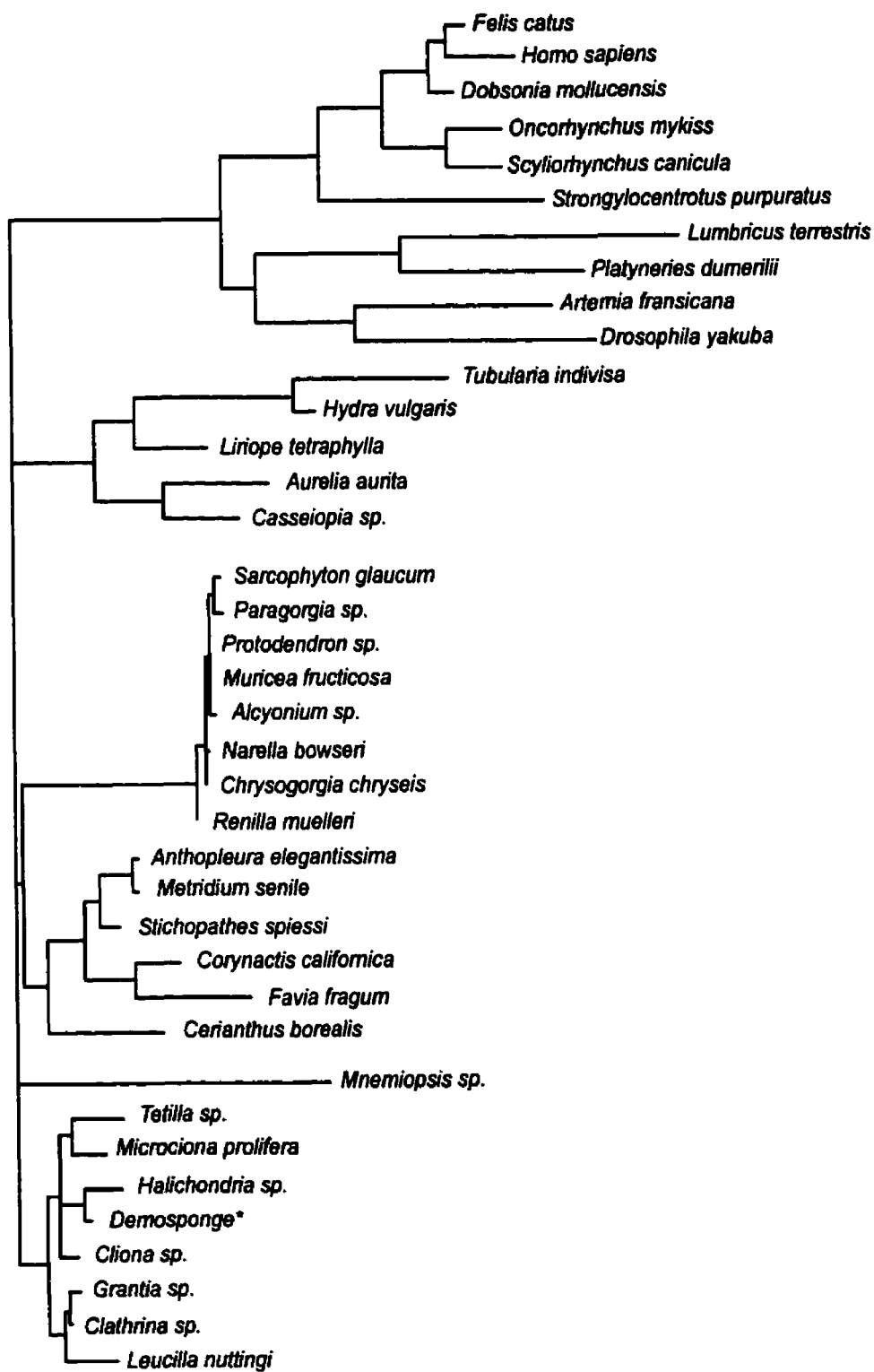


FIGURE 23

Maximum likelihood tree derived with PAUP* (Swofford 2000) from the combined diploblast + triploblast *rmL* data set. An exhaustive search on topologies was performed, subject to the constrained backbone tree in Figure 22. The tree has been arbitrarily rooted with the Porifera.



0.1

placement of the Ceriantipatharian lineage is somewhat labile, and is dependent upon the exact parameters of distance correction and tree reconstruction employed. Indeed, studies with nuclear 18S sequences have found similar results (Berntson, 1999).

One of the more striking features of any of the trees in Figures 21 and 23 is the very low degree of sequence variation in the phylum Porifera. All members of both poriferan classes appear to show a degree of sequence conservation in this section of the rRNA gene which is considerably lower than that between and even within any other examined metazoan group, with the exception of the octocorallian cnidarians. Since the data set includes members of two of the three sponge classes, this consistency of result is interesting. In addition, the representatives of the cnidarian class Anthozoa in the tree also appear to exist on a similarly short set of total branch lengths.

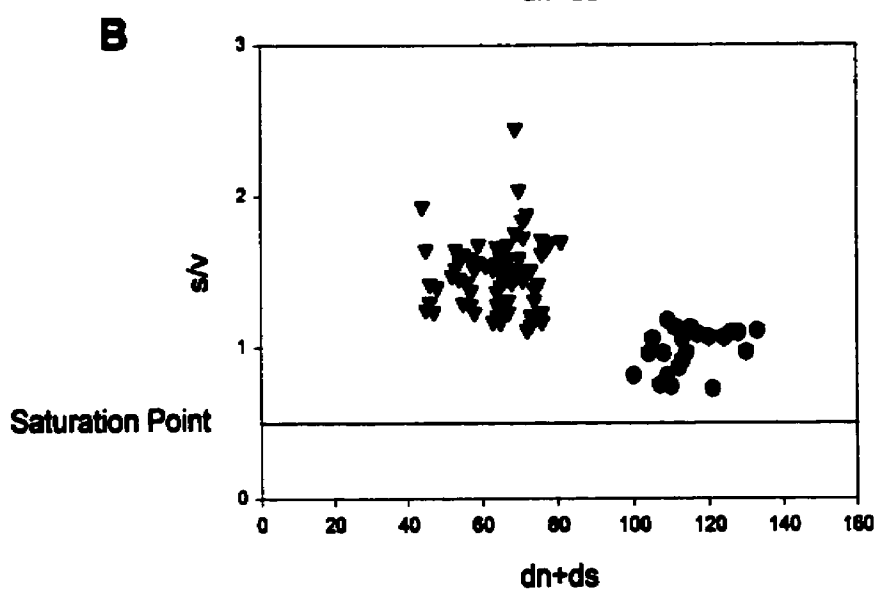
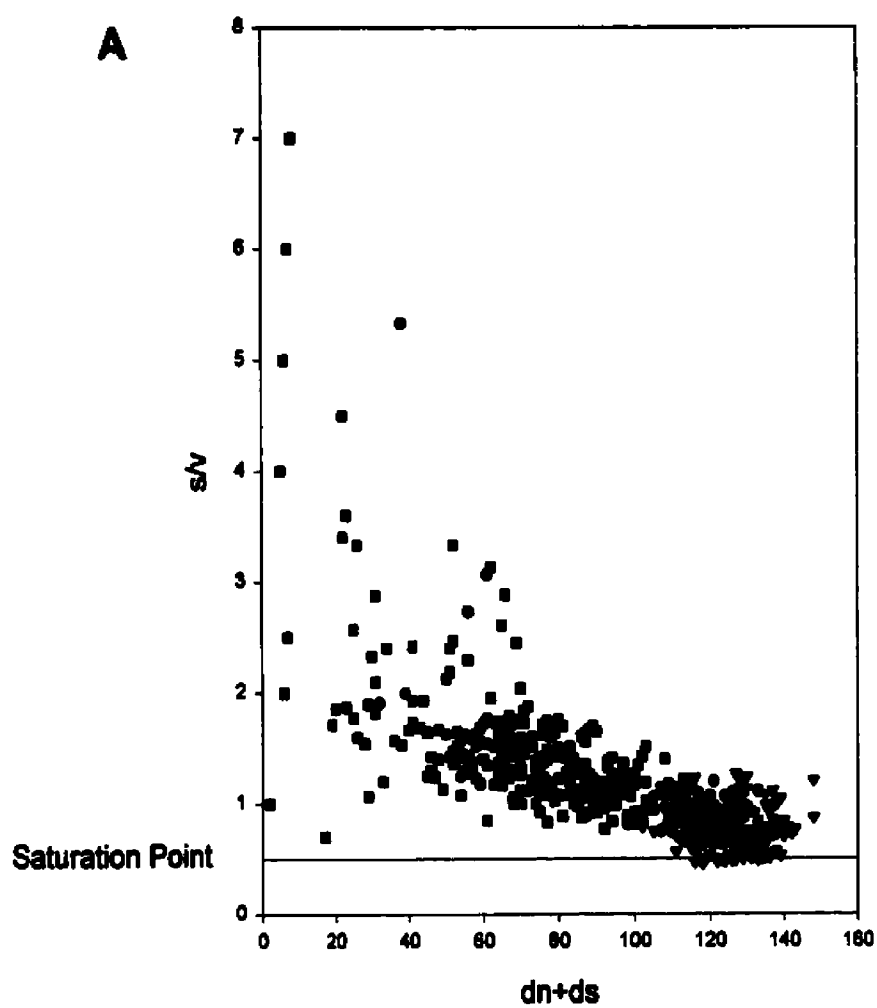
Transitional Saturation in the Sequence Data

Since the divergence times are extreme in the case of the Porifera, Cnidaria, and Ctenophora, likely at least 600 million years (Conway-Morris 1998) it is appropriate before examining rate differences between genes in these phyla in detail to first characterize the nucleotide content of the sequences themselves. Inherent biases in the data could lead to misleading inferences about the parameters of sequence evolution between the genes. In the case of the *rml* data set I first examine the degree of transitional saturation which appears in the data. More detailed analysis of the type and degree of nucleotide bias

FIGURE 24

Saturation graphs for transitions and transversions in the aligned mitochondrial *rml* data set. The transition to transversion ratio is plotted against the total number of transitions + transversions in each pairwise comparison. All data was derived with MEGA (Kumar 1993) from the original aligned data set.

- A. Filled circles represent triploblast-to-triploblast comparisons, squares represent diploblast-to-diploblast comparisons, and triangles represent diploblast-to-triploblast comparisons. Clearly the diploblast-to-triploblast comparisons show the highest degree of transitional saturation. A reference line has been plotted at an s/v value of 0.5 which corresponds to the theoretical limit of complete transitional saturation (Abouheif *et al.* 1998).
- B. A portion of the same data set representing only poriferan-to-anthozoan (triangles) and protostome-to-deuterostome (circled triangles) pairwise comparisons. The Porifera and the Anthozoa must have diverged at a time at least equal to and likely greater than the divergence between the protostomes and deuterostomes.



which exists in diploblast mitochondrial genomes will be made with the much larger protein-coding gene data set in Chapter 6.

Previous work (Mindell and Thacker 1996) has suggested that in mammalian mitochondrial DNA sequences a transitional saturation zone for mitochondrial sequences exists at greater than approximately 200 million years when the entire *mtL* sequence is considered. On the other hand this zone is more temporally extended in the slower-evolving fish lineages (Cantatore *et al.* 1994, Martin 1995) and possibly in turtles as well (Bowen *et al.* 1993). There is some indication that a similar trend of high levels of sequence divergence are found in some protostome lineages as well (Liu and Beckenbach 1992, Saito *et al.* 2000). However, we cannot make meaningful estimates of the times of origin of the diploblast phyla, in that they exist in a region of uncertainty greater than 200 million years.

Figure 24 shows the distribution of the ratio of transitional to transversional substitutions in the aligned sequence data plotted against the total number of substitutions. The bulk of the triploblast-to-diploblast pairwise comparisons in fact exist at the extreme end of the spectrum, suggesting that there is a considerable degree of transitional saturation in these comparisons, and indeed a number of these cases lie on the pure saturation line at a value of 0.5, in which case there are two transversions for every transitional difference, the maximum expected value for unrelated sequences. This is an understandable result given the extreme divergence time between the diploblast and triploblast radiations.

As well, a large number of the triploblast-to-triploblast comparisons are nearing this saturation point, in concordance with the

characterized nature of higher metazoan mitochondrial sequence evolution. What is more interesting is that the bulk of the diploblast-to-diploblast comparisons exist at much less saturated points on the curve. The divergence time between Porifera and Cnidaria must be at least as great as that between the most diverged triploblast phyla, and yet there exists a much less saturated set of sequence changes between them.

The necessity of aligning taxa from such a broad phylogenetic range has left us with a very highly conserved block of sequence in which presumably purifying selection has been a component in maintaining functionally significant sequence. However the point has recently been made (Pesole *et al.* 1999) that the case for rapid divergence between metazoan mitochondrial DNAs has often been overstated, and that many portions of the metazoan mitochondrial genome do in fact evolve at rates compatible with that of conserved nuclear DNA sequences. In the case of this particular data set the nature of the alignment criteria has left us with a particularly slowly-evolving portion of the gene, which is an excellent data set for the comparison of the relative rates of sequence evolution in these highly diverged groups.

In calculating the relative rates of change in the evolution of the mitochondrial genes, we feel that the use of both transitional and transversional data combined is justified on a number of levels. Firstly, the data does not demonstrate of complete transitional saturation for all comparisons. While the direct diploblast-to-triploblast comparisons are at the highest end of the curve and are undoubtedly subject to a high level of saturation, the characteristics of the data set as a whole are such that the application of distance-correction methods would allow a reasonable estimate of the degree of divergence to be made. Secondly,

the inclusion of the more fully saturated data set would only serve to obscure the degree of divergence between the diploblasts and triploblasts. Since we are characterizing an apparently lower rate of sequence evolution in some diploblasts taxa, its inclusion would not increase the likelihood of finding a significant difference where none existed. Nonetheless we will also analyze the transversional distance data by itself in order to serve as a comparison with the results found using the combined data.

Most sponge-sponge and sponge-anthozoan comparisons exist in an approximately the linearly accumulating portion of the graph, and appear to show that transitional saturation has not been achieved in these sequences. Given that the divergence time between these taxa is as long as the entire history of the metazoa itself, this bespeaks an extraordinarily low rate of sequence change relative to the bulk of the metazoan radiation. The root causes underlying this observation will be discussed below.

Calculation of Pairwise Genetic Distances Between Taxa

Tables 10 and 11 provide pairwise distances for the various groups of taxa involved in this study. Table 10 contains distances calculated with the gamma-corrected Kimura 2-parameter model using both transitions and transversions, whereas Table 11 shows distances calculated strictly from the transversional data. Interestingly, examination of both tables reveals similar features in both sets of results.

Table 10

Table of average pairwise gamma-corrected Kimura 2-parameter transitional+transversional distances and standard deviations between sets of taxa in the mitochondrial *rrnL* triploblast+diploblast data set. The gamma parameter has been set to an empirically derived value of 0.64. Distances were calculated with MEGA (Kumar 1993).

N		Demospongiae	Calcispongiae	Anthozoa	Scyphozoa	Hydrozoa	Ctenophora	Deuterstomes	Protostomes
5	Demospongiae	0.091 +/- 0.034							
3	Calcispongiae	0.104 +/- 0.029	0.060 +/- 0.037						
14	Anthozoa	0.284 +/- 0.065	0.269 +/- 0.061	0.256 +/- 0.171					
2	Scyphozoa	0.237 +/- 0.124	0.229 +/- 0.127	0.372 +/- 0.091					
3	Hydrozoa	0.522 +/- 0.153	0.505 +/- 0.157	0.515 +/- 0.105	0.472 +/- 0.196	0.316 +/- 0.108			
1	Ctenophora	0.483 +/- 0.077	0.450 +/- 0.029	0.534 +/- 0.059	0.495 +/- 0.086	0.289 +/- 0.044			
6	Deuterstomes	0.966 +/- 0.093	0.894 +/- 0.065	0.986 +/- 0.094	0.930 +/- 0.115	0.945 +/- 0.163	1.066 +/- 0.099	0.281 +/- 0.142	
4	Protostomes	0.804 +/- 0.051	0.776 +/- 0.051	0.864 +/- 0.111	0.801 +/- 0.036	1.037 +/- 0.221	1.241 +/- 0.240	0.784 +/- 0.151	0.734 +/- 0.211

K2P s+v gamma = 0.65

Table 11

Table of average pairwise gamma-corrected Kimura 2-parameter transversional distances and standard deviations between sets of taxa in the mitochondrial *rrnL* triploblast+diploblast data set. The gamma parameter has been set to an empirically derived value of 0.64. Distances were calculated with MEGA (Kumar 1993).

N		Demospongiae	Calcispongiae	Anthozoa	Scyphozoa	Hydrozoa	Ctenophora	Deuterstomes	Protostomes
5	Demospongiae	0.028 +/- 0.012							
3	Calcispongiae	0.038 +/- 0.013	0.012 +/- 0.008						
14	Anthozoa	0.097 +/- 0.022	0.096 +/- 0.026	0.080 +/- 0.061					
2	Scyphozoa	0.039 +/- 0.011	0.075 +/- 0.052	0.122 +/- 0.031					
3	Hydrozoa	0.147 +/- 0.086	0.182 +/- 0.046	0.204 +/- 0.046	0.174 +/- 0.050	0.122 +/- 0.044			
1	Ctenophora	0.021 +/- 0.012	0.180 +/- 0.004	0.194 +/- 0.013	0.166 +/- 0.025	0.087 +/- 0.030			
6	Deuterstomes	0.507 +/- 0.090	0.551 +/- 0.058	0.519 +/- 0.066	0.479 +/- 0.054	0.526 +/- 0.053	0.621 +/- 0.051	0.093 +/- 0.063	
4	Protostomes	0.314 +/- 0.081	0.366 +/- 0.053	0.370 +/- 0.050	0.329 +/- 0.041	0.381 +/- 0.081	0.376 +/- 0.058	0.294 +/- 0.044	0.278 +/- 0.074

K2P v gamma = 0.65

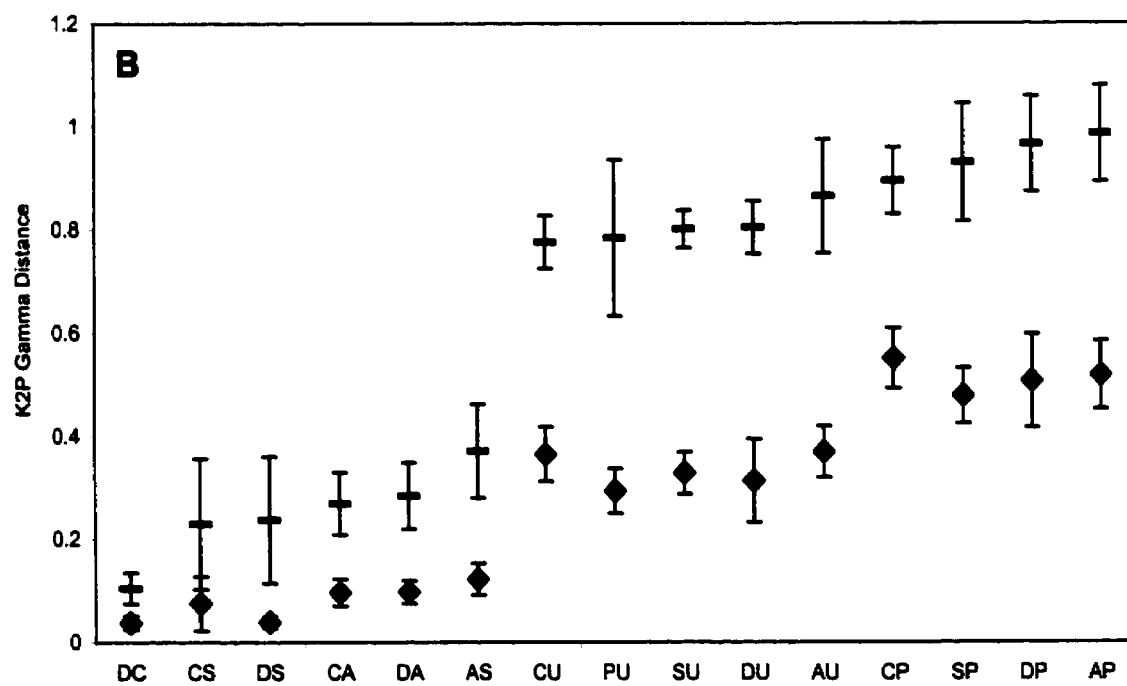
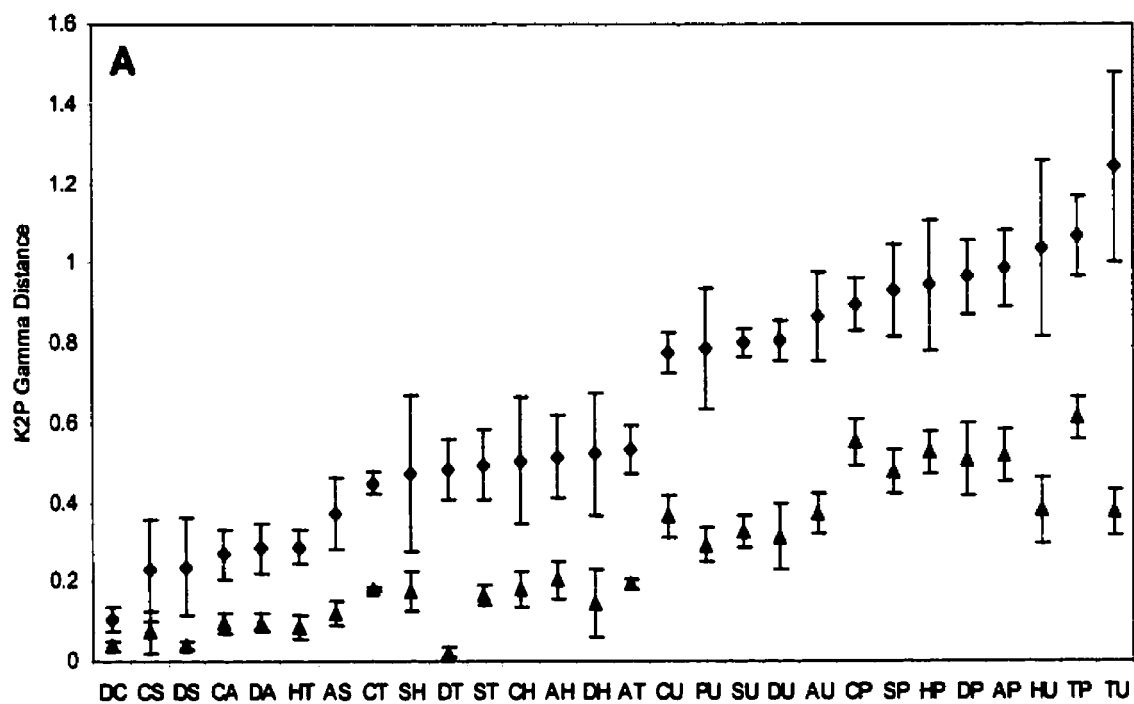
Since these are merely pairwise comparisons, the specific features of the phylogenetic trees presented in Figures 20 and 22 are not relevant, but the general trends found in both trees are apparent in this data set as well. Specifically, the Porifera show a radically reduced level of sequence divergence relative to the rest of the metazoan radiation. The average divergence of the sponge sequences is approximately 4 to 8 times less than the average divergence between triploblast taxa. This includes many pairwise comparisons for taxa which apparently diverged prior to the Cambrian explosion. As well the Cnidarian radiation appears to be divided strongly between the Anthozoa, which generally show a lower rate of divergence, and the classes Scyphozoa and Hydrozoa, which show significantly higher levels.

The distribution of genetic distances for pairwise comparisons of taxa across the metazoan radiation is shown in Figure 25A. The presentation of data points in this graph has been sorted from lowest to highest divergence level, and presents a continuum of values between all pairwise comparisons, ranging from diploblast-only to diploblast-triploblast sequence comparisons. There is an apparent upward gap in the distances for both transversional and transitional/transversional distances at the point at which the first diploblast-to-triploblast pairwise comparison is made, between calcisponges and deuterostomes. Since the hydrozoan and ctenophoran sequences are apparently the most diverged within the diploblast lineage - notwithstanding the single exceptional result described below for the Ctenophore transversional distance - these comparisons were removed and the re-plotted graph is

Figure 25

Histogram of pairwise Kimura 2-parameter gamma distances between sets of taxa as shown in Tables 10 and 11. Here the values have been sorted from lowest to highest, and highlight the distribution of inter-taxon distances. A. All inter-taxon distance means and standard deviations. Note the transition zone between the diploblast-diploblast and diploblast-triploblast comparisons. Means represented with diamonds are for the combined transitional and transversional distances, and means represented with triangles indicate the transversional distance only. B. The same data set as in Figure 25A, but with all comparisons involving the phylum Ctenophora and the cnidarian class Hydrozoa removed. The transition zone between the pairwise comparisons is now more apparent. Means represented with horizontal lines are for the combined transitional and transversional distances, and means represented with diamonds indicate the transversional distance only.

KEY: D=demosponge, C=calcisponge, A=anthozoan,
S=scyphozoan, H=hydrozoan, T=ctenophoran, P=protostome,
U=deuterostome



shown in Figure 25B. This figure emphasizes the dichotomy in inter-taxon distances between the demosponges, calcisponges, anthozoa, and scyphozoa with the triploblast radiations. Particular note should be taken of the relative genetic distances in Figure 25B labelled CA, DA, and PU as these represent the distances between the sponge and anthozoan radiations, and between the deuterostome and protostome radiations. Since the divergence time between the sponges and the Anthozoa must be at least as great if not greater than between the protostomes and deuterostomes, this result demonstrates a constitutively higher rate in the protostomes and deuterostomes (or alternately a lower rate in the diploblasts), irrespective of the long branch on which the triploblast radiation sits.

Relative Rate Tests

To confirm the apparent rates differences seen in the basal metazoa relative to the triploblast radiation we apply a 2-cluster relative rate test as described by Takezaki *et al.* (1995). This test measures the divergence of the means of two groups of sequences from one another relative to a basally diverged outgroup, which is itself the question of interest in this case. Application of the complete aligned *mmL* data set as used for construction of the phylogenetic trees in Figures 20 and 22 however would not be the ideal framework for the test itself. Although representation of larger numbers of taxa spanning the entire diploblast radiation will increase the likelihood of inferring an accurate test result (Robinson *et al.* 1998), the results of relative rate tests, and by extension of any comparison of rate differences can be adversely affected by nonindependence of branch lengths caused by an excess of extremely short internal branch lengths, for instance by oversampling of closely-

related taxa (Robinson *et al.* 1998). Our full *mL* data set may exhibit this effect in that a large number of short internal branches exist within the poriferan and anthozoan radiations. In particular the very short-branched octorallian lineage is highly oversampled. Although we regard this largely as a consequence of the slow rate of mutational accumulation in these taxa, we nonetheless examined a reduced sample set, in which 15 taxa showing short internal branches eliminated from the data set. A constrained maximum likelihood tree derived from this data set is shown in Figure 26. The resulting tree shows a more equal distribution of terminal branch lengths. The results of relative rate tests on this data set are shown in Table 12.

We cannot directly apply relative rate tests to the sponge lineages relative to the higher metazoans as they are themselves used as the outgroup to the metazoa as a whole in these tests. Attempts to align orthologous protist *mL* sequences provided such poor alignments that no trust was to be placed in the results of any tests performed with them. Since the outgroup alignment itself is absolutely critical to providing a meaningful test result, we feel there is no justification for making relative rate tests of the poriferan lineage with this data set. On the other hand, the Porifera themselves are the ideal outgroup to the metazoa as a whole for the application of this test. A high level of confidence can be placed in the results of relative rate tests for the Cnidaria and Ctenophora relative to the triploblast radiation using the sponges as an outgroup.

In order to examine the robustness of the test results we repeated the tests using a number of distance-correction methods. Table 12A shows the results of the test as applied with the Kimura 2-parameter correction on the full transitional+transversional data set, while the

Figure 26

Constrained maximum likelihood tree generated from the reduced diploblast and triploblast data set, selected for roughly equalized numbers of taxa and branch lengths, and to correct the oversampling of the octocorallian taxa. This is the data set used for most of the relative rate tests comparing the diploblast to triploblast radiations. The phylogeny was constructed with PAUP* (Swofford 2000) using an exhaustive search on the constrained backbone tree, with empirical estimation of all sequence evolution parameters from the data set.

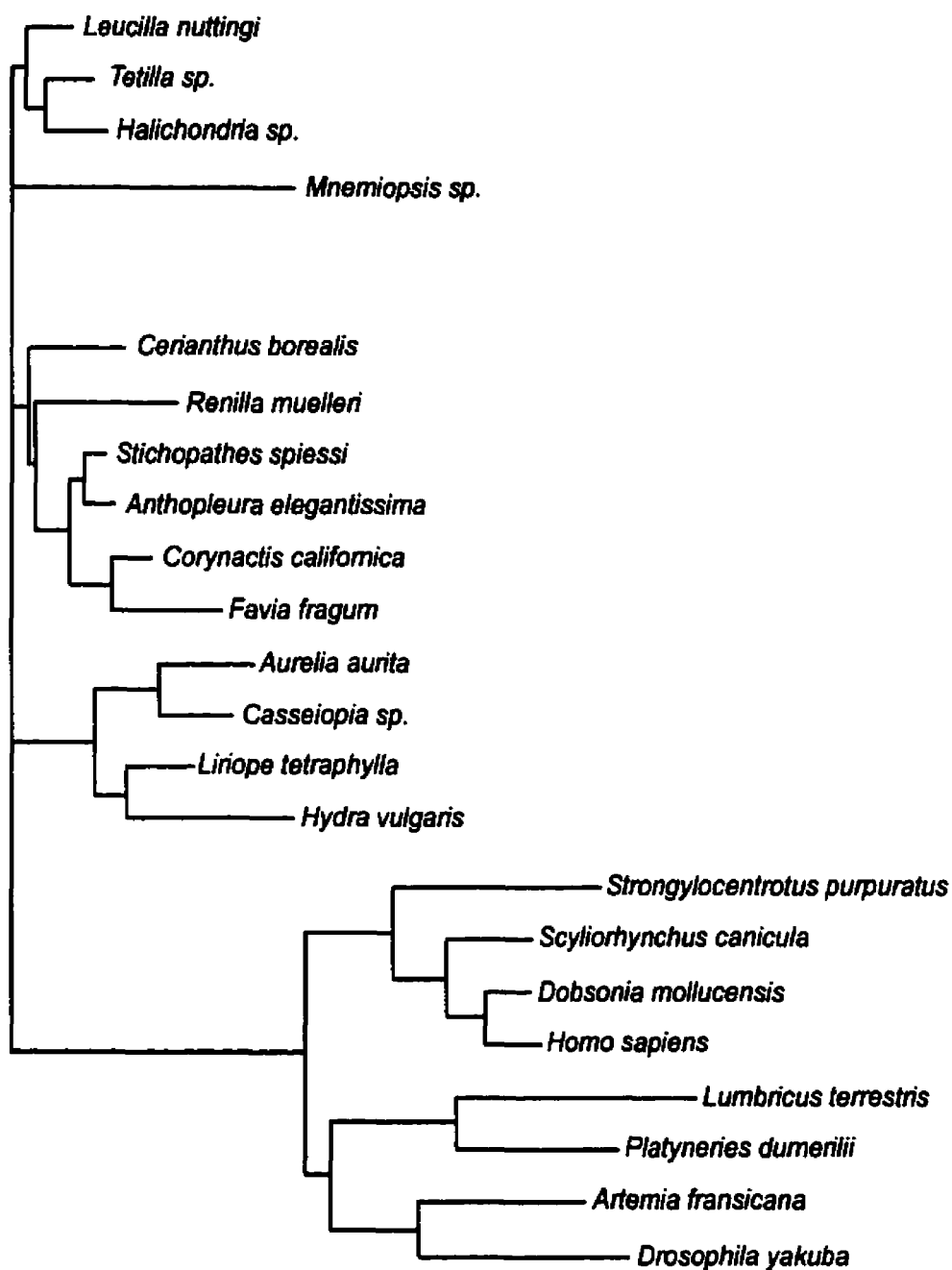


Table 12

Results of the 2-cluster relative rate test as calculated with the PHYLTEST package (Kumar 1995) on diploblast taxa compared to triploblast lineages, using the reduced data set described in the text. All tests utilized the combined demosponge and calcisponge *rml* sequence data as an outgroup for the rest of the metazoan radiation. A. Results using the Kimura 2-parameter distance correction on both transitional and transversional data. B. Results using the Kimura 2-parameter distance correction with only transversional data. C. Results using the Kimura distance correction with a gamma parameter of 0.64. D. Results using the full aligned *rml* data set (no taxa eliminated) and the Kimura 2-parameter correction on transversional data only. All test statistics which are less than the critical value of 2.86 and thus fail to show rate equivalency are shown in **bold**.

	Anthozoa	Scyphozoa	Hydrozoa	Ctenophora	Deuterostome
A					
Deuterostome	7.1571	5.2289	3.8348	2.7595	
Protostome	8.8614	7.2255	6.1374	4.4725	2.1219
B					
Deuterostome	6.2017	4.5484	4.0486	3.2469	
Protostome	7.8028	6.3889	5.8676	4.9276	2.793
C					
Deuterostome	5.3195	4.3109	3.1756	2.4683	
Protostome	6.2407	5.5357	4.8287	3.9197	1.9476
D					
Deuterostome	6.0043	4.5099	3.3499	3.3563	
Protostome	7.7816	6.4479	5.3779	5.0851	2.8801

results of the test using the Kimura 2-parameter correction using only the transversional data is shown in Table 12B, and the Kimura 1-parameter gamma correction (with a gamma parameter of 0.64) is shown in Table 12C. All tests support the same rate relationships of the diploblast with the triploblast taxa, in that significant rate differences are found.

The only exception to this trend is in the ctenophoran tests, in which equality of rates between the Ctenophora and the deuterostomes is found when using only the transversional data set. This is likely a direct effect of the observed lower transversional than transitional + transversional distance between ctenophores and sponges seen previously in the genetic distance calculations. The combined transversional and transitional distance between the Ctenophora and Demospongiae is substantial, whereas the distance from the Hydrozoa is about half of that. However based upon the purely transversional distance the Ctenophora become substantially closer to the Demospongiae, altering the distance between the Ctenophora and the outgroup used for the test. This causes the Ctenophora-to-deuterostome distance to increase. It is worth noting that when only the diploblast data set is utilized for the sequence alignments, this effect does not present itself. Rooting of the Ctenophora is problematic (Nielsen *et al.* 1996, Dewel 2000) but recent work has supported the basal placement of the phylum between the Porifera and Cnidaria (Collins 1998, Abouheif *et al.* 1998, Kim *et al.* 1999). This mitochondrial data set shows a peculiar property of the ctenophoran sequences: they are clearly very sensitive to alignment properties and to subsequence selection for

alignment. Further careful work with this phylum is called for before definitive assignment of its rooting can be made.

Table 12D shows the same 2-cluster tests with Kimura 2-parameter distances applied to the full data set as seen in Figure 23. The results of most tests are consistent with those described previously, with the exception that a significant rate difference is found between the deuterostome and protostome lineages, although at a level of bare significance. Inclusion of the full data set does indeed reduce the significance level of many of the tests, but also raises it for the ctenophoran tests, and for the Scyphozoa/protostome test as well as for the protostome/deuterostome test.

Clearly the overall results of these tests indicate that there is strong support for a significant rate difference between all of the cnidarian taxa and the triploblast lineages. Examination of the relative pairwise genetic distances between the taxa demonstrates that the Cnidaria have a lower overall rate of sequence evolution. That much of this variation is concentrated in a single long branch leading to the triploblast radiation is apparent from visual examination of the constrained maximum likelihood trees (although it is also apparent in the unconstrained trees). The relative rates between the Ctenophora and the triploblasts, and between the two primary monophyletic triploblast radiations, are more equivocal. Since the sponges appear to show a shorter set of terminal and internal branch lengths than the Cnidaria, it follows as a necessary conclusion that the Porifera are possessed of significantly lower rates of mitochondrial sequence evolution as well.

Table 13

Relative rates of mitochondrial *mtL* sequence evolution between diploblast and triploblast taxa. Average inter-taxon branch lengths relative to the last common node between each pair of taxa are presented as calculated with PAUP* (Swofford 2000) from the constrained maximum likelihood tree in Figure 26

	Porifera	Anthozoa	Scyphozoa	Hydrozoa	Ctenophora	Deuterostome
Deuterostome	0.154 +/- 0.027	0.255 +/- 0.076	0.427 +/- 0.030	0.428 +/- 0.128	0.525 +/- 0.031	
Protostome	0.133 +/- 0.024	0.221 +/- 0.67	0.370 +/- 0.031	0.371 +/- 0.095	0.455 +/- 0.034	0.757 +/- 0.132

Calculation of Relative Rates

We calculated the relative rate of sequence evolution of the *rml* genes between metazoan taxa as the ratio of their genetic distances from the last common node of each pairwise combination of taxa on the constrained maximum likelihood tree (the patristic distance). The most basal internal branch-lengths of the constrained maximum likelihood tree are zero, creating a polytomy between the Porifera, Ctenophora, Cnidaria, and the triploblast radiation. As a consequence the exact rooting of the base of the metazoan tree, in order to find an appropriate node for the inclusion of intertaxon comparisons for the Porifera, is not important for this data set. This allows us to include estimated relative rates of sequence evolution for the Porifera as well. These estimates are shown in Table 13.

The range of relative rate variation is surprisingly large – the sponges appear to evolve some 6 to 7 times slower than the triploblast lineages as a whole, while the anthozoan range is some 4 to 5 times slower. The classes Scyphozoa and Hydrozoa on the other hand demonstrate a rate difference of only some 2 to 3 times less than the triploblasts. This makes for an interesting contrast to the results presented in Figure 25, in which there is apparently a significant difference between the hydrozoan and scyphozoan rates. This is most likely attributable to the difference in the estimation of the genetic distances – maximum likelihood on a constrained tree in this case and a simple gamma-corrected Kimura 2-parameter estimate on the other. The criterion of the constrained backbone tree significantly alters the interpreted genetic distances, and will be discussed further in Chapter 6. The deuterostome lineage appears to evolve at a rate somewhat closer to

that of the diploblasts themselves, and the deuterostome and protostome radiations are very close to one another at this level of taxonomic comparison.

2. *rml* Sequence Evolution Within the Diploblast Radiation

Since the aligned *rml* data set including both diploblast and triploblast phyla contains a total of 352 nucleotides, whereas the alignments with only diploblast phyla consist of 437 nucleotides, it seems prudent to establish intra-diploblast rate variation within the context of this larger data set. Larger data sets are more reliable for the establishment of relative levels of rate variation (Tajima 1993), and the combination of a larger data set with a broad representation of all the diploblast provides a stronger estimate of these parameters (Robinson *et al.* 1998).

Alignments were constructed with the same methodology described for the diploblast/triploblast data set. ClustalW default alignments were optimized by eye, and then trimmed to a data set which was congruent almost to the base position with the conserved sequence blocks of France *et al.* (1996) which were utilized to establish the broad patterns of phylogeny within the Anthozoan radiation.

Figure 27 shows the results of 500 neighbour-joining bootstrap replicates of this data set using a gamma-corrected Kimura 2-parameter distance correction with a gamma parameter of 0.63. Overall the tree is highly similar to that of the full diploblast + triploblast data set tree shown in Figure 21. In particular the very poor resolution of the basal branching of the diploblast phyla has not improved with this larger data set, and the bootstrap values within the Porifera themselves have not increased significantly. Resolution of the relationship between the

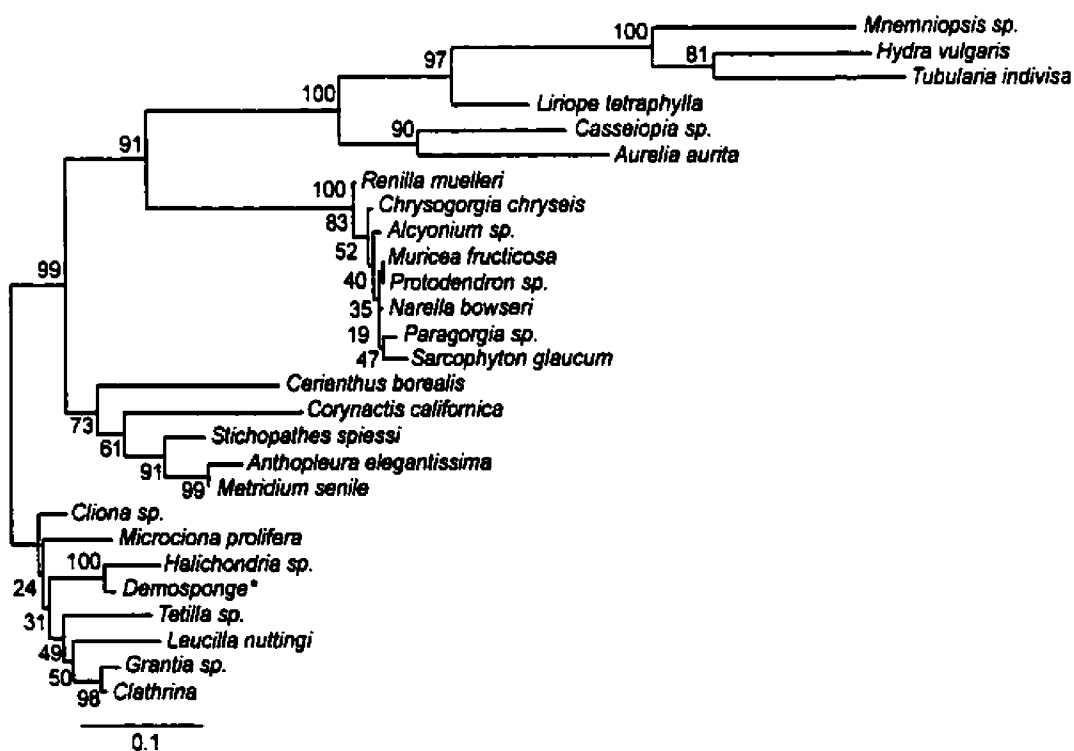
Table 14

Pairwise gamma-corrected Kimura 2-parameter transitional + transversional distances and standard deviations between sets of taxa in the mitochondrial *mtL* diploblast data set. The gamma parameter here has been set to an empirically-derived value of 0.63. Distances were calculated with MEGA (Kumar 1993).

N		Demospongiae	Calcispongiae	Anthozoa	Scyphozoa	Hydrozoa
5	Demospongiae	0.121 +/- 0.037				
3	Calcispongiae	0.128 +/- 0.026	0.074 +/- 0.049			
14	Anthozoa	0.349 +/- 0.072	0.346 +/- 0.072	0.236 +/- 0.180		
2	Scyphozoa	0.540 +/- 0.039	0.533 +/- 0.047	0.576 +/- 0.039		
3	Hydrozoa	0.752 +/- 0.198	0.749 +/- 0.220	0.683 +/- 0.128	0.514 +/- 0.112	0.327 +/- 0.085
1	Ctenophora	0.795 +/- 0.081	0.738 +/- 0.076	0.774 +/- 0.082	0.625 +/- 0.040	0.373 +/- 0.038

Figure 27

The results of 500 neighbour-joining bootstrap replicates on the 437 nucleotide aligned diploblast mt *rnl* data set, performed with MEGA2 (Kumar 2000). Kimura 2-parameter gamma distances were utilized, with a gamma parameter of 0.63, determined empirically with PAUP* (Swofford 2000), and with pairwise deletion of gaps for all sequence pairs. The tree has been arbitrarily rooted with the Porifera. Numbers above the nodes represent the percentage of bootstrap replicates supporting that node.



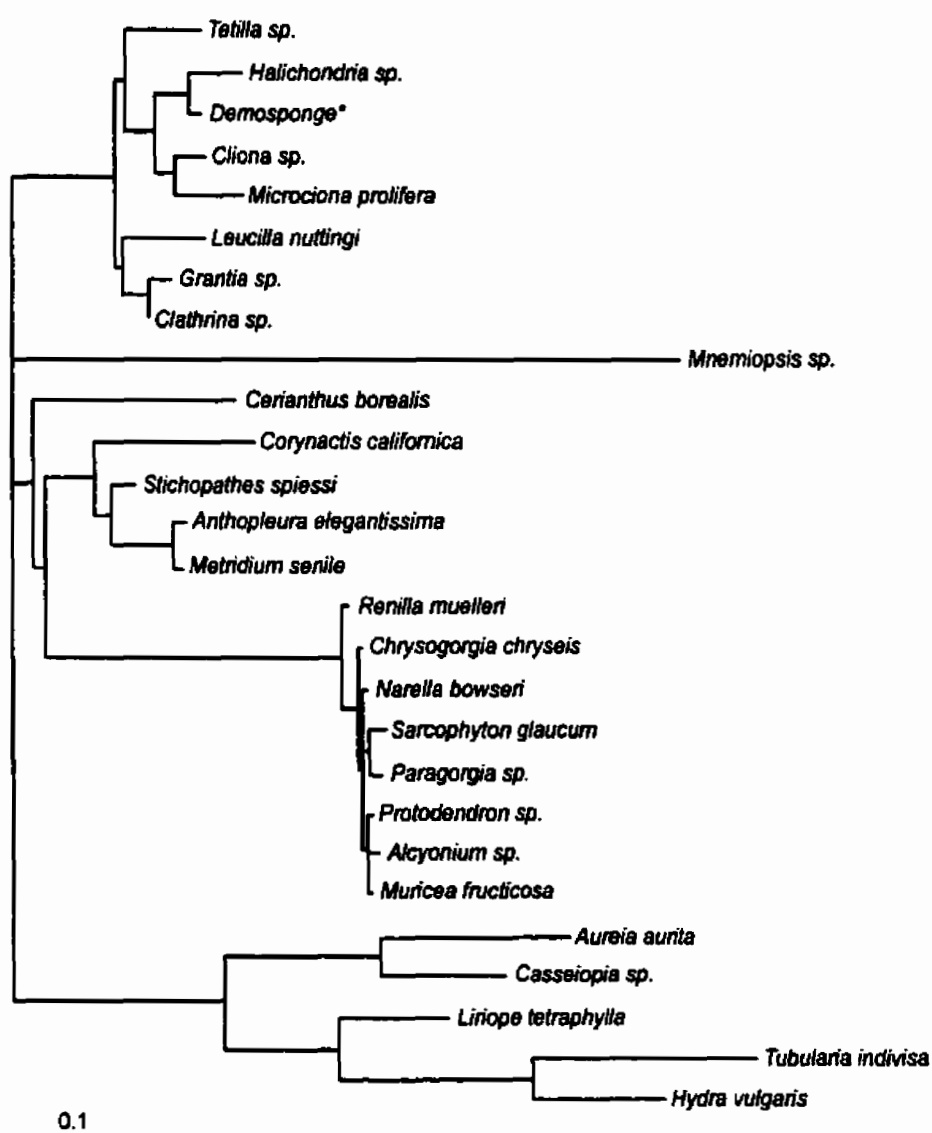
demosponges and calcisponges has if anything been reduced slightly. The Ctenophora maintain their apparent relationship with the hydrozoan cnidarians, which despite the high bootstrap value we must regard as erroneous based upon the results of multiple studies with nuclear gene phylogenies.

The relative levels of genetic distance between the various diploblast taxa are still apparent. Table 14 details the average pairwise genetic distances between taxa, all of which are congruent with the results of the larger data set shown in Tables 10 and 11. The sponges show the closest relationship with the Anthozoa, while the Ctenophora are the most divergent taxon with respect to any of the other groups. The Scyphozoa and Hydrozoa appear somewhat intermediate in this data, although the Hydrozoa are clearly the most divergent cnidarian class relative to the Anthozoa and Scyphozoa.

A constrained maximum likelihood tree is shown in Figure 28. This tree has been generated using the diploblast portion of the constraint tree previously shown in Figure 22. The relative levels of diploblast sequence divergence discussed previously are maintained in this tree. The Ctenophora, Hydrozoa, and Scyphozoa show by far the longest branches within this radiation. The strong clustering of the Octocorallia as well is maintained, and with the Hexacorallia is supportive of the phylogeny inferred by France *et al.* (1996) using the same data set. It is worth noting that the only examples of mitochondrial *mutS* homologs known to occur in Cnidarian mitochondrial genomes all occur within the octocorallian radiation (Beagley *et al.* 1995, Pont-Kingdon *et al.* 1998), which all cluster on a number of extremely short internal branches as seen in Figures 26 and 27.

Figure 28

Maximum likelihood tree derived with PAUP* (Swofford 2000) from the 437 nucleotide diploblast mt *rrnL* data set. An exhaustive search on topologies was performed, subject to the constrained backbone tree. The tree has been arbitrarily rooted with the Porifera.



In fact the primary features of the diploblast portion of the tree shown in Figure 22 have not changed significantly with the inclusion of a longer data set. Constraint to a biologically meaningful backbone tree once again produces a very short set of internal branch lengths both between phyla and in the case of the Cnidaria between classes as well. The polytomies shown in Figure 28 are in fact the maximum likelihood estimates of these internal branch lengths. That these short branch lengths are not literally the historical truth is obvious, but clearly there is a lack of informative variation for this particular taxonomic level.

Two-cluster relative rate tests (Takezaki *et al.* 1995) were performed on both the full diploblast data set and on a data set in which taxa with closely-spaced internal branches had been deleted in the manner of the two-cluster tests shown in Table 12. The results of these tests are shown in Table 15. Most notable in these results is the concordance between the results using both data sets and all distance correction methods. In every case a lack of significant rate difference is found between the Ctenophora, Scyphozoa, and Hydrozoa in all combinations. Since these are clearly the longest-branched taxa in the maximum likelihood phylogenies, this result merely confirms those results. The Anthozoa shows significant rate differences from those taxa in every test, again supporting the genetic distance differences observed in the phylogenies. Since the Porifera are used as the outgroup in all of these tests we cannot directly test the strength of support for a significant rate difference, but since their branch lengths are all shorter than those of the Anthozoans we must conclude that they are significantly shorter than those of the Ctenophora, Hydrozoa, and Scyphozoa as well.

Table 15

Results of the 2-cluster relative rate test (Takezaki *et al.* 1995) as calculated with the PHYLTEST package (Kumar 1995) on diploblast taxa compared to triploblast lineages, using the reduced data set described in the text. All tests utilized the combined demosponge and calcisponge mt *rml* sequence data as an outgroup for the Cnidaria and Ctenophora.

A. Results using the Kimura 2-parameter distance correction on both transitional and transversional data. B. Results using the Kimura 2-parameter distance correction with only transversional data. C. Results using the Kimura distance correction with a gamma parameter of 0.64. D. Results using the full aligned *rml* data set and the Kimura 2-parameter correction on transversional data only. All test statistics which are less than the critical value of 2.86 are shown in **bold**.

	Anthozoa	Scyphozoa	Hydrozoa
A			
Scyphozoa	6.0175		
Hydrozoa	6.4398	0.9966	
Ctenophora	6.2035	1.7539	1.4754
B			
Scyphozoa	5.1803		
Hydrozoa	5.0938	0.0582	
Ctenophora	4.5230	0.1462	0.1388
C			
Scyphozoa	4.7979		
Hydrozoa	4.9094	1.0917	
Ctenophora	4.5194	1.7286	1.4059
D			
Scyphozoa	5.7537		
Hydrozoa	5.1376	0.8840	
Ctenophora	5.1488	1.4075	1.1749

Relative rates of sequence evolution within the diploblast radiation calculated in the manner of Table 13 are shown in Table 16. The Porifera may have a slightly lower rate than the Anthozoan radiation, although equality is within the range of error in the estimation of their relative rates. Similar situations pertain to the Ctenophora/Hydrozoa comparison and the Scyphozoa/Hydrozoa comparison. On the other hand the Anthozoa appear to evolve from 2 to 3 times slower than the Ctenophora or either of the other two cnidarian classes. The sponges are even more extreme – the rate differences between the Porifera and the Ctenophora, Scyphozoa, and Hydrozoa ranges from some 2.8 to 3.6 times. All of these results are concordant with the relative rate test results presented in Table 15.

DISCUSSION

1. *mtL sequence evolution in the diploblast phyla has been slower overall than in the triploblast phyla*

Evidence of a slower mitochondrial sequence evolution rate in a given set of taxa is unambiguous. More problematic is the question of whether an apparently higher rate in another set of taxa represents a relatively extended and ongoing higher level of sequence evolution, or is simply evidence of an episodic increase in mutational accumulation at some point in time. In fact both of these processes may be at work in the basal metazoa relative to the triploblast radiation. Intertaxon comparisons of genetic distances in the diploblasts and the triploblasts reveal that the higher rate of sequence evolution in the triploblast is a process which crosses phylum boundaries, and which has resulted in

Table 16

Relative rates of mitochondrial *rrnL* sequence evolution between diploblast taxa. Average inter-taxon branch lengths relative to the last common node between each pair of taxa are presented as calculated with PAUP* (Swofford 2000) from the constrained maximum likelihood tree in Figure 28, and use the full data set.

	Porifera	Ctenophora	Anthozoa	Scyphozoa
Ctenophora	0.276 +/- 0.046			
Anthozoa	0.825 +/- 0.366	0.385 +/- 0.134		
Scyphozoa	0.363 +/- 0.063	0.765 +/- 0.068	0.505 +/- 0.174	
Hydrozoa	0.326 +/- 0.097	0.895 +/- 0.237	0.454 +/- 0.193	0.897 +/- 0.415

the triploblast phyla themselves being more diverged from the diploblast phyla than they are from one another.

As for understanding the parameters of sequence evolution which have led to this state, much of the issue hinges upon the relative ages of metazoan phyla themselves, as an absolute rate value and a comparison thereof cannot be made in the absence of a time span with which to correlate the genetic distances. For the base of the metazoan radiation however, this is a near impossible task at the moment. Times for the origin of the metazoa itself are under active debate (Wray *et al.* 1996, Ayala *et al.* 1998, Lynch 1999, Wang *et al.* 1999, Cutler 2000) and as a consequence the times of origin of the metazoan phyla themselves are also unclear, particularly for the deepest metazoan lineages. It is clear however, that some lineages must significantly predate the Cambrian Explosion (Conway Morris 1994, 1998). The fossil record is ambiguous and merely provides upper limits to the appearance of macroscopic metazoan organisms (Li *et al.* 1998). Even the exact ordering of the divergences of diploblast origins relative to the triploblasts is uncertain (Collins 1998, Dewel 2000).

What remains is that the Porifera are the primitive metazoan group (Conway Morris 1998, Knoll 1999, Schutze *et al.* 1999). Internal branches at the base of the metazoan tree are quite short, and suggest a geologically rapid origin to the diploblast and triploblast radiations. The triploblasts may be approximately as old as the Ctenophora and Cnidaria themselves (Dewel 2000), but must at least predate the Cambrian Explosion 547 million years ago (Conway Morris 1998, Li *et al.* 1998). Thus the lower overall degree of intra-diploblast sequence evolution relative to that of the triploblasts supports a higher overall rate of

mitochondrial sequence evolution at least in this portion of the mitochondrial *rnl* gene. Nonetheless it is clear that much of the rate differential between the two groups is concentrated in a single long branch. It is also clear that this long branch does not represent the source of all of the observed rate difference. The Porifera and the cnidarian class Anthozoa are both found to have significantly lower rates than the Scyphozoa, Hydrozoa, and Ctenophora. Since this rate difference is detected in the absence of any triploblast comparisons, we must conclude that the apparent slowness of the former two taxa is not attributable solely to the long triploblast branch.

The significance of this rate differential is not clear. Rate differentials are ultimately rooted in differences in either the rate of mutation or the rate of fixation, but these two factors can be broken up into an extensive set of underlying causes (Mindell and Thacker 1996) including rate of replication, base composition, repair efficiency, purifying selection, and effective population size. The data set at hand is not sufficient to address all possible causes for this observation.

The triploblasts as a whole appear on a very long branch relative to the diploblasts. If we posit a single origin for the triploblast phyla, which is universally supported by nuclear gene data (Wainwright *et al.* 1993, Kobayashi *et al.* 1996, Collins *et al.* 1998, Borchellini *et al.* 1998, Kim *et al.* 1999), then the very extended single branch leading to these phyla could be indicative of a period of rapid sequence evolution. An extended internal branch suggests either that this rate differential is ancient, or that there was also an episodic increase in the triploblast rate at some point. The possible influence of population-level factors in causing an apparent rate increase at some point early in the history of the

triploblasts should not be ignored. The lack of fossil evidence for truly ancestral triploblasts could represent a very limited geographic distribution and effective population size for these animals. This is particularly relevant in that the long internal branch leading to the triploblasts is supported by the nuclear data and is not merely a mitochondrial peculiarity.

*2. The poriferan classes Demospongiae and Calcispongiae, and the Cnidarian class Anthozoa show extremely slow sequence *rrnL* divergence rates. Even transitional saturation has not been fully achieved.*

The generally low rate of mitochondrial sequence evolution in the diploblast taxa is at its most extreme in the demosponges and calcisponges, cnidarian class Anthozoa. This correlation is interesting on a number of levels. Firstly, it has been reasonably well established that the Anthozoa do in fact represent the most basal cnidarian class (Bridge *et al.* 1992, Bridge *et al.* 1995, Kim *et al.* 1999). Since the sponges are also undoubtedly the most basally diverged metazoans, the most parsimonious interpretation is that metzoa as a whole had a very low primitive rate of mitochondrial sequence evolution, and that the higher triploblast rates are a derived condition. A transition between rate levels has occurred at some point. That this transition may not have occurred all at once, or is all due to an acceleration event as seen in the single long triploblast branch, may be supported by the relatively higher rates within the Ctenophora, Scyphozoa, and Hydrozoa.

The Octocorallia have been placed near the base of the Anthozoan clade (Hyman 1945, Chen 1995, Dewel 2000) and their mitochondrial

rDNA sequence evolution rates are strikingly low (Best and Thomas 1994, France *et al.* 1996, Romano and Palumbi 1996, Romano and Palumbi 1997, van Oppen *et al.* 1999), although characterization of rates in the Porifera at similar levels of taxonomic divergence could reveal similar patterns given their conservation at high taxonomic levels. The octocorals appear to exist on a long branch relative to the poriferan radiation, but this could represent a case of incomplete taxonomic sampling. The place of octocorals is relevant to an understanding of the causes of rate variation within the metazoa, in that they are the only taxon known specifically to possess an apparent mitochondrial DNA repair mechanism.

3. *The cnidarian classes Scyphozoa and Hydrozoa, and the Phylum Ctenophora show the highest rates within the diploblasts, but the Cnidaria as a whole still appear on shorter branches than does the triploblast radiation.*

The Scyphozoa and Hydrozoa appear to have the highest overall rates within the Cnidaria, even when the highly conserved octocorallian sequences are not included. Based upon the limited set of taxa presented here, there appears to be something of a gradient in rates running from Anthozoa to Scyphozoa to Hydrozoa as found in the more limited previous analysis with the diploblast and triploblast data set. That the Hydrozoa have higher rates than the Anthozoa has been supported by data previously published (Cunningham 1993, Pont-Kingdon *et al.* 2000) but not directly analyzed.

The phylogenetic placement of the Ctenophora is more problematic, but in any case appears to show a higher rate than the

demosponge, calcisponge, or anthozoan radiations. This result must be regarded as preliminary in the absence of multiple *rml* sequences. However, in the case of the data from the tentaculate Ctenophore *Pleuobranchia bacheii*, which was eliminated prior to this study, it is worth noting that both sequences in fact clustered on long branches near the *Beroe* sequence. Whether or not one of them was a mitochondrially-inserted nuclear sequence, both supported the long ctenophoran branch. As well, the exact placement of the Ctenophora within the diploblast radiation does not affect our interpretation of the results. Whether they are placed as sister taxa to the Cnidaria or to the Bilateria, they maintain their long relative branch lengths.

Chapter 6

Mitochondrial Sequence Evolution Rate Variation Across the Diploblast Radiation II: Protein-Coding Genes

Introduction

There are many advantages of utilizing protein-coding gene sequences over ribosomal RNA sequences for direct comparison of the relative mitochondrial sequence evolution rates. To begin with, realistic alignment of purely orthologous positions within the sequence is difficult in ribosomal RNA genes owing the complex interplay of sequence evolution with secondary and tertiary structure conservation (Gutell 1993). A purely algorithmic approach to the primary sequence data may leave doubt as to the veracity of the sequence alignments produced in the absence of concordant secondary structure data, particularly in less well-conserved regions. Doubt has recently been raised about the utility of even secondary structure data in these alignments (Hancock and Volger 2000).

Protein-coding genes provide a number of advantages for studying the parameters of mitochondrial sequence evolution. Alignment by codon is typically unambiguous in the more conserved mitochondrial genes, as opposed to the substantial levels of alignment ambiguity which are often present with ribosomal RNA sequences. This is particularly cogent in the case of this study as the degree of sequence divergence between the diploblasts and triploblasts is often substantial. Breakdown of the data by codon position allows for the study of regions which are under some more clearly understood selective constraints. In addition it

provides the ability to make comparisons of the rate of evolution at the nucleotide level with that at the amino acid level, and for consequent examination of selective constraints at multiple levels sequence evolution. As well, rates of mitochondrial sequence variation are more clearly quantified for the protein-coding genes than for the ribosomal RNA genes owing to the relative ease with which orthologous sites can be directly aligned.

Conserved *mtL* primers have been published which allow for reliable amplification of this gene from a wide range of taxa, whereas little published data exists for the protein-coding genes. Design of conserved primers is problematic as the triploblast sequences are sufficiently divergent from the diploblast ones as to make their contribution to primer design unclear. Since there are as of this writing only three extant complete cnidarian mtDNA sequences in GenBank and one sponge sequence which can be utilized for the design of protein-coding sequence primers, there is substantially less data available for judging the degree of variability in different portions of the mtDNA sequences.

I chose to amplify portions of the mitochondrial *cox2*, *cox3*, and *atp6* largely on the basis of sequence availability. Several sets of conserved primers were designed and tested in combinations with one another. In most cases different sets of primers worked in different combinations in each taxon.

Van Oppen *et al.* (1999) followed up on the work of Romano and Palumbi (1996,1997) by studying the evolution of the mitochondrial Cytochrome b gene in the scleractinian corals. As with the mitochondrial rRNA data, the rate of sequence evolution in this gene was found to be

strikingly low. In fact within the scleractinian genus *Acropora*, the cytochrome b gene was found to be evolving some 10 to 20 times slower than the vertebrate mitochondrial sequence evolution rate – one of the slowest metazoan mitochondrial rates found to date. A brief description of sequence variation in Anthozoan mitochondrial *cox1* sequences (Best and Thomas 1993) also supported a generalized lowering of the mitochondrial sequence evolution rate within this class relative to the triploblast radiation. These combined results suggest that the sequence conservation seen in *mL* throughout the cnidarian radiation may represent a common feature of cnidarian mitochondrial genes, rather than a specific feature of ribosomal RNAs in this phylum. It is worth noting however that another brief analysis (van Oppen *et al.* 1999) found a lower intraclass level of sequence identity in a very small sample of cnidarian *cob* and *nad6* sequences than for comparable interclass variation in the Arthropoda.

What is clear from the literature is that the anthozoan Cnidaria have provided some evidence of a slower mitochondrial rate of sequence evolution. Data gathered from both ribosomal RNA genes and protein-coding genes has shown similar results. That this rate difference exists, and that it has a specific taxonomic distribution has been shown in the previous chapter, but due to the nature of the rRNA gene alignments it was not possible to extract a great deal of information about the parameters of sequence evolution by gene and by types of sequence change. The protein-coding genes allow us to dissect this rate difference in considerably more detail.

Amplification, Alignment, Nucleotide Composition

PCR amplification of all genes was as described previously (see Chapters 3 and 4). All sequence alignments were performed again in order to not introduce any systematic errors which might have been introduced from previous work. Sequences of diploblast and triploblast mitochondrial genes were obtained from GenBank using the *Entrez* browser and are listed in Table 17. All genes utilized in this study are encoded on the same strand as the *Tetilla* sp. and other sponge and cnidarian sequences, so as not to introduce artifacts derived from differential nucleotide composition of the heavy and light strands. Genes encoded on an opposite strand from the rest of the mitochondrial genome can be easily detected by their aberrant nucleotide composition in a sequence alignment, or placement in a phylogenetic tree.

Alignment of all genes was performed upon the translated amino acid sequences using the CLUSTALW package (Thompson *et al.* 1994). All alignments were performed using the default weighting parameters for introduction and extension of gaps, but variation of these weight parameters resulted in no significant change in sequence alignments in regions of well-conserved sequence alignment such as in *cox2* and *cox3*. In some poorly-conserved regions of *atp6*, alteration in the gap opening and extension parameters altered the sequence alignment significantly, but these regions were eliminated from the final analysis (see below). Following CLUSTALW alignment, all amino acid sequence alignments were examined by eye and adjusted to

Table 17

Species, genes, and accession numbers utilized in this study. All sequences not amplified and sequenced in this laboratory were retrieved from GenBank using the web-based Entrez system for sequence retrieval. Sequences were downloaded and then edited by hand using the BioEdit package (Hall 1999) and then assembled into FASTA, PHYLIP, or MEGA formatted files using ESEE3 (Cabot and Beckenbach 1993) for further processing with other phylogenetic analysis software packages.

Species	Genes	Accession Number
Porifera		
<i>Tetilla</i> sp.	<i>cox2, cox3, atp6</i>	AF285262
<i>Halichondria</i> sp.	<i>cox2</i>	AF362016
	<i>cox3</i>	AF362011
	<i>atp6</i>	AF362013
<i>Cliona</i> sp.	<i>atp6</i>	AF362012
Unclassified demosponge	<i>cox2</i>	AF362017
	<i>cox3</i>	AF362009
	<i>atp6</i>	AF362014
<i>Clathrina</i> sp.	<i>cox2</i>	AF362018
	<i>cox3</i>	AF362015
	<i>atp6</i>	AF362010
Cnidaria		
<i>Metridium senile</i>	<i>cox2, cox3, atp6</i>	AF000023
<i>Sarcophyton glaucum</i>	<i>cox2, cox3, atp6</i>	AF064823
<i>Acropora tenuis</i>	<i>cox2, cox3, atp6</i>	AF152244
<i>Hydra vulgaris</i>	<i>cox2</i>	AF100773
Ctenophora		
<i>Beroe</i> sp.	<i>cox2</i>	AF362019
Protostomes		
<i>Platyneries dumerilii</i>	<i>cox2, cox3, atp6</i>	NC_000931
<i>Katharina tunicata</i>	<i>cox2, cox3, atp6</i>	KTU09810
<i>Artemia franciscana</i>	<i>cox2, cox3, atp6</i>	X69067
<i>Lumbricus terrestris</i>	<i>cox2, cox3, atp6</i>	U24570
<i>Drosophila yakuba</i>	<i>cox2, cox3, atp6</i>	X03240
Deuterostomes		
<i>Balanoglossus carnosus</i>	<i>cox2, cox3</i>	AF051097
<i>Strongylocentrotus purpuratus</i>	<i>cox2, cox3, atp6</i>	X12631
<i>Scyliorhinus canicula</i>	<i>cox2, cox3, atp6</i>	NC_001950
<i>Homo sapiens</i>	<i>cox2, cox3</i>	J01415
<i>Bos taurus</i>	<i>atp6</i>	NC_001567
Outgroups		
<i>Allomyces macrogynus</i>	<i>cox2, cox3</i>	AMU41288
<i>Schizosaccharomyces pombe</i>	<i>atp6</i>	X54421
<i>Prototheca wickerhamii</i>	<i>cox3, atp6</i>	U02970
<i>Chondrus crispus</i>	<i>cox2, cox3</i>	Z47547

eliminate any algorithm-induced errors. FASTA-format output files of the amino acid alignments were used to generate DNA sequence alignments using small BASIC programs written for that purpose.

For the *cox2* and *cox3* sequences the alignments thus produced were judged sufficiently accurate by themselves to be used directly in both phylogeny inference and in measurement of pairwise comparisons of sequence properties. Some regions of uncertain or missing DNA sequence data were left in the data set for *cox2*, but this was not felt to be a drawback inasmuch as pairwise comparisons were being made and the inclusion of these regions would simply bolster to pairwise information content of those species for which the sequence was available. All relative rate tests were performed using complete deletion of gaps.

Atp6 being the least conserved of the three genes at both the amino acid and nucleotide sequence level showed the most regions in which proper sequence alignment was uncertain. As a result, these regions were eliminated from the data set. Thus the data set consists of bases 4 to 12, 32 to 46, 93 to 113, 175 to 201, 352 to 366, and 523 to 570 of the *Tetilla* sp. *atp6*.

The nucleotide composition of *cox2*, *cox3*, and of *atp6* are shown in Table 18. Most protein-coding gene sequences demonstrate the AT preference commonly found in metazoan mitochondrial genomes (Wolstenholme 1992). The diploblast genes appear to show a higher frequency of G and C overall, and at codon positions 1, 2, and 3 in general, but visual inspection of the tables of nucleotide frequencies shown here reveals no obvious clustering of nucleotide biases between the two groups.

Table 18

Nucleotide composition of mitochondrial protein-coding genes in diploblasts and triploblasts. Compositional data was calculated with MEGA (Kumar 1993).

Nucleotide composition

All values in per cent (%) except Totals

cox2

Diploblasts

	A	T	C	G	Total
Spinosa	29.3	36.8	12.6	21.2	744
Halichondria	27.9	32.8	15.1	24.1	555
D1	30.3	33.5	15.5	20.6	669
Beroe	29.2	35.5	16.2	19.1	524
Hydra	35.1	43.3	11.2	10.4	756
Metridium	29.3	34.9	16.2	19.5	747
Sarcophyton	28.6	34.9	16.9	19.6	762
Acropora	26.7	36.6	13.3	23.4	744
TOTAL	29.6	36.2	14.6	19.6	5501

Triploblasts

	A	T	C	G	Total
Purpuratus	26.4	30.6	24.8	18.3	690
Platynereis	31.8	29.6	21.4	17.2	692
Balanoglossus	25.6	25.4	31.3	17.7	753
Homo	28.7	25.1	31.3	14.9	684
Dogfish	31.3	30.3	23.5	14.9	693
Katharina	25.7	39.4	12.3	22.6	690
Artemia	26.8	35.7	19.8	17.8	687
Drosophila	33.6	40.3	12.5	13.5	687
Lumbricus	31.7	29.3	22.3	16.7	687
TOTAL	29.0	31.7	22.2	17.1	6263

cox3

Diploblasts

	A	T	C	G	Total
Metridium	23.3	37.3	18.4	21.0	789
Sarcophyton	26.5	35.6	17.2	20.7	786
Acropora	20.4	42.7	13.1	23.8	789
Tetilla	26.1	38.0	13.8	22.1	687
Demo2	27.3	37.8	13.7	21.2	693
UnkCalcD1	27.3	33.6	17.5	21.6	699
Halichondria	25.7	31.6	18.0	24.7	693
TOTAL	25.1	36.8	16.0	22.2	5136

Triploblasts

	A	T	C	G	Total
bovine	26.5	29.4	29.0	15.0	785
urchin	26.7	31.3	22.5	19.5	783
Balanoglossus	23.0	27.6	31.5	17.9	782
yakuba	30.8	40.4	14.2	14.6	789
Artemia	24.3	38.0	17.1	20.6	773
Katharina	21.7	41.3	14.5	22.6	780
Platynereis	24.6	33.7	22.0	19.8	778
lumbricus	22.5	31.8	26.3	19.4	779
Dogfish	27.1	30.9	25.7	16.3	786
TOTAL	25.2	33.8	22.5	18.4	7035

atp6

Diploblasts

	A	T	C	G	Total
Acropora	21.3	41.3	14.4	22.9	699
Metridium	24.1	39.7	17.1	19.1	690
Sarcophyto?	31.5	36.2	15.7	16.7	708
Tetilla	27.6	40.8	11.7	19.9	735
Clio?a	26.0	41.5	12.0	20.6	651
Halicho?dria	25.6	36.5	15.2	22.7	735
Demo2	26.9	43.7	11.4	18.0	650
CalcU	28.9	37.0	15.0	19.2	714
TOTAL	26.5	39.5	14.1	19.9	5582

Triploblasts

	A	T	C	G	Total
Artemia	26.1	39.2	19.5	15.2	660
Kathari?a	22.9	47.3	9.8	19.9	693
Lumbricus	28.3	31.9	27.4	12.4	696
Dyakuba	31.9	44.0	14.1	10.1	675
Platy?ereis	28.7	34.3	21.4	15.5	696
Dogfish	29.7	34.9	24.1	11.3	684
Homo	30.2	25.6	33.8	10.4	681
Spurp	26.7	33.6	23.6	16.1	690
TOTAL	36.4	35.2	19.5	8.9	1825

Nucleotide Bias

A principal component analysis of nucleotide content in the diploblast and triploblast mitochondrial genes (not shown) demonstrated a strong correlation between %A and %G, and between %T and %C in each strand. This trend is typical of metazoan mitochondrial DNAs (Jermin *et al.* 1995, Reyes *et al.* 1998, Gissi *et al.* 2000) and merely reflects strand-specific compositional preferences.

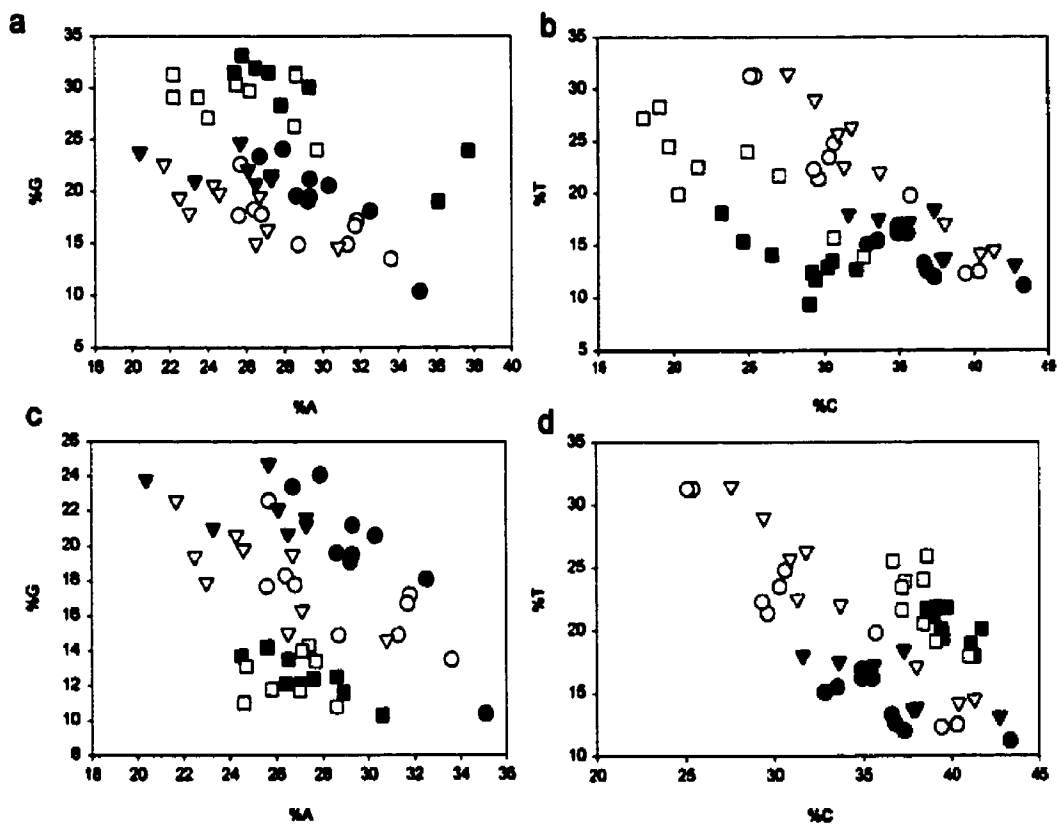
The degree of nucleotide bias in the diploblast sequences falls within the range of variation observed for the triploblast sequences. Figure 29 shows %G plotted against %A and %T plotted against %C for codon positions 1 and 2 of all three genes. While the diploblast sequences cluster together more tightly than do the triploblast sequences, they cluster near or within the triploblast distribution. It seems apparent that the degree of nucleotide bias in the metazoa is not sufficient to strongly separate the diploblast and triploblast radiations, although the diploblasts may demonstrate a somewhat elevated level G and T relative to the triploblasts on average.

This GT bias is demonstrated as well in Table 19. This table shows the cumulative codon usage statistics for each gene and relative synonymous codon usage values (RSCU) (Li 1997), compiled for all the protein-coding genes in that group. In cases in which a given codon is involved in a genetic code change, the RSCU values have not been calculated or included for that codon. In almost every case the RSCU values of codons with terminal G's or C's is higher in diploblasts than in triploblasts. The only consistent exception to this trend is for CUN leucine codons, which have lower RSCU values for the G and C-terminal codons for all three genes.

Figure 29

Correlations between content of purines and pyrimidines in one strand of mitochondrial protein-coding genes in the metazoa. All data were generated from the aligned nucleotide sequences with MEGA (Kumar 1993). In all charts filled symbols represent diploblast sequences, and open circles represent triploblast sequences. Circles represent *cox2* sequences, triangles represent the *cox3* sequences, and squares represent the *atp6* sequences. In all cases the diploblast values fall within the range of variation found in the triploblast data.

- A. Percent A versus percent G for nucleotide position 1.
- B. Percent C versus percent T for nucleotide position 1.
- C. Percent A versus percent G for nucleotide position 2.
- D. Percent C versus percent T for nucleotide position 2.



Partial codon adaptation indices (CAI) were calculated for the three sets of protein-coding genes as well (Sharp and Li 1987). Calculations did not include data for any amino acid which is involved in a genetic code change in any of the taxa analyzed. RSCU_{max} tables for *E. coli* were utilized in the calculation. These values are presented in Table 20, and show that despite the nucleotide bias present in the RSCU values, the total CAI values for diploblasts and triploblasts do not exhibit any striking variation from one another, and certainly not in a taxon-specific manner. The most clearly divergent values were for *Hydra vulgaris* within the diploblasts, and for *Drosophila yakuba* within the triploblasts, but all values appear to be within a normal range of variation, and there is no apparent trend either between genes or between taxa. Use of alternate RSCU_{max} values calculated from mitochondrial sequence data did not show significant differences from this pattern.

In total these results indicate a slight nucleotide bias in the diploblast phyla towards G and T in their protein-coding gene sequences. Although the bias itself is consistent throughout the three genes analyzed, we do not regard it as a sufficiently large effect to by itself have a detrimental impact upon any phylogenetic work performed using these genes. In particular the distribution of the primary bias shows little or no taxon-specific variation. There is apparently significant variation in nucleotide bias from species to species, but is linked largely to strand-specific purine and pyrimidine content. It must be emphasized that this is primarily a difference in strand bias – the overall AT and GC contents are very similar between diploblasts and triploblasts. The total (double-stranded) data falls within reasonable ranges for the metazoa as a whole

Table 19

Relative synonymous codon usage table for mitochondrial mitochondrial protein-coding genes. RSCU tables for the combined diploblast and triploblast data sets were calculated with the General Codon Usage Analysis V1.0 package (McInerny 1998). In order to allow for a meaningful comparison of RSCU values, values were deleted from the tables for all amino acids in which one or more codons is labile in any of the taxa examined. Effectively this excludes serine, asparagine, isoleucine, methionine, lysine, and arginine. Labile codons are indicated in the tables with a **Lab** label.

.

Cumulative Codon Usage

AA Codon	cox2			cox3			atp6			cox2			cox3			atp6		
	DIP			DIP			DIP			DIP			DIP			DIP		
	N	RSCU	TRIP	N	RSCU	TRIP	N	RSCU	TRIP	N	RSCU	TRIP	N	RSCU	TRIP	N	RSCU	TRIP
Ser UCU	51	(---)	47	(---)	27	(---)	33	(---)	33	(---)	33	(---)	33	(---)	33	(---)	39	(---)
UCC	18	(---)	25	(---)	2	(---)	21	(---)	8	(---)	8	(---)	21	(---)	8	(---)	21	(---)
UCA	16	(---)	35	(---)	18	(---)	43	(---)	30	(---)	30	(---)	43	(---)	30	(---)	29	(---)
UCG	7	(---)	5	(---)	18	(---)	5	(---)	15	(---)	15	(---)	5	(---)	15	(---)	6	(---)
Cys UGU	14	(1.40)	11	(0.96)	20	(1.74)	15	(1.03)	13	(1.06)	8	(1.33)	15	(1.03)	13	(1.06)	8	(1.33)
UGC	6	(0.60)	12	(1.04)	3	(0.26)	14	(0.97)	1	(0.14)	4	(0.67)	14	(0.97)	1	(0.14)	4	(0.67)
Tyr UGA	18	(0.84)	44	(9.78)	23	(0.74)	93	(1.58)	8	(0.80)	25	(4.55)	93	(1.58)	8	(0.80)	25	(4.55)
Trp UGG	25	(1.16)	9	(2.00)	39	(1.26)	25	(0.42)	12	(1.20)	11	(2.00)	25	(0.42)	12	(1.20)	11	(2.00)
Pro CCU	38	(1.71)	35	(1.30)	32	(1.78)	30	(1.28)	30	(1.82)	41	(1.62)	30	(1.28)	30	(1.82)	41	(1.62)
CCC	8	(0.36)	33	(1.22)	8	(0.44)	21	(0.89)	8	(0.48)	24	(0.95)	21	(0.89)	8	(0.48)	24	(0.95)
CCA	22	(0.99)	34	(1.26)	17	(0.94)	39	(1.66)	20	(1.21)	31	(1.23)	39	(1.66)	20	(1.21)	31	(1.23)
CGG	21	(0.94)	6	(0.22)	15	(0.83)	4	(0.17)	8	(0.48)	5	(0.20)	4	(0.17)	8	(0.48)	5	(0.20)
Arg CGU	5	(---)	17	(---)	3	(---)	12	(---)	5	(---)	5	(---)	12	(---)	5	(---)	5	(---)
CGC	1	(---)	10	(---)	2	(---)	7	(---)	1	(---)	3	(---)	7	(---)	1	(---)	3	(---)
CGA	11	(---)	23	(---)	5	(---)	27	(---)	11	(---)	19	(---)	27	(---)	11	(---)	19	(---)
CGG	4	(---)	6	(---)	0	(---)	3	(---)	5	(---)	3	(---)	3	(---)	5	(---)	3	(---)
Thr ACU	37	(1.61)	38	(1.27)	58	(1.83)	61	(1.38)	36	(1.52)	39	(1.22)	61	(1.38)	36	(1.52)	39	(1.22)
ACC	7	(0.30)	31	(1.03)	22	(0.38)	43	(0.97)	12	(0.51)	37	(1.16)	43	(0.97)	12	(0.51)	37	(1.16)
ACA	38	(1.65)	45	(1.50)	44	(1.39)	68	(1.54)	35	(1.47)	49	(1.53)	68	(1.54)	35	(1.47)	49	(1.53)
ACG	10	(0.43)	6	(0.20)	13	(0.41)	5	(0.11)	12	(0.51)	3	(0.09)	5	(0.11)	12	(0.51)	3	(0.09)
Ser AGU	17	(---)	9	(---)	44	(---)	14	(---)	33	(---)	15	(---)	14	(---)	33	(---)	15	(---)
AGC	4	(---)	5	(---)	13	(---)	19	(---)	6	(---)	12	(---)	19	(---)	6	(---)	12	(---)
Leu AGA	26	(---)	19	(---)	22	(---)	26	(---)	13	(---)	40	(---)	26	(---)	13	(---)	40	(---)
Leu AGG	7	(---)	4	(---)	9	(---)	5	(---)	9	(---)	9	(---)	5	(---)	9	(---)	9	(---)
Ala GCU	44	(1.78)	49	(1.52)	40	(1.47)	70	(1.58)	59	(1.72)	44	(1.42)	70	(1.58)	59	(1.72)	44	(1.42)
GCC	16	(0.65)	27	(0.84)	22	(0.81)	56	(1.27)	32	(0.93)	39	(1.26)	56	(1.27)	32	(0.93)	39	(1.26)
GCA	16	(0.65)	44	(1.36)	26	(0.95)	37	(0.84)	21	(0.61)	32	(1.03)	37	(0.84)	21	(0.61)	32	(1.03)
GCG	23	(0.93)	9	(0.28)	21	(0.77)	14	(0.32)	25	(0.73)	9	(0.29)	14	(0.32)	25	(0.73)	9	(0.29)
Gly GGU	33	(1.50)	23	(0.96)	37	(1.06)	26	(0.55)	24	(0.83)	21	(0.99)	26	(0.55)	24	(0.83)	21	(0.99)
GGC	9	(0.41)	20	(0.83)	13	(0.37)	52	(1.11)	11	(0.38)	15	(0.71)	52	(1.11)	11	(0.38)	15	(0.71)
GGA	24	(1.09)	34	(1.42)	16	(1.04)	72	(1.53)	21	(0.72)	30	(1.41)	72	(1.53)	21	(0.72)	30	(1.41)
GGG	22	(1.00)	19	(0.79)	53	(1.53)	36	(0.81)	60	(2.07)	19	(0.89)	36	(0.81)	60	(2.07)	19	(0.89)

Table 20

Incomplete codon adaptation indices for mitochondrial protein-coding genes. All values were calculated by hand with from RSCU tables generated with the GCUA package (McInnery 1998) and RSCU values for *E. coli* (Gouy and Gautier 1986) highly expressed genes. The codon adaptation indexes are regarded as incomplete in that they did not include data from any codons of amino acid for which there is an alteration of the codon specificity in any taxon. This eliminates serine, asparagine, isoleucine, methionine, lysine, and arginine.

CAI*	<i>atp6</i>	<i>cox2</i>	<i>cox3</i>
Porifera			
<i>Tetilla sp.</i>	0.185	0.231	0.216
<i>Halichondria sp.</i>	0.328	0.253	0.307
<i>Cliona sp.</i>	0.216	-	
<i>Clathrina sp.</i>	0.328	0.236	0.284
Demosponge**	0.158	-	0.173
Cnidaria			
<i>Metridium</i>	0.228	0.241	0.360
<i>Sarcophyton</i>	0.211	0.259	0.338
<i>Acropora</i>	0.249	0.265	0.208
<i>Hydra</i>	-	0.095	-
Ctenophora			
<i>Beroe</i>	-	0.229	-
Protostomes			
<i>Platyneries</i>	0.370	0.290	0.295
<i>Katharina</i>	0.141	0.195	0.255
<i>Artemia</i>	0.234	0.310	0.381
<i>Lumbricus</i>	0.275	0.243	0.327
<i>Drosophila</i>	0.118	0.135	0.125
Deuterostomes			
<i>Balanoglossus</i>		0.329	0.283
<i>Urchin</i>	0.241	0.322	0.345
<i>Scyliorhynchus</i>	0.201	0.266	0.254
<i>Homo</i>	0.169	0.315	-
<i>Bovine</i>	-	-	0.227

*Incomplete CAI values calculated against *E. coli* RSCU_{max} values. All codons with labile translations were eliminated from consideration.

**Unclassified tetractinomorph demosponge.

(Wolstenholme 1992) and shows the typical AT-bias of mitochondrial genomes.

Construction of Phylogenies with Nucleotide and Amino Acid Data

Phylogenetic trees for *cox2* are shown in Figure 30. Figure 30A shows the results of 500 bootstrap neighbour-joining replicates on the *cox2* nucleotide positions 1 and 2 data. All distances estimates were made with the Kimura 2-parameter correction, using an empirically-derived gamma parameter of 0.55 calculated by PAUP* (Swofford 1999). The most notable features of this tree are (1) the clear misrooting of the outgroups between the diploblast and triploblast radiations, (2) the inversion of the relative sponge and cnidarian radiations (an “inverted” sponge tree), (3) the poor levels of bootstrap support for deep internal branchings within both the diploblast and triploblast radiations, and (4) the misrooting of both the *Beroë* sequence with the Cnidaria and the *Hydra* sequence closer to the triploblast lineage.

Figure 30B shows a constrained maximum likelihood tree generated with PAUP* using an exhaustive search algorithm and the constraint backbone tree shown in Figure 22. Forcing the data into a tree consistent with the structure of the metazoan radiation as it is currently understood (Cavalier-Smith *et al.* 1996, Conway-Morris 1998) reveals an interesting trend in the data: while the triploblast radiation as a whole remains on a fairly long internal branch, and the Porifera themselves are well separated from the rest of the metazoan radiation, there is a near-polytomy between the Ctenophora, Cnidaria, and the triploblasts. In fact constraint of the maximum likelihood trees derived from nuclear gene data (Kobayashi *et al.* 1996, Van de Peer *et al.* 1995, Borchellini *et al.* 1998) to the same tree results in a similar decrease in

these internal branch lengths (not shown). The only exception to this trend has been found in nuclear 18S rRNA data which has been carefully selected for low and equal divergence rates (Kim *et al.* 1999). The precise positioning of the Ctenophora within the constraint tree does not affect this result – a similar tree is found if the Ctenophora are placed as a sister taxon to the triploblasts as well. While the branch lengths in the constrained tree are distinctly shorter for the Porifera, the position of the Cnidaria is more equivocal. The extreme divergence of the *Hydra vulgaris* sequence is readily apparent.

Figure 30C shows the results of 500 neighbour-joining bootstrap replicates nucleotide positions 1 and 2 of the *cox3* sequence data using a gamma value of 0.45. In the case of this gene and ATPase Subunit 6 we have no representative sequences from the Hydrozoa, Schyphozoa, or Ctenophora. Nonetheless the tree demonstrates many of the same issues evident in the *cox2* data. In particular the misplacement of the nonmetazoan outgroup between the diploblast and triploblast lineage is identical, as is the inversion of the relative branching order of the Anthozoa and Porifera. The poor level of bootstrap support found throughout the tree leads to significant differences from the *cox2* tree with regard to the internal branches of the triploblasts, anthozoans, and cnidarians.

Figure 30D shows a maximum likelihood tree constructed from the *cox3* data using the same backbone constraint tree as in Figure 30B. Constraint of the inferred tree to a biologically meaningful structure leads again to a dramatic decrease in the internal branch lengths separating the anthozoan and triploblast lineages. In this case as well there is very short internal branch separating the divergence between the

sponge and anthozoan lineages. The triploblast radiation appears again to exist on a very long branch relative to the diploblast lineages.

Figure 30E shows a bootstrap neighbour-joining tree constructed from the ATPase Subunit 6 nucleotide 1 and 2 data using a gamma parameter of 0.92. The pairwise genetic distances for all taxa are greater for ATPase Subunit 6 than for the other two protein-coding genes, and interestingly this greater divergence appears to result in the highest bootstrap support levels for almost all the internal nodes of the inferred tree. Nonetheless the reversal of the sponge/anthozoan branching order when rooted against the nonmetazoan outgroup is still apparent, as is the poor internal resolution of the anthozoan lineage with respect to the Porifera.

Constraint of the data to the backbone tree in the maximum likelihood calculations (Figure 30F) shows the same essential elements as are seen in Figures 30B and 30D. The near-zero length of the internal branches separating the Porifera, Anthozoa, and the triploblasts is evident, as is the very long branch separating the triploblast lineage from the diploblasts. As a whole the *atp6* data set appears to provide support for a tree much more congruent with the currently understood structure of the metazoan radiation than does either of the two more highly conserved genes. In particular the structure of the deuterostome radiation is identical between the constrained and unconstrained trees – this is the only case in which this is true for all three genes.

The phylogenetic tree shown in Figure 31 is derived from the same data as for Figure 30E, but with the nonmetazoan outgroup sequences eliminated from consideration. Figure 31A shows the results of 500 neighbour-joining bootstrap replicates of the *atp6* data, while Figure 31B

shows the results of an unconstrained maximum likelihood solution on the same data set. While these are unrooted trees, the root has been placed for convenience between the Porifera and Cnidaria. It is interesting that in the absence of a nonmetazoan outgroup the overall structure of the metazoan radiation is reconstructed with some faithfulness. The same can be said in broad terms for the Porifera and Anthozoa for the more highly conserved genes, albeit with an overall lower level of bootstrap support and with significant misrooting of the Ctenophora and the Hydrozoa in the case of *cox2*. The cnidarian radiation is the only strongly problematic point in the *atp6* tree. There is clearly a lack of resolution of the relationship between the anthozoan Cnidaria and the triploblasts.

The principal and outstanding error in all of the trees generated from the protein-coding gene data is the misrooting of the nonmetazoan outgroup taxa. In every case a tree freely generated from this data places the root between the diploblasts and triploblasts. This effect has also been noted in numerous nuclear gene phylogenies (Kobayashi *et al.* 1993, Kobayashi *et al.* 1996, Van de Peer *et al.* 1997, Borchellini *et al.* 1998). The effect is not clearly a case of simple long branch attraction, as methods used to detect long-branch attraction effects such as Relative Apparent Synapomorphy Analysis (Lyons-Weiler and Hoelzer 1997) do not detect any evidence of such effect (not shown). It is more likely a due to a combination of very long external with quite short internal branches (Huelsenbeck and Hillis 1993). If we compare Figure 30E and Figure 31A we can see that the outgroup misrooting has a particular consequence: it results in a decrease in the relative distance between the triploblast and diploblast lineages. The effect is not as pronounced for

the *atp6* sequences, but comparison of the neighbour-joining trees with the constrained maximum likelihood trees in Figures 30B and 30D will reveal the same trend. We will discuss this below in context of the relative rate tests. Constrained maximum likelihood trees for the translated amino acid sequences of the three protein-coding genes are shown in Figure 32. Since the constraint option necessarily imposes the same overall backbone of relationships between the taxa as in the nucleotide sequences, the most important point to be considered is inferred relative internal and terminal branch lengths. All three genes share the long internal branch leading to the triploblast radiation, and as well all three genes show very short internal branches between the Porifera and Cnidaria. The internal branch is longer for *cox2*, likely as a consequence of the extremely long branch leading to the *Hydra vulgaris* *cox2* sequence. The nucleotide composition of *Hydra attenuata* *cox2* is aberrant relative to the rest of the metazoa (Pont-Kingdon 2000) and it is not clear whether this gene is encoded on the same strand as in the rest of the metazoans examined here. If it is not then we might expect to see aberrant branch lengths as found here from both the nucleotide and amino acid sequence data.

Most but not all of the variation between the two groups appears to be concentrated in the single long branch leading to the triploblast radiation. The *cox3* data appears to provide the least distinction between the diploblasts and triploblast radiations, in that the long internal triploblast branch is short compared to that of the other two protein-coding genes, and the deuterostome radiation exists on a particularly short set of internal branches

Figure 30

Phylogenetic trees derived from mitochondrial protein-coding DNA sequences, nucleotide positions 1 and 2.

A,C,E: Results of 500 neighbour-joining bootstrap replicates of the *cox2*, *cox3*, and *atp6* data performed with MEGA2 (Kumar *et al.* 2000). Distances were estimated with the Kimura 2-parameter gamma correction, with a gamma value of 0.45, 0.55, and 0.85 derived empirically from the same data set with PAUP* (Swofford 2000).

B, D, F: Maximum likelihood phylogenies constrained to a biologically meaningful interphylum backbone tree. The phylogeny was constructed with PAUP*, using exhaustive tree searches.

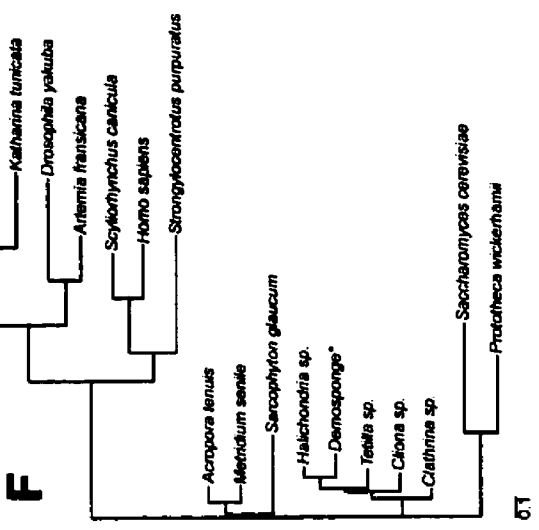
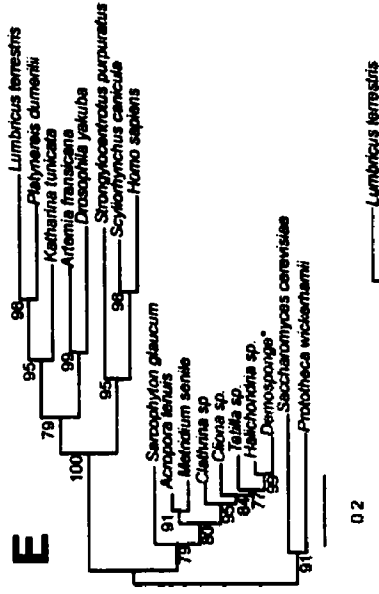
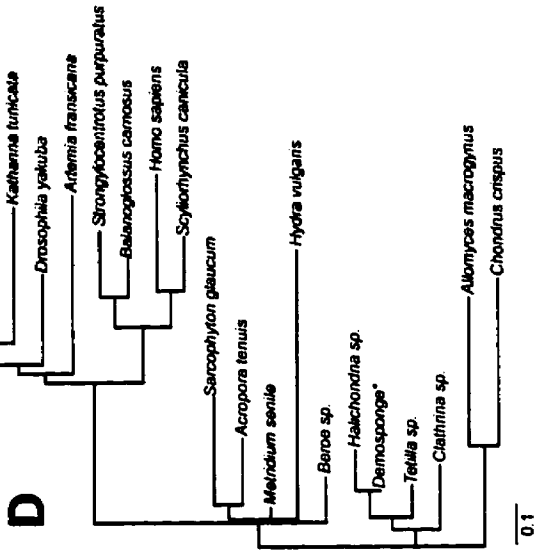
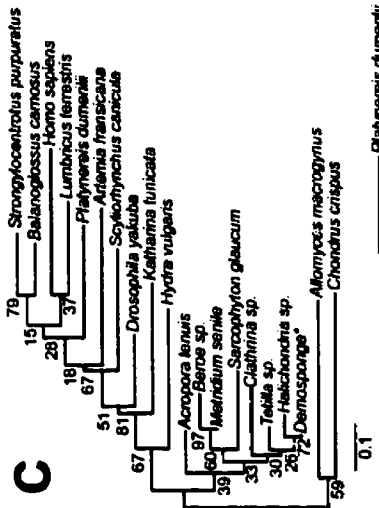
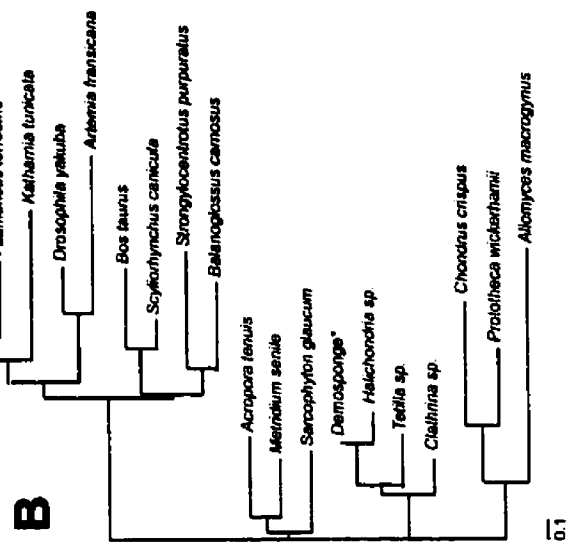
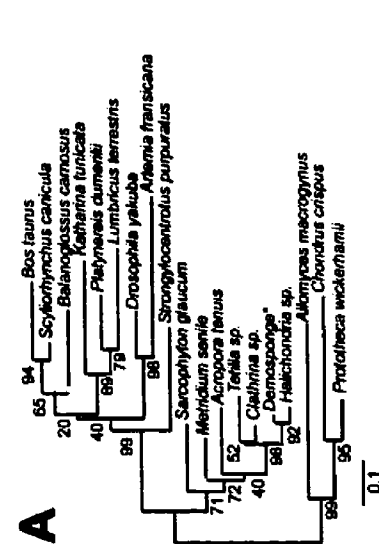


Figure 31

Neighbour-joining tree of *atp6* nucleotide sequence using nucleotide positions 1 and 2. Distances were derived by the Kimura 2-parameter gamma correction using an empirically-derived gamma value of 0.85. A. Neighbour-joining tree, with proportions of 500 bootstrap replicates are shown for nodes in the tree. This tree has been rooted with the demosponges as a whole as an outgroup, and no nonmetazoan taxa are included. Note the similarity of this tree to the constrained maximum likelihood tree shown in Figure 30F. B. Unconstrained maximum likelihood tree derived with PAUP* (Swofford 2000).

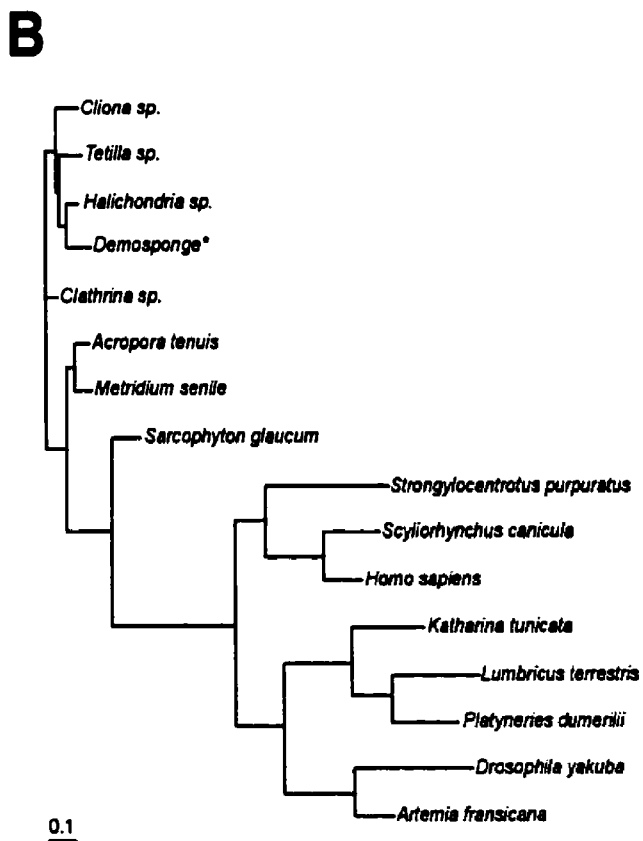
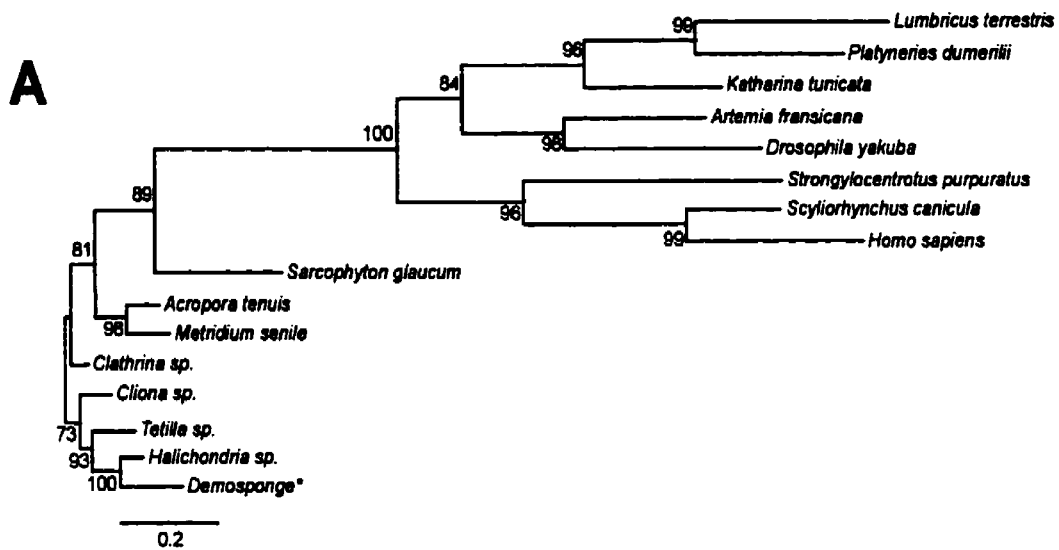
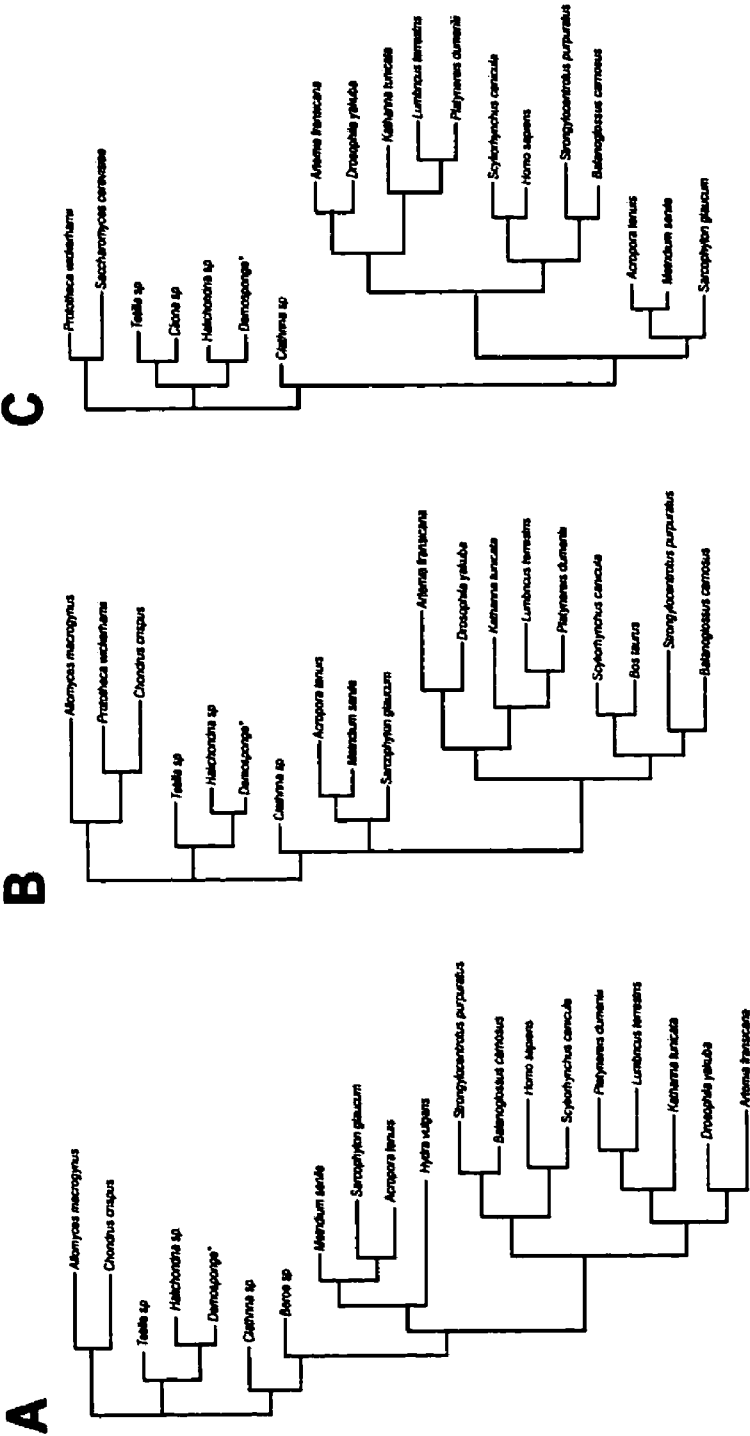


Figure 32

Constrained maximum likelihood tree of translated amino acid sequence data translated from mitochondrial protein-coding gene DNA sequences. Phylogeny was inferred by exhaustive search using the ProtML program from the MOLPHY package (Adachi and Hasegawa 1996), using the same constrained backbone tree as in the nucleotide analyses. The mitochondrial transition probability matrix was used for the analysis. The single highest-scoring maximum likelihood tree was retained as the solution. A. *cox2* B. *cox3* C. *atp6*



Relative Rates Of Sequence Evolution

Relative rate tests on both the nucleotide sequence data and the amino acid sequence data were performed using the two-cluster test of Takezaki *et al.* (1995). All analyses were performed using the Phyltest package (Kumar 1995). Table 21 columns 1 and 2 show the results of the tests on all three genes for both nucleotide positions 1 and 2, and amino acid distances respectively. The two-cluster provides a measure of the divergence of the means of the distances of two groups away from one another relative to outgroup sequences. This is an appropriate test for the question of inter-taxon differences in rate of sequence evolution within the metazoa. In every case for the application of this test we are examining the relative rates of monophyletic groups which diverged at or before the Cambrian explosion. The two-cluster test was originally designed for use in conjunction with the branch length test for the construction of linearized (clocklike) trees, but that test is inappropriate for this data set in that it will merely provide a measure of the divergence of a given sequence away from the mean of all sequences relative to the outgroup. This measure is not useful for testing the inter-group variation.

In all tests involving the Porifera, a fungal sequence was used as the outgroup for the test. Use of the very long-branched nonmetazoan outgroups has some drawbacks however: the sensitivity of the test itself is reduced proportionately to the distance to the outgroup itself (Robinson *et al.* 1998). The tests involving Porifera with a nonmetazoan outgroup must therefore be regarded as conservative compared to the tests which utilize a single poriferan lineage, the demosponges, as a metazoan outgroup. All other relative rate tests utilized the Porifera as a

whole as an outgroup for the test, and this is a much more appropriate outgroup.

For the nucleotide sequence data the two-cluster test produces concordant results for all taxon comparisons with the exception of the Deuterostome/Protostome tests. Universal support is found for significant rate differences between the diploblast and triploblast lineages, while there is no support for significant differences between the Porifera and Anthozoa themselves. The deuterostome and protostome lineages show a lack of support for significant rate differences in *cox3* and *atp6*, while *cox2* shows a significant rate difference (likely due to the longer sponge-cnidarian branch length inferred when the hydrozoan and ctenophoran lineages are included in the analysis).

Of course there are significant rate differences between many taxa both within and between the deuterostome and protostome radiations. The Arthropods, and in particular the insects, have very high rates of mitochondrial sequence evolution relative to the metazoa as a whole (Liu and Beckenbach 1992) and some deuterostome lineages, notably the sharks, have been found to have very low rates (Martin *et al.* 1992). The test as applied here is not sensitive to this order of rate difference owing to (1) the inclusion of small numbers of representative taxa from the different triploblast radiations, and (2) the scale of the tests themselves. All of the tests presented here examine rate differences between representative members of monophyletic groups which diverged at least 550 million years ago. Inclusion of multiple sharks and dipterans, for instance, would likely show strong support for rate differences between these two groups. This difference would not strongly alter the results of the relative rate tests for the poriferan or cnidarian radiations relative to

the triploblasts themselves, as this result is a comparison across a different scale. In particular the long branch leading to the triploblasts would not change in any case, and the results would be not significantly different from those found with the inclusion of a smaller number of triploblast taxa.

Table 21 also shows the results of the two-cluster test applied to the translated amino acid sequence data. The results are in agreement with those of the nucleotide sequence data with two exceptions. Firstly, the *cox2* amino acid data fails to detect significant rate difference between the deuterostome and protostome radiations. As discussed above, these particular results must only be regarded as comparisons of the average rates of two very divergent groups, and must be read in context with the results of the Tajima tests discussed below. Secondly, in the case of the amino acid data there is no evidence for a significant rate difference between the poriferan and deuterostome radiations in the case of *cox3*. This is the first indication in any data set of a lack of significant rate difference between a diploblast group and a triploblast one.

In order to quantify the level of support within the clusters previously tested for the rate differences detected, and as a framework within which to examine the results of tests by monophyletic group, relative rate tests were performed in a pairwise manner using the method of Tajima (1993), as implemented in the MEGA2 package (Kumar *et al.* 2000). The results of these tests are detailed in columns 3 and 4 of Table 21. All results presented are from direct pairwise tests on single sequences for nucleotide positions 1 and 2. Gamma-corrected Kimura distances were used for the nucleotide tests, and Poisson-corrected

distances for the amino acid rate tests. Use of other distance correction methods provided congruent sets of results. The nucleotide-based tests all utilized the 2D method (separate analysis of transitions and transversions) as this has previously been found to be the most appropriate test on mitochondrial data. The *Tetilla* sp. poriferan sequence was used as the outgroup for all relative rate tests which did not include sponge comparisons, whereas a fungal outgroup was used for all tests that included sponges. A drawback of the Tajima test is that it utilizes only data points which present specific patterns of change between lineages – specifically only a total of two changes between the ingroup sequences relative to the outgroup. As a result the test is less powerful for smaller data sets, and the results must be regarded as conservative relative to those of the two-cluster test.

The results of the Tajima tests on the nucleotide sequence data are somewhat surprising. While the Poriferan/Anthozoan, Anthozoan/Triploblast, and Deuterostome/ Protostome results largely support the results found with the two-cluster tests, the *cox3* data shows a strong lack of support for a significant rate difference between the sponges and the triploblast lineages. In the case of both the deuterostomes and the protostomes less than 50% of the tests support a significant rate difference between them and the sponges. The deuterostome results are particularly strong. These results support the results for the two-cluster test discussed above.

As discussed above, phylogenetic trees constructed by neighbour-joining from pairwise genetic distance values suffer from a common rooting problem – this misrooting of nonmetazoan outgroups between the diploblast and triploblast radiations has been widely described in the

literature and has been termed the “diploblast monophyly” result. In particular the misrooting of the tree in the case of the mitochondrial data typically takes place closer to or within the Cnidaria, leaving the Porifera “inverted” on the final tree (see Figures 30A, 30C, and 30E). This result is a direct reflection of biases in the original genetic distance estimates and is found in both nucleotide and amino acid-based phylogenies – the long-branched triploblasts and the long-branched outgroups tend to cluster, whereas the diploblasts branch lengths exhibit more coherence as a monophyletic group and thus cluster closer to one another. Since the relative rate tests themselves are ultimately little more than comparisons of genetic distances, either pairwise or in groups, the results of the tests must reflect the same biases present in the original genetic distance estimates, and thus in the absence of a predetermined tree backbone will reflect a more equivalent rate of diploblast and triploblast sequence evolution than is actually the case. This effect will be particularly strong for the Porifera versus the triploblasts, in that the Cnidaria themselves may cluster closer to the outgroup in terms of simple genetic distances. In essence the unconstrained pairwise relative rate tests will be descriptive of the results seen in unconstrained distance-based phylogenetic trees.

The two-cluster relative rate test was reapplied to the nucleotide and amino acid sequence data using the constrained maximum likelihood tree results seen in Figures 30 and 31. The two-cluster test was performed in this case with the *tpcv* program from the *lintr* package (Takezaki 1995). The program itself is more restrictive in terms of which tests between groups may be applied, but the results of these constrained relative rate tests are presented in Table 22. Results of

these tests support the same results, although the analytical framework is not identical. Most remarkably, the results of the Poriferan rate test against the triploblasts with a fungal outgroup fail to find a significant rate difference between the taxa. Since examination of the constrained maximum likelihood trees shows that the Cnidaria exist at distances roughly midway between those of the Porifera and those of the triploblasts it is possible that the combination of the Cnidaria and the triploblasts in the tests is biasing the result against significant rate differences. The results of removal of the Cnidaria from the test are also shown in Table 22. This results in an increase in the significance of the test (from 79.6% to 91.98%) but not enough to match a 5% significance level. Further breakdown of the constrained trees into strictly deuterostome-to-sponge distances and protostome-to-sponge distances shows that this bias exists in the deuterostomes alone (not shown). The results of the deuterostome/protostome relative rate tests in the case of the constrained tree also show strong support for a significant rate difference between the two lineages for *cox3*.

We are left with strong evidence of rate differences between the Anthozoa and the triploblast lineages no matter what analysis is chosen. The sponges, despite the very frequent occurrence of even shorter internal and terminal branches than in the Anthozoa, appear to show the same trend with the single exception of the *cox3* amino acid data. There remains the strong possibility that were it possible to utilize a more closely related outgroup the test would provide a significant result due solely to a lower resulting variance in distance estimates

Relative rates of mitochondrial sequence evolution for the three genes were calculated from the constrained maximum likelihood trees.

For each pairwise rate comparison the combined branch distance from the last common node of the two taxa was used to calculate a rate ratio. Calculation of rate ratios in a strictly pairwise manner relative to an outgroup (Gissi *et al.* 2000) provided a congruent set of results. These relative rates are detailed in Table 23. The Porifera as a group appear to evolve from 5 to 10 times slower than the triploblast lineages, and the Anthozoa about 3 to 7 times slower than at the nucleotide level. The Porifera also seem to evolve somewhat slower than the Anthozoa by a factor of about 2, although *cox2* exhibits a wider range of variation. Much but not all of this rate difference is concentrated in a single long branch leading to the bilateral animal lineage, and this is consistent with the long branch observed widely in phylogenetic studies of the basal metazoa. Nonetheless the high triploblast rate is not merely the consequence of a single long branch, as shown in Figure 33. For all genes the Porifera and Anthozoa have lower intertaxon distances than do the protostomes and deuterostomes. Since the former divergence must precede the latter temporally (by at least tens of millions of years if not more), this result is likely the consequence of a consistent rate bias. The amino acid results are compatible with the nucleotide results, albeit with a lower average rate ratio. These results are compatible with the *mL* rate differences between the sponge/anthozoan divergence and the deuterostome/protostome divergence noted in Figures 24 and 25. The slower overall rate of amino acid sequence evolution in the deuterostome phyla relative to the protostomes has been noted in the literature (Ayala *et al.* 1998).

FIGURE 33

Intertaxon distances between early-diverging primary metazoan lineages, illustrating the lower diploblast rate of sequence evolution in the absence of the long triploblast branch. For the nucleotide data, gamma-corrected Kimura 2-parameter distances were estimated for all poriferan to cnidarian (filled circles) and deuterostome to protostome (open circles) combinations of identical taxa using nucleotide positions 1 and 2, transversions only. Gamma parameters of 0.45, 0.55, and 0.85 were used for *cox2*, *cox3*, and *atp6* respectively, the values being derived empirically from the data by PAUP*. For the translated amino acid sequence data the Poisson correction was used. Complete deletion of gaps was utilized in all cases. a. *cox2* and *atp6* nucleotide sequences. b. *cox2* and *cox3* nucleotide sequences. c. *cox2* and *atp6* amino acid sequences. d. *cox2* and *cox3* amino acid sequences. All other combinations of genes show the same pattern, as does the nucleotide data when transitions are included.

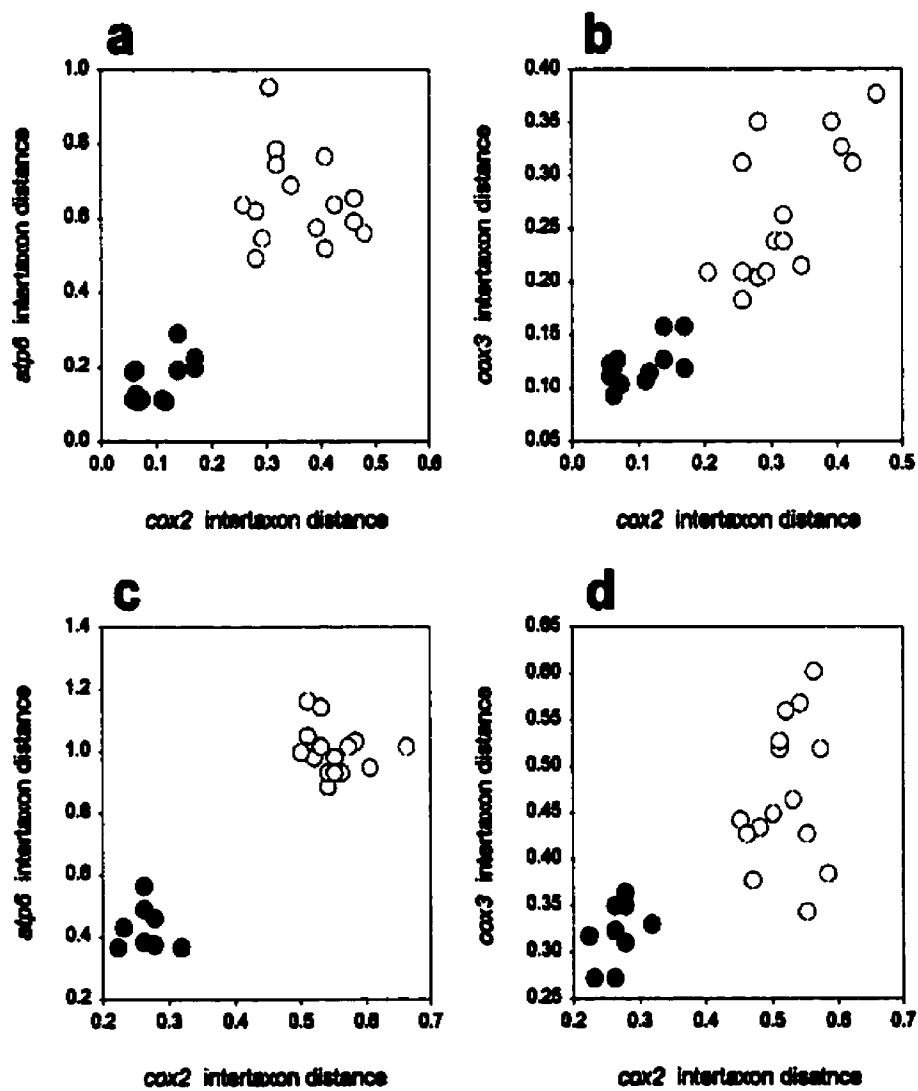


Table 21

Results of relative rate tests for mitochondrial protein-coding genes.

Columns A and B show the results of the unconstrained two-cluster test on nucleotide and amino acid data, and columns C and D show the results of strict pairwise Tajima 2D relative rate test on nucleotide and amino acid data. Two-cluster tests were performed with the Phyltest package (Kumar 1995), using the Kimura 2-parameter correction on nucleotide 1 and 2 data. Congruent results were obtained using gamma-corrected distances on transversions only. The Tajima tests were all performed with MEGA2 (Kumar *et al.* 2000), using nucleotide position 1 and 2 data with gamma-corrected Kimura 2-parameter distances.

Z-values for test results which do not reject the null hypothesis of rate equality are printed in **bold text**.

	Two-Cluster RRT Z-value (nt)	Two-Cluster RRT Z-value (amino acid)	Tajima 2D Tests Supporting Rate Difference (nt)	Tajima Tests Supporting Rate Difference (amino acid)
Porifera/Anthozoa				
<i>cox2</i>	0.9115	0.2266	3/12	1/12
<i>cox3</i>	0.1671	0.9196	0/12	0/12
<i>atp6</i>	0.8149	0.6749	0/15	3/15
Porifera/Deuterostomes				
<i>cox2</i>	3.6987	3.4573	12/16	12/16
<i>cox3</i>	2.4129	0.4853	2/16	0/20
<i>atp6</i>	3.8982	5.1029	14/15	11/15
Porifera/Protostomes				
<i>cox2</i>	2.2106	3.5343	13/20	13/20
<i>cox3</i>	3.4632	2.2650	9/20	7/20
<i>atp6</i>	4.5674	6.0939	24/25	25/25
Anthozoa/Deuterostomes				
<i>cox2</i>	5.2642	5.0987	12/12	12/12
<i>cox3</i>	5.3312	5.8507	12/12	12/12
<i>atp6</i>	6.2365	8.7179	9/9	9/9
Anthozoa/Protostomes				
<i>cox2</i>	5.6229	5.9602	15/15	15/15
<i>cox3</i>	5.9452	6.4689	15/15	15/15
<i>atp6</i>	7.0237	9.8736	15/15	15/15
Deuterostomes/Protostomes				
<i>cox2</i>	2.5099	1.5908	6/20	6/20
<i>cox3</i>	1.8037	1.4142	1/20	1/20
<i>atp6</i>	0.2045	1.4933	0/15	0/15

Table 22

Results of the 2-cluster relative rate test on a constrained maximum likelihood tree for mitochondrial protein-coding genes. Relative rate tests were performed using the tpcv program from the lintr package (Takezaki 1997). Input files for the program were constructed by hand from the output files of the constrained maximum likelihood trees generated with PAUP* (Swofford 2000) in the case of nucleotide sequence data, and from the output files of the protml program in the Molphy package (Adachi and Hasegawa 1996) in the case of the amino acid sequence data. They are therefore relative rate tests of the trees shown in Figures 30B, 30D, 30F, 31A, 31B and 31C. The only exceptions to this is for the additional *cox3* amino acid sequence test in which the Cnidaria were removed and the phylogenies again reconstructed and tested.

Results for tests which fail to reject the null hypothesis of rate equality are shown in **bold text**.

Test (Nonmetazoan outgroup)	Nucleic Acid (nt1+nt2)	Amino Acid
Sponge/(Anthozoa/Triploblast)		
<i>cox2</i>	3.3572 (99.92%)	3.6578 (99.96%)
<i>cox3</i>	2.9607 (99.70%)	1.2703 (79.60%)**
<i>atp6</i>	4.8279 (99.96%)	5.3569 (99.96%)
Sponge/(Ctenophora/Anthozoa/ Hydrozoa/Triploblast)		
<i>cox2</i>	3.3572 (99.92%)	3.7199 (99.96%)
Anthozoa/Triploblast		
<i>cox2</i>	4.6439 (99.96%)	6.1606 (99.96%)
<i>cox3</i>	6.8021 (99.96%)	6.2546 (99.96%)
<i>atp6</i>	7.0299 (99.96%)	9.2882 (99.96%)
Deuterostome/Protostome		
<i>cox2</i>	1.1943 (76.60%)	0.7869 (56.46%)
<i>cox3</i>	2.2369 (97.42%)	2.3927 (98.32%)
<i>atp6</i>	0.8672 (61.02%)	1.7916 (92.66%)

** - with Cnidaria removed = 1.7957 (91.98%)

Table 23

Relative rates of mitochondrial protein-coding gene evolution in the metazoa. Relative rates were calculated as the mean and standard deviation of the ratio of pairwise distances between taxa from the last common node between them. Results shown in bold indicate comparisons of taxa which have failed the pairwise two-cluster relative rate test of Table 21. Other methods of relative rate calculation which include distance to an outgroup (i.e., Gissi *et al.* 2000) produce compatible results. The columns labelled F1 and F2 are merely the reciprocal of the relative rate values for columns 1 and 3, to provide a more intuitive value for the degree of difference in rate of sequence evolution.

Values for comparisons which have failed to reject the null hypothesis of rate constancy are shown in **bold text**.

	Rate Pairwise (Constrained ML, nt)	F1	Rate Pairwise (Constrained ML, amino acid)	F2
(Demospongia / Calcarea)/Anthozoa				
<i>cox2</i>	1.017 +/- 0.689	1.0	0.996 +/- 0.505	1.0
<i>cox3</i>	1.148 +/- 0.242	0.9	1.166 +/- 0.417	0.9
<i>atp6</i>	1.174 +/- 0.532	0.9	0.995 +/- 0.336	1.0
(Demospongia / Calcarea)/Deuterostomes				
<i>cox2</i>	0.232 +/- 0.055	4.3	0.306 +/- 0.084	3.3
<i>cox3</i>	0.394 +/- 0.081	2.5	0.505 +/- 0.142	2.0
<i>atp6</i>	0.175 +/- 0.043	5.7	0.201 +/- 0.044	5.0
(Demospongia / Calcarea)/Protostomes				
<i>cox2</i>	0.232 +/- 0.054	4.3	0.295 +/- 0.076	3.4
<i>cox3</i>	0.330 +/- 0.071	3.0	0.372 +/- 0.098	2.7
<i>atp6</i>	0.142 +/- 0.037	7.1	0.164 +/- 0.038	6.1
Anthozoa/Deuterostomes				
<i>cox2</i>	0.252 +/- 0.162	4.0	0.303 +/- 0.144	3.3
<i>cox3</i>	0.348 +/- 0.057	2.9	0.452 +/- 0.165	2.2
<i>atp6</i>	0.180 +/- 0.092	5.6	0.201 +/- 0.060	5.0
Anthozoa/Protostomes				
<i>cox2</i>	0.252 +/- 0.160	4.0	0.291 +/- 0.134	3.4
<i>cox3</i>	0.291 +/- 0.050	3.4	0.330 +/- 0.114	3.0
<i>atp6</i>	0.146 +/- 0.075	6.9	0.164 +/- 0.051	6.1
Deuterostomes/Protostomes				
<i>cox2</i>	1.033 +/- 0.301	1.0	0.985 +/- 0.312	1.0
<i>cox3</i>	0.731 +/- 0.229	1.4	0.623 +/- 0.208	1.6
<i>atp6</i>	0.692 +/- 0.140	1.4	0.740 +/- 0.195	1.4

Synonymous and Nonsynonymous Rates

The factor which most strongly separates the diploblast from the triploblast sequences is their genetic distance. This is apparent from the results of the relative rate test, in which we see significant differences from the triploblast taxa across the diploblast radiation. Although the possible causes for significant rate differences between lineages are manifold, purifying selection acting upon the mitochondrial genomes of diploblasts can be tested directly by examination of the relative rates of synonymous and nonsynonymous codon changes.

A measure of the selective constraint acting upon a gene is omega, the ratio of nonsynonymous changes per nonsynonymous site (dN) to synonymous changes per synonymous site (dS) (Li 1997). For relatively low divergence levels the estimate of omega is independent of divergence when maximum likelihood or the approximate methods of Yang and Nielsen (2000) are utilized. In the case of extremely low divergences omega cannot be estimated reliably owing to the very low number of nonsynonymous mutations, and in the case of very high divergences such as those observed between metazoan phyla estimation of the number of synonymous changes becomes problematic as the actual number becomes much greater than two. Indeed the method of Nei and Gojobori (1986), and that of Ohta (1995) show very strong overestimates of the number of synonymous changes even at relatively low relative distances (Yang and Nielsen 2000). Maximum likelihood methods, and the approximate method of Yang and Nielsen are linear for a much wider range of relative divergences (well over 1 synonymous change per synonymous site), but the inherent error of the estimates increases greatly for extremely long divergences. For this reason most of the data

in this study cannot reasonably be utilized to estimate the omega value and thus we cannot distinguish between functional constraint and changes in the rate of mutation or fixation for the broad sweep of the metazoan radiation. However since the rate of nonsynonymous change is much lower, particularly in the case of mitochondrial genes, we can make reasonable estimates of the dN value across this range.

In order to meaningfully compare the rate of nonsynonymous substitutions the ideal analysis framework is to make comparisons between taxa with common times of divergence (Li 1997). This is a difficult task in the case of the overall metazoan radiation, as the timing of origins of the basal radiations is currently under debate (ie, Wray *et al.* 1996, Ayala *et al.* 1998, Wang *et al.* 1999). However, comparison of taxa representing several of the major metazoan phyla can be utilized to characterize at least an approximately common time of divergence. The poriferan, cnidarian, and ctenophoran lineages are clearly the most ancient of the extant phyla, and there is little doubt that the branches leading to the annellids, molluscs, and arthropods date back to a rapid enough divergence that their respective relationships are obscured in molecular data (Kumar and Rzhetsky 1996, Abouheif *et al.* 1998).

An additional problem is that of nonconcordant genetic codes. Methods of estimating synonymous and nonsynonymous codon changes rely upon estimation of the probabilities of various indirect pathways between two codons, based upon the probability of change between amino acids in a pairwise manner. Since a genetic code change between two taxa shifts the relative probabilities in amino acid change matrix in a manner that is uncharacterized temporally and in a site-specific manner, meaningful estimation of the rates of synonymous and nonsynonymous

codon change is not practical with current methods. No statistical framework currently exists for the analysis of this problem (Yang, pers. com.).

In order to approach the problem we chose only to analyze pairwise rates between taxa which share a common mitochondrial genetic code. When combined with the analysis of data which share at least an approximately common divergence time the result should reveal lineage-specific trends in nonsynonymous codon change rates, which accumulate more slowly than do synonymous changes for most genes. Figure 34 shows estimates of the number of nonsynonymous changes per nonsynonymous site for all three protein-coding genes. The estimates are shown as calculated by the methods of Nei and Gojobori (1986), Yang and Nielsen (2000), and maximum likelihood (Yang and Nielsen 1998) and are plotted against one another in order to examine the fidelity and concordance of their estimated values. Overall concordance of the results is apparent for *cox3*, with the diploblasts exhibiting a substantially lower rate of nonsynonymous substitutions per nonsynonymous site relative to any comparison of triploblast taxa with similar divergence times. Since the diploblast radiation certainly preceeds the triploblast one temporally (Conway Morris 1993) any bias induced by a longer divergence time between diploblast taxa will tend to obscure any difference, making the implications of apparent differences more rather than less significant.

In the case of *cox2* and *atp6* we see a substantial degree of discordance between the nonsynonymous rate estimates made by maximum likelihood with those made by the method of Nei and Gojobori and that of Yang and Nielsen. The relative rate tests do seem to indicate

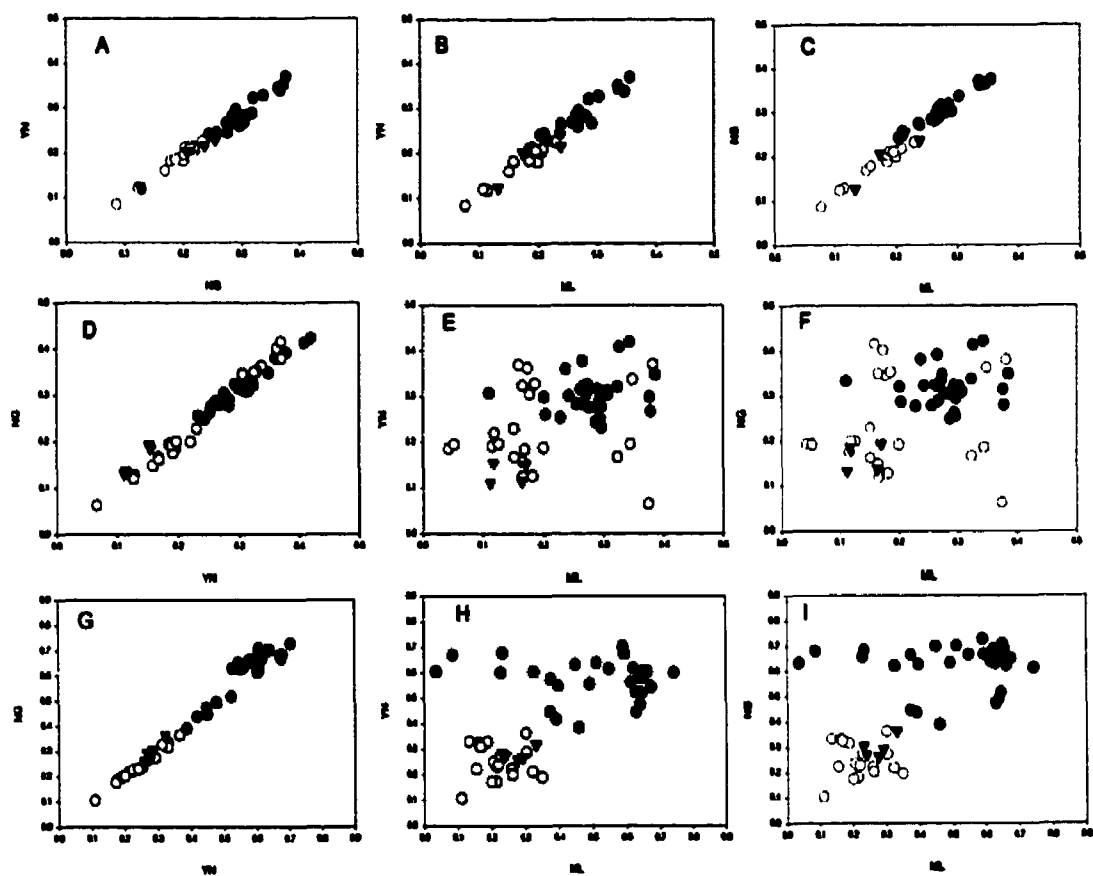
that the level of sequence divergence between the diploblasts and triploblasts are more nearly equal in *cox3*, and it seems possible that the higher levels of divergence seen in the former two genes could lead to more divergent nonsynonymous rate estimates. Therefore we will only examine the results of dN calculations derived from *cox3*, as the discordance between methods seen for the other two genes suggests that the level of sequence divergence between taxa in these genes is resulting in unreliable estimates of this parameter.

The calculation of nonsynonymous rate is based primarily on the utilization of pairwise nucleotide frequencies to quantify the probability of change between different codons, and in the case of higher nucleotide divergence discordance between the different methods could result. In particular the maximum likelihood approach relies upon estimation of the likelihood of various codon changes derived from the empirical nucleotide frequencies in an explicitly optimized model, and in the case of more extreme pairwise distances the average frequencies in the two sequences will not reliably estimate the frequencies in either sequence.

In our conservative approach to nonsynonymous rate estimation only the *cox3* results will be accepted for nonsynonymous rate comparison as they are concordant between widely different algorithms. We see a substantial (although not overwhelming) reduction of the nonsynonymous rate of codon change in the diploblasts relative to the triploblasts. In the absence of reasonable estimates of the rate of synonymous changes we cannot utilize an omega value to contrast the effects of purifying selection with those of mutation rate. A stronger set of constraints upon change in the diploblast phyla would be reflected in a lower omega value relative to the triploblasts, independent of

Figure 34

Concordance between estimates of nonsynonymous substitutions per nonsynonymous site (dN) for mitochondrial protein-coding genes. All parameter estimates were made using the yn00 program from the PAML package (Yang 2000). The F1x4 codon frequency model was utilized for all estimates in order to minimize the number of estimated parameters, although the use of other models was found to give compatible results. The package estimated the dN value via the approximate methods of Nei and Gojobori (1986) and Yang and Nielsen (2000) as well as by a strict maximum likelihood method (Yang and Nielsen 2000). A,B,C. Concordance between dN estimates for *cox3*. D,E,F. Concordance between dN estimates for *cox2*. G,H,I. Concordance between dN estimates for *atp6*. ML= maximum likelihood method, YN = Yang and Nielsen method, NG = Nei and Gojobori method.



divergence time. For the triploblasts, a relatively more relaxed set of functional or structural constraints would be reflected in a higher omega value (bias towards nonsynonymous mutations). On the other hand, a lower rate of mutational events or a more efficient error repair mechanism itself could lead to the same lowering both of overall rate and of nonsynonymous rate relative to the (unseen) lower synonymous rate. We cannot distinguish between these cases with this data set.

Figure 35 shows the dN values plotted against the Kimura 2-parameter distance for each sequence pair. We see a strong linear relationship between dN and genetic distance, with an R^2 value of almost 0.95. Clearly there must be a correlation between distance and dN whatever the case, but there is no evidence for significantly more or less nonsynonymous substitutions in the diploblasts relative to the triploblasts. The relationship between dN and genetic distance is found to be curvilinear by simulation (not shown) and a steeper curve results from more stringent constraints upon nonsynonymous changes (a higher dN/dS ratio). This is intuitively obvious as well – a lower dN value must exist on a line of smaller slope. In the case of this data the two sets do not apparently exist upon different slopes, but there is insufficient data to make any definitive statement.

Discussion

- 1) *There is little or no significant difference in nucleotide composition or codon usage between diploblasts and triploblasts.*

Analysis of these genes suggests that the overall parameters of sequence evolution in the diploblast radiation – those of nucleotide composition, and of codon usage biases, are not significantly different from those of

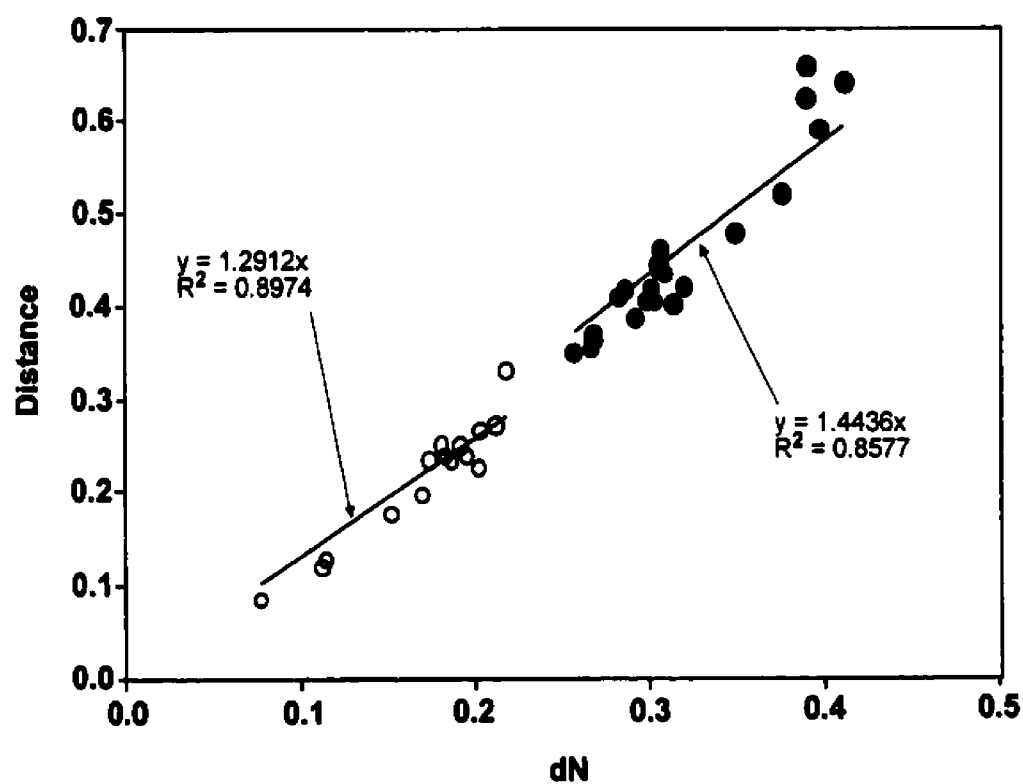
the triploblasts. For measures of nucleotide composition and bias the taxa described here do appear to cluster somewhat more closely to one another than do the triploblast taxa, but there is nonetheless no indication of a fundamentally different regime of evolutionary constraints upon the evolution of these genes in a strictly compositional sense. In all cases the diploblast values fall within the range of variation seen in the triploblasts. The previously described characteristics of metazoan mitochondrial DNA composition and bias (Wolstenholme 1992) are not altered.

2) *The Porifera and Anthozoa all occur on very short set of internal branches on constrained trees.*

Constraint of the inferred maximum likelihood tree to one which is compatible with a biologically meaningful interpretation produced a consistent reduction in the internal branch lengths of the diploblast radiation to very low values. The terminal branch lengths of the diploblasts and triploblasts are not strongly altered by this constraint, and the result itself seems largely to involve a redistribution of internal branches towards a longer single branch leading to the triploblasts. Inclusion of the very long branched *Hydra vulgaris* sequence in the *cox2* data lengthens the Porifera-Cnidaria internal branch significantly, but the branch length between the Ctenophora and Cnidaria is still very short. Although we have not shown it here, constraint of nuclear protein-coding gene data for the HSP 70 gene (Borchiellini *et al.* 1998), the Elongation Factor 1- α gene (Kobayashi *et al.* 1996), and for the data of some published 18S rRNA sequences (Kobayashi *et al.* 1993) to the same backbone tree produces an identical reduction of the interphylum distances.

Figure 35

Pairwise nonsynonymous changes per nonsynonymous site (dN) for Cytochrome Oxidase Subunit III plotted against pairwise Kimura 2-parameter gamma distance. An empirically-derived gamma parameter of 0.55 was used for the distance calculations in MEGA (Kumar 1993). The maximum likelihood estimates of the dN parameter were calculated with the yn00 program from the PROTML package (Yang 2000) for data sets divided into diploblasts and triploblasts. Open circles represent sponge and anthozoan values, filled circles represent triploblast values.



Whether this short internal branch length is one of the sources of bias in the data which leads to the inference of misleading trees remains an open question. The bootstrap values typical for the neighbour-joining trees derived from the mitochondrial nucleotide sequence data show that there is little justification for a high level of confidence in any particular inference of branching order or branch length derived from the data, and the same has held true for the results of studies on nuclear genes (Lafay *et al.* 1998, Wainwright *et al.* 1993, Kobayashi *et al.* 1993, Kobayashi *et al.* 1996, Van de Peer and Wachter 1997, Borchellini *et al.* 1998).

Whether this result represents a truly rapid ancient divergence between these phyla cannot currently be deciphered, but some recent work on nuclear 18S data (Kim *et al.* 1999) using lower metazoan taxa carefully selected for equal evolutionary rates suggests that this result could be artifactual, and a consequence of the analysis of genes with divergent evolutionary rates.

3) *The triploblasts appear to occur on a single very long branch in the metazoan radiation.*

On the other hand, the same work described above (Kim *et al.* 1999) has provided strong support for the result that the triploblast radiation has undergone an extreme genetic divergence relative to the lower metazoan phyla. Nuclear 18S data selected for low and equal evolutionary rates provided evidence for the branch between the diploblasts and the triploblasts being some three to four times longer than any other interphylum distance in any of the taxa examined. This strongly supports the results of this study which also place the triploblasts on a very long internal branch in the metazoan radiation. That this is one of the primary sources of the bias which leads to the

common finding of “diploblast monophyly” by misrooting of long-branched outgroups can hardly be doubted. The results of every study of the structure of the metazoan radiation published to date which have included a significant number of basal metazoan taxa have supported the existence and length of this branch.

What this means in terms of the early evolution of the triploblasts themselves is less clear. The existence of a single long branch similar to the one found here for triploblasts has been interpreted as evidence for a burst of accelerated sequence evolution in the dipteran insects (Friedrich and Tautz 1997), and indeed has been put forward as evidence of a short but intense acceleration of sequence evolution in the basal triploblast taxa (Cavalier-Smith *et al.* 1996) based upon a similar case for the nuclear 18S rRNA data, but care must be taken in making such inferences when faced with divergences this ancient. The extinction of multiple taxa which might have existed on transitional branches between the diploblasts and triploblasts, for instance the Archaeocyathans (Boardman *et al.* 1987), could just as easily produce a similarly long branch when the surviving taxa are examined. Nonetheless one cannot help but be struck by the congruence between the high level of both genetic, morphological, and behavioural divergence between the diploblast and triploblast phyla.

- 4) *There is strong evidence for a lower rate of sequence evolution in the Porifera and Anthozoa relative to the triploblasts at the nucleotide level.*

The rate of nucleotide sequence evolution in the Porifera and Anthozoa is lower than that of the triploblasts. The work described here leaves little doubt that these diploblasts are on shorter branches than

the triploblasts at multiple taxonomic levels. Much of the apparent rate difference could be accounted for by the long triploblast branch itself, but this is by no means the only source of the difference. Examination of both the constrained and unconstrained DNA trees reveals that the sponges in particular are on extremely short internal branches. This difference is only contrasted more sharply against the triploblasts when one notes that the sponges themselves represent the most ancient metazoan phylum, and that the degree of divergence between the demosponges and calcisponges, for instance, must be at least as long as if not greater than the length of the rest of the metazoan radiation itself.

5) There is evidence for a lower rate of amino acid sequence evolution in diploblasts in at least some protein-coding genes.

The degree of amino acid sequence divergence reflects that of the nucleotide sequence divergence, albeit at a lower level. This is a particularly relevant issue for studies which have attempted to date the origin of the metazoa as a whole using linear regression on amino acid sequence divergence data which have been correlated with the fossil record (Wray *et al.* 1996, Ayala *et al.* 1998, Lynch *et al.* 1999). If the basal metazoa exhibit a consistent bias towards lower rates of amino acid sequence evolution than do the triploblasts themselves, then the results of any such studies must necessarily be biased in favour of finding a more recent divergence than is the case (see Figure 1A). Attempts to account for the existence of overdispersed molecular clocks in the data have been made (Cutler 2000) but the effect of this systematic bias upon the results remains to be characterized.

The failure to find a significant difference between the poriferan and deuterostome amino acid sequence evolution rates is peculiar,

particularly in the presence of significance for the rate difference between the Anthozoa and the triploblasts. I believe this result is an artifact that results from the difficulty of finding an appropriate outgroup for tests of sponges against the rest of the metazoan radiation. Since the nonmetazoan outgroups are universally on very long branches, the sensitivity of the test is proportionately reduced specifically for the tests involving the sponges.

6) *There is no strong evidence for difference in nonsynonymous rate when corrected for genetic distance.*

The current data set cannot be used to distinguish between alternate hypotheses about the source of the lower rate of sequence evolution in the Porifera and Anthozoa. In the single case in which reasonable estimates of even one parameter of selective constraint can be made, that of the nonsynonymous rate ratio in *cox3*, there is no evidence that the Porifera and Anthozoa arose and remain under a stricter selective regime than the triploblast taxa. All the differences in nonsynonymous rates found between the taxa can be reasonably accounted for by a difference in genetic distance. This argument is weak, however, and a more appropriate standard by which to judge the effects of selection would be the nonsynonymous to synonymous rate ratio. A broader sample of more closely related taxa would be the appropriate data set with which to examine this question.

Bibliography

- Abouheif E, Zardoya R, Meyer A (1998) Limitations of metazoan 18S rRNA sequence data: Implications for reconstructing a phylogeny of the animal kingdom and inferring the reality of the Cambrian explosion. *J. Mol. Evol.* 47:394-405
- Adachi J, Hasegawa M (1994) MOLPHY Version 2.3: Programs for Molecular Phylogenetics Based On Maximum Likelihood. Institute of Statistical Mathematics, Tokyo.
- Adoutte A, Balavoine G, Lartillot N, Lespinet O, Prud'homme B, de Rosa R (2000) The new animal phylogeny: Reliability and implications. *Proc. Nat. Acad. Sci.* 97:4453-4456
- Aguinaldo AM, Turbeville JM, Linford LS, Rivera MC, Garey JR, Raff RA, Lake JA (1997) Evidence for a clade of nematodes, arthropods, and other moulting animals. *Nature* 387:489-493
- Ayala FJ, Rzhetsky A, Ayala FJ (1998) Origin of the metazoan phyla: Molecular clocks confirm paleontological estimates. *Proc. Nat. Acad. Sci.* 95:606-611
- Baldauf SL (1999) A Search for the Origins of Animals and Fungi: Comparing and Combining Molecular Data. *Am. Nat.* 154(4):178-188
- Baldauf S, Palmer J (1993) Animals and fungi are each other's closest relatives: Congruent evidence from multiple proteins. *Proc. Nat. Acad. Sci.* 90:11558-11562
- Beagley CT, Macfarlane JL, Pont-Kingdon GA, Okimoto R, Okada NA, Wolstenholme DR (1995) Mitochondrial genomes of Anthozoa (Cnidaria). *Prog. Cell Res.* 5:149-153

- Beagley CT, Okada NA, Wolstenholme DR (1996) Two mitochondrial group I introns in a metazoan, the sea anemone *Metridium senile*: One intron contains genes for subunits 1 and 3 of NADH dehydrogenase. *Proc. Nat. Acad. Sci.* 93:5619-5623
- Beagley CT, Okimoto R, Wolstenholme DR (1998) The mitochondrial genome of the sea anemone, *Metridium senile* (Cnidaria): introns, a paucity of tRNA genes, and a near standard genetic code. *Genetics.* 148(3):1091-1108
- Beaton MJ, Roger AJ, Cavalier-Smith T (1998) Sequence analysis of the mitochondrial genome of *Sarcophyton glaucum*: Conserved gene order among Octocorals. *J. Mol. Evol.* 47:697-708
- Berntson EA, France SC, Mullineaus LS (1999) Phylogenetic relationships within the class Anthozoa (phylum Cnidaria) based on nuclear 18S rDNA sequences. *Mol. Phylo. Evol.* 1999 13(2):417-33
- Best BA and Thomas WK (1993) Anthozoan mtDNA: A fundamentally different rate of evolution. *Am. Zool.* 34:459
- Boardman RS, Cheetham AH, Rowell AJ (eds) *Fossil Invertebrates*. Blackwell Scientific Publications, Palo Alto USA 1987
- Boore JL and Brown WM (1994) Complete DNA sequence of the mitochondrial genome of the black chiton, *Katharina tunicata*. *Genetics* 138(2):423-443
- Boore JL, Brown WM (1995) Complete sequence of the mitochondrial DNA of the annelid worm *Lumbricus terrestris*. *Genetics* 141(1):305-319
- Boore JL (1999) Animal mitochondrial genomes. *Nucl. Acids Res.* 27(8):1767-1780

- Boore JL (2000) The duplication/random loss model for gene rearrangement exemplified by mitochondrial genomes of deuterostome animals. *Comparative Genomics. Kluver Academic Publishers Computational Biology Series Volume 1.* Kluver Academic Publishers, London.
- Borchiellini C, Boury-Esnault N, Vacelet J, Le Parco Y (1998) Phylogenetic analysis of the Hsp70 sequences reveals the monophyly of the metazoa and specific phylogenetic relationships between animals and fungi. *Mol. Biol. Evol.* 15(6):647-655
- Bridge D, Cunningham CW, Schierwater B, DeSalle R, Buss LW (1992) Class-level relationships in the phylum Cnidaria: Evidence from mitochondrial genome structure. *Proc. Nat. Acad. Sci.* 89:8750-8753
- Bridge D, Cunningham CW, DeSalle R, Buss LW (1995) Class-level relationships in the phylum Cnidaria: molecular and morphological evidence. *Mol. Biol. Evol.* 12(4):679-689
- Cabot EL, Beckenbach AT (1989) Simultaneous editing of multiple nucleic acid and protein sequences with ESEE. *Comput. Appl. Biosci* 5:233-234.
- Cavalier-Smith T, Allsopp MTEP, Hao EE, Boury-Esnault N, Vacelet J (1996) Sponge phylogeny, animal monophyly, and the origin of the nervous system: 18S rRNA evidence. *Can. J. Zool.* 74:2031-2045
- Chen JY, Oliveri P, Li CW, Zhou GQ, Gao F, Hagadorn JW, Peterson KJ, Davidson EH (2000) Precambrian animal diversity: Putative phosphatized embryos from the Doushanto formation of China. *Proc. Nat. Acad. Sci.* 97:4457-4462

- Chi NW, Kolodner RD (1994) Purification and characterization of MSH1, a yeast mitochondrial protein that binds to DNA mismatches. *J. Biol. Chem.* 269(47):29984-29992
- Christen R, Ratto A, Baroin A, Perasso R, Grell KG, Adoutte A (1991) An analysis of the origin of metazoans, using comparisons of partial sequences of the 28S RNA, reveals an early emergence of the triploblasts. *EMBO J.* 10(3):499-503
- Clark-Walker GD (1992) Evolution of mitochondrial genomes in fungi. *International Review of Cytology.* 141:89-121
- Clary DO, Wolstenholme DR (1985) The mitochondrial DNA molecular of *Drosophila yakuba*: nucleotide sequence, gene organization, and genetic code. *J. Mol. Evol.* 22(3):252-71
- Clayton DA (1991) Replication and transcription of vertebrate mitochondrial DNA. *Annu. Rev. Cell Biol.* 7:453-478
- Collins AG (1998) Evaluating multiple alternative hypotheses for the origin of Bilateria: An analysis of 18S rRNA molecular evidence. *Proc. Nat. Acad. Sci.* 95:15458-15463
- Conway-Morris S (1993) The fossil record and the early evolution of the metazoa. *Nature* 361:219-225
- Conway-Morris S (1994) A paleontological perspective. *Curr. Opin. Genet. Devel.* 4:802-809
- Conway-Morris S (1998) Metazoan phylogenies: falling into place or falling to pieces? A paleontological perspective. *Curr. Opin. Genet. Devel.* 8:662-667

- Conway-Morris S (2000) The Cambrian “explosion”: slow fuse or megatonnage? *Proc. Nat. Acad. Sci.* 97:4426-4429
- Cunningham CW, Buss LW (1993) Molecular evidence for multiple episodes of paedomorphosis in the family Hydractiniidae. *Biochem. Syst. Ecol.* 21(1):57-69
- Cummings DJ (1992) Mitochondrial genomes of ciliates. *International Review of Cytology.* 141:1-65.
- Cutler DJ (2000) Estimating divergence times in the presence of an overdispersed molecular clock. *Mol. Biol. Evol.* 17(11):1647-1660
- de Bruijn MH (1983) *Drosophila melanogaster* mitochondrial DNA, a novel organization and genetic code. *Nature* 304(5923)234-241
- Dewel RA (2000) Colonial origin for eumetazoa: Major morphological transitions and the origin of bilaterian complexity. *J. Morphol.* 243:35-74
- Doolittle RF, Feng Df, Tsang Sm Cho G, Little E (1996) Determining divergence times of the major kingdoms of living organisms with a protein clock. *Science.* 271:470-477
- Elzanowski A, Ostell J (2000) The genetic codes. National Center for Biotechnology Information. URL: <http://www.ncbi.nlm.nih.gov/htbin-post/Taxonomy/wprintgc?mode=c> (2001/07/15)
- Felsenstein J (1993) PHYLIP (Phylogeny Inference Package) version 3.5c. Distributed by the author. Department of Genetics, University of Washington, Seattle.

- Feng Df, Cho G, Doolittle RF (1997) Determining divergence times with a protein clock: Update and reevaluation. *Proc. Nat. Acad. Sci.* 94:13028-13033
- France SC, Rosel PE, Agenbroad JE, Mullineaux LS, Kocher TD (1996) DNA sequence variation provides support for a two-subclass organization of the Anthozoa (Cnidaria) *Mol. Marine Biol. Biotech.* 5(1):15-28
- Friedrich M and Tautz D (1997) An episodic change of rDNA nucleotide substitution rate has occurred during the emergence of the insect order Diptera. *Mol. Biol. Evol.* 14(6):644-653
- Gamulin V, Skorokhod A, Kavsan V, Muller IM, Muller WEG (1997) Experimental indication in favour of the introns-late theory: The receptor tyrosine kinase gene from the sponge *Geodia cydonium*. *J. Mol. Evol.* 44(3):242-252
- Gehling JG (1991) The case for Ediacaran fossil roots to the metazoan tree. *Mem. Geol. Soc. India* 20:181-223
- Gehling J, Rigby J (1996) Long expected sponges from the neoproterozoic Ediacara fauna of South Australia. *J. Paleontol.* 70(2):185-195
- Gissi C, Reyes A, Peole G, Saccone C (2000) Lineage-specific evolutionary rate in mammalian mtDNA. *Mol. Biol. Evol.* 17(7):1022-1031
- Gouy N, Gautier C (1986) Codon usage in bacteria: correlation with gene expressivity. *Nucl. Acids Res.* 10(22):7055-7074
- Grotzinger JP, Bowring SA, Saylor BZ, Kaufman AJ (1995) Biostratigraphic and geochronologic constraints on early animal evolution. *Science.* 270:598-604

- Gutell RR (1993) Collection of small subunit (16S- and 16S-like) ribosomal RNA structures. *Nucleic Acids Res.* 21(13):3051-3054
- Halanych KM, Bacheller JD, Aguinaldo AM, Liva SM, Hillis DM, Lage JA. (1995) Evidence from 18S ribosomal DNA that the lophophorates are protostome animals. *Science.* 267:1641-1643.
- Hall T (1999) BioEdit – A biological sequence editor. Department of Microbiology, North Carolina State University.
<http://jwbrown.mbio.ncsu.edu/BioEdit/bioedit.html> (2001/07/15)
- Hancock JM, Volger AP (2000) How slippage-derived sequences are incorporated into rRNA variable-region secondary structure: implications for phylogeny reconstruction. *Mol. Phylo. Evol.* 14(3):366-374
- Hatta M, Fukami H, Wang W, Omori M, Shimoike K, Hayashibara T, Ina Y, Sugiyama T (1999) Reproductive and genetic evidence for a reticulate evolutionary history of mass-spawning corals. *Mol. Biol. Evol.* 16(11):1607-1613
- Hillis DM, Dixon MT (1991) Ribosomal DNA: molecular evolution and phylogenetic inference. *Q. Rev. Biol.* 1991 66(4):411-53
- Higgins DG (1994) CLUSTAL V: multiple alignment of DNA and protein sequences. *Methods Mol. Biol.* 25:307-318
- Huelsenbeck JP, Hillis DM (1993) Success of phylogenetic methods in the four-taxon case. *Syst. Biol.* 42(3):247-264
- Hyman L (1940) *The Invertebrates: Protozoa Through Ctenophora.* McGraw Hill. New York.

- Jacobs HT, Elliott DJ, Math VB and Farquharson A (1988) Nucleotide sequence and gene organization of sea urchin mitochondrial DNA. *J. Mol. Biol.* 202(2):185-217
- Jermin LS, Graur D, Crozier RH (1995) Evidence from analyses of intergenic regions for strand-specific directional mutation pressure in metazoan mitochondrial DNA. *Mol. Biol. Evol.* 12(4):558-563
- Johnson PH, Grossman LI (1977) Electrophoresis of DNA in agarose gels. Optimizing separations of conformational isomers of double and single-stranded DNA. *Biochemistry* 16:4217-4224
- Kim J, Kim W, Cunningham CW (1999) A new perspective on lower metazoan relationships from 18S rDNA sequences. *Mol. Biol. Evol.* 16(3):423-427
- Knoll AH and Carroll SB (1999) Early animal evolution: Emerging views from comparative biology and geology. *Science.* 284:2129-2137
- Kobayashi M, Takahashi M, Wada H, Satoh N (1993) Molecular phylogeny inferred from sequences of small subunit ribosomal DNA, supports the monophyly of the Metazoa. *Zool. Sci.* 10:827-833
- Kobayashi M, Wada H, Satoh N (1996) Early evolution and the phylogenetic status of diploblasts as inferred from amino acid sequence of elongation factor 1-alpha. *Mol. Phylo. Evol.* 5(2):414-422
- Kozloff EN (1996) *Marine Invertebrates of the Pacific Northwest.* University of Washington Press, Seattle.
- Kumar S, Tamura K, Nei M (1993) *MEGA:Molecular Evolutionary Genetics Analysis, version 1.0.* The Pennsylvania State University, University Park, PA 16802

- Kumar S, Rzhetsky A (1996) Evolutionary Relationships of Eucaryotic Kingdoms. *J. Mol. Evol.* 42:183-1937
- Kumar S (1995) PHYLTEST: A Program for Testing Phylogenetic Hypotheses. The Pennsylvania State University, University Park, PA 16802
- Kumar S, Tamura K, Jakobsen I, Nei M (2000) MEGA:Molecular Evolutionary Genetics Analysis, version 2.0. The Pennsylvania State University, University Park, PA 16802
- Lafay B, Boury-Esnault N, Vacelet J, Christen R (1992) An analysis of partial 28S ribosomal RNA sequences suggests early radiations of sponges. *BioSystems* 28:139-151
- Leblanc C, Boyen C, Richard O, Bonnard G, Grienberger JM and Kloareg B (1995) Complete sequence of the mitochondrial DNA of the rhodophyte *Chondrus crispus* (Gigartinales). Gene content and genome organization. *J. Mol. Biol.* 250(4):484-495
- Li, W.H. (1997) Molecular Evolution. Sinauer Associates, Sunderland, MA, USA
- Li CW, Chen JY, Hua TE (1998) Precambrian sponges with cellular structures. *Science*. 279:879-804
- Liu H, Beckenbach AT (1992) Evolution of the mitochondrial cytochrome oxidase subunit II gene among 10 orders of insects. *Mol. Phylo. Evol.* 1(1):41-52
- Liu H (1993) Molecular evolution among several orders of insects based on mitochondrial DNA analysis. Ph.D. Thesis. Simon Fraser University

- Lowe TM, Eddy SR (1997) tRNAscan-SE: A program for improved detection of transfer RNA genes in genomic sequence. *Nucl. Acids Res.* 25:955-964
- Lynch M (1999) The age and relationships of the major animal phyla. *Evolution* 53(2):319-325
- Lyons-Weiler J, Hoelzer G (1997) Escaping from the Felsenstein Zone by detecting long branches in phylogenetic data. *Mol. Phylo. Evol.* 8(3):375-384
- Martin AP, Naylor GJ, Palumbi SR (1992) Rates of mitochondrial DNA evolution in sharks are slow compared with mammals. *Nature* 357(6374):153-5
- McInnerny J (1998) GCUA:General Codon Usage Analysis v1.0. The Natural History Museum, Cromwell Road, London
- Mindell DP and Thacker CE (1996) Rates of molecular evolution: Phylogenetic issues and applications. *Ann. Rev. Ecol. Syst.* 27:279-303
- Muse SV, Weir BS (1992) Testing for equality of evolutionary rates. *Genetics.* 132:269-276
- Nei M, Gojobori T (1986) Simple methods for estimating the numbers of synonymous and nonsynonymous nucleotide substitutions. *Mol. Biol. Evol.* 3(5):418-426
- Nielsen C (1996) Cladistic analysis of the animal kingdom. *Biol. J. Linn. Soc.* 57:385-410
- Nikoh N, Hayase N, Iwabe N, Kuma K, Miyata T (1994) Phylogenetic relationship of the kingdoms Animalia, Plantae, and Fungi inferred from 23 different protein species. *Mol. Biol. Evol.* 11(5):762-768

- Nikoh N, Iwabe N, Kuma K, Ohno M, Sugiyama T, Watanabe Y, Yasui K, Shicui Z, Hori K, Shimura Y, Miyata T (1997) An estimate of the divergence time of Parazoa and Eumetazoa and that of Cephalochordata and Vertebrata by aldolase and triose phosphate isomerase clocks. *J. Mol. Evol.* 45:97-106
- OGMP (2000) The Organelle Genome Megasequencing Project.
<http://megasun.bch.umontreal.ca/ogmpproj.html> (2001/07/15)
- Ohta T (1995) Synonymous and nonsynonymous substitutions in mammalian genes and the Nearly Neutral Theory. *J. Mol. Evol.* 40:56-63
- Ojala D, Montoya J, Attardi G (1981) tRNA punctuation model of RNA processing in human mitochondria. *Nature* 290:470-474
- Okimoto R, Macfarlane JL, Clary DO and Wolstenholme DR (1992) The mitochondrial genomes of two nematodes, *Caenorhabditis elegans* and *Ascaris suum*. *Genetics* 130(3):471-498
- Paquin B, Lang BF (1996) The mitochondrial DNA of *Allomyces macrogynus*: the complete genomic sequence from an ancestral fungus. *J. Mol. Biol.* 255(5):688-701
- Perez ML, Valverde JR, Batuecas B, Amat F, Marco R, Garesse R (1994) Speciation in the *Artemia* genus: mitochondrial DNA analysis of bisexual and parthenogenetic brine shrimps. *J. Mol. Evol.* 38(2):156-16
- Page RDM (2000) TREEVIEW. Institute of Biomedical and Life Sciences.
 University of Glasgow, Glasgow, UK

- Paquin B, Lang BF (1996) The mitochondrial DNA of *Allomyces macrogynus*: the complete genomic sequence from an ancestral fungus. *J. Mol. Biol.* 255(5):688-701
- Pearson WR, Lipman DJ (1988) Improved tools for biological sequence analysis. *Proc. Nat. Acad. Sci.* 85:2444-2448
- Pesole G, Gissi C, de Chirico A, Saccone C (1999) Nucleotide substitution rate of mammalian mitochondrial genomes. *J. Mol. Evol.* 48:427-434
- Pont-Kingdon GA, Beagley CT, Okimoto R, Wolstenholme DR (1994) Mitochondrial DNA of the Sea Anemone *Metridium senile* (Cnidaria): Prokaryote-like genes for tRNA(fmet) and small-subunit ribosomal RNA, and standard genetic code specificities for AGR and ATA codons. *J. Mol. Evol.* 39:387-399
- Pont-Kingdon GA, Okada NA, Macfarlane JL, Beagley CT, Wolstenholme DR, Cavalier-Smith T, Clark-Walker GD (1995) A coral mitochondrial *mutS* gene. *Nature* 375:109-111
- Pont-Kingdon GA, Okada NA, Macfarlane JL, Beagley CT, Watkins-Sims CD, Cavalier-Smith T, Clark-Walker GD, Wolstenholme DR (1998) Mitochondrial DNA of the coral *Sarcophyton glaucum* contains a gene for a homologue of bacterial MutS: A possible case of gene transfer from the nucleus to the mitochondrion. *J. Mol. Evol.* 46:419-431
- Pont-Kingdon GA, Vassort CG, Warrior R, Okimoto R, Beagley CT, Wolstenholme DR (2000) Mitochondrial DNA of *Hydra attenuata* (Cnidaria): A sequence that includes an end of one linear molecular and the genes for 1-rRNA, tRNA^{f-met}, tRNA^{Trp}, COII, and ATPase 8. *J. Mol. Evol.* 51:404-415

- Retallack GJ (1994) Were the Ediacaran fossils lichens? *Paleobiology*. 20(4):523-544
- Reyes A, Gissi C, Pesole G, Saccone C (1998) Asymmetrical directional mutation pressure in the mitochondrial genome of mammals. *Mol. Biol. Evol.* 15(8):957-966
- Rich A, RajBhandary UL. (1976) Transfer RNA: molecular structure, sequence, and properties. *Ann. Rev. Bioch.* 45:805-860
- Robinson M, Gouy M, Gautier C, Mouchiroud D (1998) Sensitivity of the relative-rate test to taxonomic sampling. *Mol. Biol. Evol.* 15(9):1091-1098
- Robitzki A, Schroder HC, Ugarkovic D, Pfeifer K, Uhlenbruck G, Muller WG (1989) Demonstration of an endocrine signalling circuit for insulin in the sponge *Geodia cydonium*. *EMBO J.* 8(10):2905-2909
- Romano SL and Palumbi SR (1996) Evolution of Scleractinian corals inferred from molecular systematics. *Science* 271:640-642
- Romano SL and Palumbi SR (1997) Molecular evolution of a portion of the mitochondrial 16S ribosomal gene region in Scleractinian corals. *Mol. Evol.* 45:397-411
- Rychlik W, Rhoads RE (1989) A computer program for choosing optimal oligonucleotides for filter hybridization, sequencing and in vitro amplification of DNA. *Nucl. Acids Res.* 17:8543-8551
- Saito M, Kojima S, Endo K (2000) Mitochondrial COI sequences of brachiopods: genetic code shared with protostomes and limits of utility for phylogenetic reconstruction. *Mol. Phylogenet. Evol.* 15(3):331-344

- Sankoff D, Leduc G, Antoine N, Paquin B, Lang BF, Cedegren R (1992) Gene order comparisons for phylogenetic inference: Evolution of the mitochondrial genome. *Proc. Nat. Acad. Sci.* 89(14):6575-6580
- Schutze J, Krasko A, Custodio MR, Efremova SM, Muller IM, Muller WEG (1999) Evolutionary relationships of Metazoa within the eukaryotes based on molecular data from Porifera. *Proc. R. Soc. Lond. B.* 266:63-73
- Seilacher A (1992) Vendobionta and Psammocorallia: lost constructions of Precambrian evolution. *J. Geol. Soc. Lond.* 149:607-613
- Sharp PM, Li WH (1987) The codon adaption index - a measure of directional synonymous codon usage bias, and its potential applications. *Nucleic Acids res.* 15(3):1281-1295
- Simpson TL (1984) The cell biology of sponges. Springer-Verlag, NY pp.114-123
- Sprinzi M, Steegborn C, Hubel F, Steinberg S (1996) Compilation of tRNA sequences and sequences of tRNA genes. *Nucleic Acids Res.* 24(1):68-72
- Stiller JW and Hall BD (1999) Long-branch attraction and the RDNA model of early eukaryotic evolution. *Mol. Biol. Evol.* 16(9):1270-1279
- Swofford DL (1993) PAUP:Phylogenetic analysis using parsimony, version 3.1.1. Illinois Natural History Survey, Champaign
- Swofford, D.L. 2000. PAUP*. Phylogenetic analysis using parsimony (and other methods), Version 4.0. Sinauer Associates, Sunderland, Massachusetts.
- Tajima F (1993) Simple methods for testing the molecular evolutionary clock hypothesis. *Genetics.* 135:599-607

- Takezaki N (1995) **lintr** – a package for creating linearized trees. National Institute of Genetics. Mishima, Japan.
<http://iubio.bio.indiana.edu/soft/molbio/evolve/lintr/> (2001/07/15)
- Takezaki N, Rzhetsky A, Nei M (1995) Phylogenetic test of the molecular clock and linearized trees. *Mol. Biol. Evol.* 12(5):823-833
- Thompson JD, Higgins DF, Gibson TJ (1994) **CLUSTAL W**: improving the sensitivity of progressive multiple sequence alignment through sequence weighting, position-specific gap penalties and weight matrix choice. *Nucl. Acids Res.* 22(22):4673-4680
- Van de Peer Y, de Wachter R (1997) Evolutionary relationships among the eukaryotic crown taxa taking into account site-to-site rate variation in 18S rRNA. *J. Mol. Evol.* 45(6):619-30
- Van de Peer Y, Baldauf SL, Doolittle WF, Meyer A (2000) An updated and comprehensive rRNA phylogeny of (crown) Eukaryotes based on rate-calibrated evolutionary distances. *J. Mol. Evol.* 51:565-576
- Van Oppen MJH, Hislop NR, Hagerman PJ, Miller DJ (1999) Gene content and organization in a segment of the mitochondrial genome of the Scleractinian coral *Acropora tenuis*: Major differences in gene order within the Anthozoan subclass Zoantharia. *Mol. Biol. Evol.* 16(12):1812-1815
- Van Oppen MJH, Willis BL, Miller DJ (1999) Atypically low rate of cytochrome *b* evolution in the scleractinian coral genus *Acropora*. *Proc. R. Soc. Lond. B* 266:179-183

- Waggoner BM (1996) Phylogenetic hypotheses of the relationships of arthropods to Precambrian and Cambrian problematic fossil taxa. *Syst. Biol.* 45(2):190-222
- Wainwright P, Hinkle G, Sogin M, Stickel S (1993) Monophyletic origins of the metazoa: An evolutionary link with fungi. *Science* 260:340-342
- Wang DY, Kumar S, Hedges SB (1999) Divergence time estimates for the early history of animal phyla and the origin of plants, animals, and fungi. *Proc. R. Soc. Lond. B.* 266:163-171
- Watkins RF, Beckenbach AT (1999) Partial sequence of a demosponge mitochondrial genome reveals sequence similarity to Cnidaria in cytochrome oxidase subunit II and the large ribosomal RNA subunit. *J. Mol. Evol.* 48:542-554
- Wei Y (1992) Gene organization and evolution of mitochondrial genomes from two invertebrate phyla: Vestimentifera and Chaetognatha. Ph.D. Thesis. Simon Fraser University
- Willmer PG (1990) *Invertebrate Relationships: Patterns in Animal Evolution.* Cambridge University Press, Cambridge.
- Wolff G, Plante I, Lang BF, Kueck U and Burger G (1994) Complete sequence of the mitochondrial DNA of the chlorophyte alga *Prototheca wickerhamii*. Gene content and genome organization. *J. Mol. Biol.* 237, 75-86
- Wolstenholme DR (1992) *Animal Mitochondrial DNA: Structure and Evolution.* International Review of Cytology. 141:173-216.
- Wray GA, Levinton JS, Shapiro LH (1996) Molecular evidence for deep Precambrian divergences among metazoan phyla. *Science.* 274:568-573

- Xiao S, Zhang Y, Knoll AH (1998) Three-dimensional preservation of algae and animal embryos in a Neoproterozoic phosphorite. *Nature*. 391:553-558.
- Yang Z, Nielsen R (1998) Synonymous and nonsynonymous rate variation in nuclear genes of mammals. *J. Mol. Evol.* 46:409-418
- Yang Z, Nielsen R (2000) Estimating synonymous and nonsynonymous substitution rates under realistic evolutionary models. *Mol. Biol. Evol.* 17(1):32-43
- Zuker M. (1994) Prediction of RNA Secondary Structure by Energy Minimization. in *Computer Analysis of Sequence Data*. A.M. Griffin and H.G. Griffin eds. *Methods in Molecular Biology*, Humana Press Inc., 267-294 (1994)